US Army Corps
of Engineers
Baltimore District

CONSTRUCTION SPECIFICATIONS AND DESIGN CRITERIA

PATTON HALL STRATEGIC ENGAGEMENT PLATFORM
JOINT BASE MYER-HENDERSON HALL, VA

REQUEST FOR PROPOSAL. **xxxxx-xx-x-xxx**
CONTRACT NO. **xxxxxxx**
DATE: **NOVEMBER 2025**

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ACRONYM LIST

ACRONYM	MEANING
ABA	Architectural Barriers Act
AC	Air conditioning
ADA	Americans with Disabilities Act
A/E	Architectural and Engineering
AHC	Architectural Hardware Consultant
AV	Audiovisual
Ave	Avenue
AWI	Architectural Woodwork Institute
AWWA	American Water Works Association
BEAP	Base Exterior Architectural Plan
BMP	Best Management Practices
BOD	Beneficial Occupancy Date
BPA	Blanket Purchase Agreements
BVA	Best value analysis
BVD	Best value determination
C	Celsius
CAC	Ceiling Attenuation Class
CCTV	Closed-Circuit-Televisions
CDC	Certified Door Consultant
CF/CI	Contractor Furnished, Contractor Installed
CH	Chapter
CID	Comprehensive Interior Design
CMU	Concrete Masonry Units
COR	Contracting Officer Representative
CRF	Condensation Resistance Factor
CRM	Cultural Resources Manager
CRREL	Cold Regions Research and Equipment Laboratory
dBA	A-weighted decibels
DBT	Design Basis Threat
DDC	Direct Digital Control
DLA	Defense Logistics Agency
DOAS	Dedicated Outdoor Air System
DoD	Department of Defense
DOR	Designer of Record
DPW	Directorate of Public Works
DWV	drain waste vent
e.g.	for example
EHC	Electrified Hardware Consultant

ACRONYM	MEANING
LAN	Local Area Network
lbs	pounds
LCCA	Life-Cycle Cost Analysis
LEED	Leadership in Energy and Environmental Design
LID	Low Impact Development
LOW	Limits of Work
m	meters
MAU	Make-up air unit
MDF	Medium-density fiberboard
mils	one-thousandth of an inch
mm	millimeter
MPI	Master Painter Institute
MUTCD	Manual on Uniform Traffic Control Devices
NACE	National Association of Colleges and Employers
NCIDQ	National Council for Interior Design Qualification
NCPC	National Counterproliferation Center
NEC	National Electrical Code/Network Enterprise Center
NIPRNET	Non-classified Internet Protocol Router Network
NLEB	Northern long-eared bat
NRC	Noise Reduction Coefficient
OA	Outside air
OSP	Outside Plant
PCASE	Pavement-Transportation Computer Assisted Structural Engineering
PCC	Portland cement concrete
PDA	Pile Driver Analyzer
pH	potential of hydrogen
POC	point of contact
PROD	Permits of Decision
psf	pounds per square foot
psig	Pounds per square inch gauge
PVC	polyvinyl chloride
PVF2	polyvinylidene fluoride
Rd	Road
RFP	Request for proposal
RMSD	Roofing materials and systems directory
SDDCTEA	Military Surface Deployment and Distribution Command Transportation Engineering Agency
sf	square feet
SHGC	Solar Heat Gain Coefficients

EIFS	Exterior Insulation Finish System	SHS	State Highway specifications
EISA	Energy Security and Independence Act	SID	Structural Interior Design
ER	Engineering Regulation	STC	Sound Transmission Class
ESA	Endangered Species Act	SWRI	Sealant, Waterproofing & Restoration Institute
etc.	and so forth	TAA	Trade Agreement Act
EUI	Energy Use Index	TBD	to be determined
FF&E	Furniture, Fixtures & Equipment	TCB	tri-colored bat
FM	Factory Mutual	TCNA	Tile Council of North America
ft	feet	TR	Telecommunications room
Ft-C	foot-candle	TSS	Total Suspended Solids
GFGI	Government Furnished, Government Installed	TV	Television
gpm	gallons per minute	UNICOR	Federal Prison Industries, Inc
GSA	General Services Administration	US	United States
GWB	Gypsum Wallboard	USACE	United States Army Corps of Engineers
HC	Heavy Commercial	VAV	Variable Air Volume
HIPAC	High Performance Architectural Coatings	VDEQ	Virginia Department of Environmental Quality
HPL	High Pressure Laminate	VIP	very important person
ICF	Insulated Concrete Forms	VLT	Visible Light Transmission
IDG	Installation Design Guide	VTC	Video teleconferencing
IEQ	Indoor environmental quality	VOC	volatile organic compounds
IT	Information/Technology	VPDES	Virginia Pollutant Discharge Elimination System
ITN	Information Technology Node	WG	Washington Gas
JBM-HH	Joint Base Myer – Henderson Hall		

1. PROJECT DESCRIPTION

1.1 OBJECTIVES

MILITARY FACILITY	CIVILIAN FACILITY
STRATEGIC ENGAGEMENT PLATFORM	CONFERENCE CENTER

U.S. Army's objective is that the building will have a 50-year useful design life before a possible reuse, re-purpose or renovation requirement, to include normal sustainment, restoration, and modernization activities. Therefore, the design and construction should provide an appropriate level of quality to ensure continued use of the facility over that time-period with the application of reasonable preventive maintenance and repairs that would be industry-acceptable to a major civilian sector project OWNER.

Site infrastructure will have at least a 50-year life expectancy with industry-accepted maintenance and repair cycles.

The government is required by Public Law 102-486, Executive Order 13123, and Federal Energy Act 2005 to design and construct facilities in an energy conserving manner while considering life cycle cost over the life of the facilities.

Develop project site for efficiency and to convey a sense of unity or connectivity with the adjacent buildings and with the Installation as a whole.

Requirements stated in this contract are minimums. Innovative, creative, and life cycle cost effective solutions, which meet or exceed these requirements are encouraged. Further, the OFFEROR is encouraged to seek solutions that will expedite construction and shorten the schedule. The intent of the Government is to emphasize the placement of funds into functional/operational requirements. Materials and methods should reflect this by choosing the lowest Type of Construction allowed by code for this occupancy/project allowing the funding to be reflected in the quality of interior/exterior finishes and systems selected.

1.2 SECTION ORGANIZATION

This Section is organized under 5 major chapters.

- Chapter 1 - Defines project objectives and provide a comparison between military facilities and comparable “civilian” type buildings.
- Chapter 2 - Describes scope of the project.
- Chapter 3 - Describes site analysis.
- Chapter 4 - Lists applicable industry and government design criteria, generally applicable to all facility types, unless otherwise indicated in the Section. It is not an all-inclusive list. Other industry and government standards may also be used, where necessary to produce professional designs, unless they conflict with those listed.
- Chapter 5 - Provides general room requirements.

2. SCOPE

The project consists of designing and constructing the new Patton Hall – Strategic Engagement Platform (SEP). This project replaces the existing Patton Hall which has been damaged beyond repair. The new facility square footage will be equal to or less than the building square footage to be demolished which is 70,600 square feet (SF). Minimum program space requirements are defined in the Program Requirement provided in Appendix A. The facility configuration and placement must be within the defined Limits of Work (LOW) as shown on CS100 – Site Plan Limits of Work in Appendix B.

The new facility will serve as the heart and embodiment of the U.S. Army's longstanding tradition of leadership excellence. Patton Hall, for years, provided a gathering place for the Army. This new center will continue to be both an integral hub for internal community engagement, and a prominent platform for external strategic outreach in support of U.S. Army leadership goals.

Within the U.S. Army, the facility will provide a distinguished, ceremonial setting for significant milestones in the lives of soldiers and officers, including promotions, commissions, weddings, retirements, and memorial services. The facility must have a design language that respects both joyous and somber occasions, and which can flexibly accommodate a constantly rotating range of event types.

External to the service, the venue will serve as the central showcase of U.S. Army leadership. It will be the U.S. Army's primary center to build inter-agency goodwill while promoting U.S. Army leadership's strategic vision, policies, and culture. As such, the new facility must present an environment of hospitality, respect, cooperation, and unity that extends to other branches and agencies.

Where possible, these goals should be accomplished through thoughtful, comprehensive consideration of user experience, procession, programming, massing, and daylighting. The building's heritage, permanence, and respectability should be embodied in the building itself rather than relying solely on transitory elements such as finishes and artificial lighting. This attentive approach to fundamentals will ensure that the facility remains hallowed ground for generations to come.

Provide compliance with Department of Defense (DoD) Minimum Antiterrorism for Building standards. Comprehensive building and furnishings related interior design services are required. Follow the guidelines for the design and treatment of historic properties. Provide Access for individuals with disabilities Architectural Barriers Act (ABA). Incorporate Cyber Security Measures. Design and achieve Sustainability/Energy measures. Provide facilities for a minimum life span of 50 years in accordance with DoD's Unified Facilities Criteria (UFC 1-200-02) including energy efficiencies, building envelope and integrated building systems. All structures must comply with the UFCs, Federal, State, installation and local standards. Where conflict exists between the standards, the most stringent requirements will govern.

2.1 PROJECT BUILDINGS

2.1.1 BUILDING 214 – PATTON HALL

Building 214 at Joint Base Myer-Henderson Hall was constructed in 1896, with expansions in 1957 and 1968. It currently serves as a consolidated club for personnel of all ranks and their guests at Joint Base Myer-Henderson Hall. The building is a contributing element to the Fort Myer National Historic District and a National Historic Landmark. The building is a historic property per 36CFR§800.16 as documented in the Program Agreement (PA) due to its association with individuals such as former base commander George Patton. Its proximity to Washington DC enables Patton Hall to be used as a conferencing center and events venue for the Greater Washington DC Area, including Arlington National Cemetery and senior military leadership.

The original building 214 was constructed as an administration building in 1896 with additions to the building completed in 1910 and 1955. The administration building was enjoined with Building 215, an adjacent late 19th century guard house/jail house, in the 1960s, to form what is today referred to as 'Patton Hall. It is a two-story, masonry-constructed building that is capped by a slate-clad hipped roof. The foundation is clad with stone veneer, and the exterior walls are faced with brick laid in running bond. The exterior walls are pierced by 2/2, double-hung wood windows. The windows all have stone sills and brick lintels.

The south (front) elevation has a two-story projecting entry pavilion addition constructed in the 1990s that is capped by a slate clad hipped roof. The first floor of the pavilion contains modern mechanical glass doors with one-light, wood sidelights. The main entrance is sheltered under a hipped roof porch and covered walk, with Tuscan columns, which wraps around the entry pavilion. On the second story of the pavilion above the main entrance is a paired window with a semicircular transom.

A two-story, brick-faced, flat-roof addition was constructed onto the northeast side of the building in the 1950s. A two-story brick, flat roof addition and a one-story brick, flat-roof addition was constructed between the historic buildings and onto the south end of the building's original main block in the 1960s, west of the entry pavilion. The west end of the 1960s addition is enjoined with Building 215 a one-and-half story 1896 building that originally functioned as a "Guardhouse" and forms what is today referred to as "Patton Hall".

A site visit and concept design charrette conducted in 2023 identified that many facility systems were either failed or nearing failure, and that spatial arrangements did not comply with current codes. Work classifications and alternative solutions were considered, initially focusing on repairing the existing facility. During the charrette, programmatic changes progressed, but cost estimates were significantly above the expected budget. Due to the limited site area, the proposed three-story building required compliance with UFC 4-010-01, Standard 6, Progressive Collapse, which applied to new construction or major alterations and contributed to higher costs. Historic preservation requirements also added to expenditures.

After evaluating expenses and benefits related to preserving parts of the historic structure, renovation plans were discontinued. A new facility will be built to replace the original building, intended to serve as a modern venue for very important people (VIP) events and functions.

2.1.2 BUILDING 214 – GUARDHOUSE

The Guardhouse is a one-story building clad with brick laid in running bond, and is capped by a slate-clad timber-framed hipped roof with overhanging eaves and exposed rafter ends. The elevations are pierced by 2/2 double-hung, wood-sash windows located in round arched window openings with concrete sills. The roof contains hipped roof dormers with paired louvered openings. The south (front) elevation has a hipped roof porch with wood post supports. The main entrance is a double-leaf wood paneled door located inside the porch. The south half of the building includes a basement with wood joist framed first floor supported by timber girders and steel posts. The basement exterior walls are cast-in-place concrete with shallow foundations.

2.2 PROJECT DESIGN

Provide overall project design per Section 01 33 16.00 10, Design Data (Design After Award).

2.3 SITE DESIGN

The project site is located within Joint Base Myer – Henderson Hall (JBM-HH) in the National Capital Region, Arlington County, Virginia. The existing Patton Hall building is located on Buffalo Soldier Avenue near the intersection with Arlington Road. The proposed Patton Hall building will be a multi-use building with associated parking facilities. The building will need to have a Drop-Off area for VIP's and 20-passenger buses. The Drop-Off area will be sheltered via Porte Cochere or other canopy cover, installed high enough to allow bus transportation to pass through. Because the building will include a full kitchen facility, an external receiving area sized to accommodate a large box truck for delivery of goods is required for this facility.

The proposed site should be designed to meet the requirements of UFC 3-201-01 Civil Engineering as well as the Installation Design Guide for Joint Base Myer-Henderson Hall.

2.4 BUILDING DESIGNS

Contractor and Designer of Record must be responsible and provide oversight, inspection and overall project coordination throughout. Design and construction approvals are required through review, comment, resolution, and concurrence by Government. Design and construct all structures to comply with UFC 3-301-01 "Structural Engineering" and other criteria listed in Chapter 4.

Design Submittals must be in accordance with Design Submittal Procedures in the Specifications, and other sections of this request for proposal (RFP). Delegated designs and proprietary materials specified in design documents must comply with DoD regulations.

2.5 TECHNICAL AND DESIGN REQUIREMENTS FOR ROOMS, PRODUCTS, MATERIALS, AND FINISHES

For technical and design requirements of rooms, products, materials, and finishes refer to Room Data Sheets located in Appendix A. As a Strategic Engagement Platform – this facility is expected to be a world-class conference meeting center to be attended by high-ranking officials and dignitaries.

2.6 INTERIOR DESIGN

Excellence in design is a core objective for maximizing mission efficiency and long-term facility performance. Achieving this goal requires commitment to elevated quality standards, where interior design is fully integrated with architectural and engineering disciplines. The Designer of Record will be responsible for delivering Comprehensive Interior Design (CID) that demonstrates seamless coordination between Structural Interior Design (SID) and Furniture, Fixtures & Equipment (FF&E), in alignment with applicable Unified Facilities Criteria (UFC), and sustainability goals consistent with Leadership in Energy and Environmental Design (LEED) certification targets. Finishes, materials, and detailing must reflect a high level of design sophistication, contributing to operational efficiency, aesthetic coherence, livability, functionality, life-cycle value, and productivity. Interior design solutions should incorporate ergonomic principles, sustainable attributes, and a streamlined inventory of components. The quality of finishes is expected to reinforce the project's identity and mission, while supporting compliance with UFC standards and contributing to LEED credits where applicable.

2.7 COMPREHENSIVE INTERIOR DESIGN (CID)

Comprehensive Interior Design is required for spatial layout of Furniture, Fixtures & Equipment throughout and coordination of overall color scheme of building finishes through development for SID. Assure coordination of color, patterns, & finishes throughout between SID & FF&E.

2.7.1 STRUCTURAL INTERIOR DESIGN (SID)

SID includes building-related finishes, colors, and materials that are physically part of the building itself, such as walls, ceilings, floor finishes, and coverings, window treatments (blinds, shades), signage, and built-in casework.

2.7.2 FURNITURE, FIXTURES & EQUIPMENT (FF&E)

FF&E are defined as all movable / relocatable property and equipment within a building, room, or space. Designer of Record will develop room designs with FF&E layouts, demonstrating that design integrates physical sizes of FF&E and achieve necessary clearance, access, operability, and coordination with building systems. Development of FF&E includes layout, selection of finishes and colors to be coordinated with SID selections, and creation of specifications and procurement documentation required to procure and install FF&E under a separate contract.

Procurement and installation of FF&E is NOT included in this contract. With separate funding, Government must procure and install FF&E under a separate contract. Contractor will accommodate that effort with allowance for entry of FF&E vendor/installer onto this project site at appropriate time to permit completion of installation for a complete and usable facility coinciding with Beneficial Occupancy Date (BOD). The furniture vendor/installer contract will include all electrical pre-wiring and whips for final connection to the building electrical systems; however, Contractor will make final connections to building systems under this contract. Furthermore, General Contractor will provide all Information/Technology (IT) wiring (i.e. Local Area Network (LAN), phone, etc.) up to and including faceplates. Contractor to perform connections for all freestanding systems furniture, desktops, and equipment as applicable, with services to install cable and face plates in coordination with FF&E vendor/installer to accomplish installation at the appropriate time, along with final IT connections to building systems under this contract.

Preliminary FF&E information, including dimensions and sizes, are provided on Room Data Sheets; refer to Appendix A. Scheduled items of FF&E is not considered an all-inclusive list.

2.8 AMERICANS WITH DISABILITIES ACT (ADA) / ARCHITECTURAL BARRIERS ACT (ABA)

Project designs will achieve full compliance with ABA as required by federal law. ABA contains scoping and technical requirements for accessibility to sites, facilities, buildings, and elements by individuals with disabilities. Apply requirements during design for new construction, additions, alterations, and lease of sites, facilities, buildings, and elements to the extent required by regulations issued by Federal agencies under the ABA.

2.9 ANTITERRORISM AND FORCE PROTECTION REQUIREMENTS (AT/FP)

There is a Design Basis Threat (DBT) for the building design that requires threat enhancements to the site and the building. The applicable Level of Protection for the new building is indicated in the DBT which is greater than the minimum AT standards outlined in UFC 4-010-01. Contractor's design solution has to comply with UFC 4-010-01 and UFC 4-020-01 and will incorporate items listed in the DBT. The DBT document is categorized as Controlled Unclassified Information (CUI). A copy of the DBT will be provided by the Contracting Officer upon request by the Contractor. Contractor must manage and protect the DBT in accordance with DoDI 5200.48.

2.10 INSTALLATION DESIGN GUIDE / INSTALLATION PLANNING STANDARD

Built environments develop a sense of community, which affect overall safety, visual appeal, sense of place, and provides visual orientation across the installation. JBM-HH, Installation Planning Standard (IPS) provides guidance for features of Site Planning, Building Standards, Circulation (pedestrian, vehicles, & trails), Signage, Road Networks, Building Material & Colors, Utilities, and Landscaping. Project design is required to comply with JBM-HH, Installation Planning Standards (IPS). and following design guidelines outlined in the agency Programmatic Agreement (PA)) found in Appendix H. JBM-HH, IPS identifies the project location within the Fort Myer Historic District for building type use being a strategic engagement platform (assembly).

The Government initiated the Section 106 process (36 CFR Part 800) with consulting parties. The Programmatic Agreement that was executed during the development of the RFP can be referenced for more information (see Appendix H). The selected Contractor and their designer of record are responsible for providing the Government supporting design documents as described in the PA to finalize the Section 106 process. As part of this RFP, the Contractor is responsible for adhering to all the requirements stipulated in the PA.

In addition, this project requires coordination with, review and approval by Commission of Fine Arts (CFA) and National Capital Planning Commission (NCPC). During RFP development, the Government initiated contact with the agency staff to describe the project scope and establish design guidelines. The selected Contractor and their designer of record are responsible for understanding and accepting requirements for design approval by these agencies. The Contractor is responsible all aspects of the review process in order to receive project concurrence from NCPC and CFA. Reference the agency web sites for submission requirements. Both agencies necessitate different submittal requirements, timeline, and/or lead-time for concurrence. Contractor is responsible for developing and coordinating various submittal criteria with appropriate lead-time, and documents necessary to achieve required approvals prior to start of construction. Contractor will coordinate requirements for each agency and provide all necessary submittal documents to the Contracting Officer for review prior to submitting to the outside agency. The Government will support the submissions with required signatures or cover letters from the owner, JBM-HH DPW.

2.11 SUSTAINABLE DESIGN AND CONSTRUCTION

Design and construct project to comply with sustainability requirements identified in Part 2 Section 01 33 29, Sustainability Reporting for Design-Build. Design and construct all scope items in accordance with UFC 1-200-02, High Performance and Sustainable Building Requirements.

2.12 LEADERSHIP IN ENERGY AND ENVIRONMENTAL DESIGN (LEED)

Project is required to achieve, at a minimum, a LEED rating of Silver under Version 4 of LEED. Offerors will not assume LEED credit points, propose strategies or decisions not within control of the Contractor. Develop Sustainability strategies during design and achieve objective through management of construction processes. The project has been registered with USGBC. The Project ID is 1000222332. Upon award contractor has to request access/control of the project from the Contracting Officer.

2.13 CYBERSECURITY

Design, acquire and execute all control systems (including systems separate from management utility monitoring and control system) in accordance with UFC 4-010-06 Cybersecurity of Facility-Related Control Systems. The DOR is required to create a tailored CCI and Controls list for each identified FRCS that is applicable to the project. Coordination between the DOR and each system’s Authorizing Official and Systems Owner is required to establish the Confidentiality-Integrity-Availability (CIA) of each FRCS.

Several Facility Related Control Systems have been identified for the new building. A smart electric meter will be installed by Dominion Electric to track power consumption. Additionally, a Monaco-based fire alarm reporting system will receive a new extension card at the D21 console. Both of these systems will require cybersecurity controls per UFC 4-010-06.

2.14 OPERATION

The facility will have normal operating hours; however, due to the nature of the facility, there will be times when the facility is occupied after hours for special events such as military balls, conferences, and weddings.

2.15 STAFFING/OCCUPANCY

The number of occupants specified in the RFP are identified for programming purposes. DOR is to determine the final occupancy necessary for designing standard building features such as structural, egress, and plumbing fixtures as specified in applicable building or life safety codes.

2.16 SPECIAL DESIGN CHALLENGES

Pool must remain open and accessible during normal summer operations (Memorial Weekend to Labor Day; daily from 1100 to 1900). Access to a pool house (gender specific restrooms, shower, locker room, pool storage) must remain available during pool operations. Once the Patton Hall building has been demolished, the pool area will have a gap in perimeter security. The pool area perimeter security will need to be preserved at all times with permanent measures IAW UFC 4-750-07 Recreational Aquatic Facilities paragraph 4-2.4.

Traffic cannot be hindered or blocked along Buffalo Soldier, Custer Road, and Johnson Lane without prior approval and coordination with Contracting Officer. Request must be submitted to Contracting Officer a minimum of 14 calendar days. Request has to be made by submitting JBM-HH Form 7 (JBM-HH_Request_for_Support_Form_7_Blank.pdf).

During demolition of Building 214, the Guardhouse must not be structurally compromised or degraded beyond current condition.

Phasing requirements of the project are to be determined and proposed for approval by the Contractor.

2.17 GOVERNMENT-FURNISHED GOVERNMENT INSTALLED EQUIPMENT (GFGI)

Contractor must coordinate with Contracting Officer for requirements of Government Furnished Government Installed (GFGI) items scheduled, and provide design and deliver suitable structural support, brackets for projectors/ TVs, with all utility connections and space with required clearances for all GFGI items. Fire extinguishers, fire extinguisher brackets, and cabinets are Contractor furnished and installed (CF/CI). All Computers and related hardware, copiers, faxes, printers, video projectors, and TVs are GFGI. For scheduled GFGI items, refer to Room Data Sheets.

2.18 BUILDING COMMISSIONING

Provide commissioning to meet requirements identified in Section 01 91 00.15 Total Building Commissioning, ER 1110-345-723 Total Building Commissioning and UFC 1-200-02 High Performance and Sustainable Building Requirements. A draft copy of the Owner's Project Requirement (OPR) can be found in Appendix A.

3. SITE ANALYSIS

Site work consists of all aspects of a complete project site. This includes site planning and design of all new site features (buildings, roads, parking, walkways, fencing, etc.), demolition, earthwork / grading, storm drainage, utilities, signage, landscaping, force protection, stormwater management and erosion and sediment control, and obtaining necessary permits.

The existing site consists of a 77,600 square foot (SF) building that consists of several building additions from the original 1896 building. The building was expanded between the years 1957-1968. Building 214 will be demolished under this project with the exception of the "Guardhouse" located on the west side of Patton Hall. The Guardhouse will be preserved by this project; in addition to restoring the original exterior condition of the Guardhouse, all utilities services and accessibility will need to be restored. The Guardhouse must be occupiable once Building 214 is demolished. The current building orientation has multiple entrances with a main entrance and drop off along Buffalo Soldier Ave on the south. There is an additional entrance on the south side of the building at the historic guardhouse and an entrance on the east side of the building adjacent to Custer Road and the parking lot. The rear of the building that is connected to the swimming pools is accessed from the lower level and secured with fencing.

Custer Road is situated along the east side of the building. It provides access to the facilities located to the north of Building 214 as well as provide an area for parents to drop children off and access the swimming pools from the exterior. The existing parking lot has two entrances on Custer Road. The boomerang-shaped parking lot has 107 existing parking spaces with five ABA accessible parking spaces positioned closest to the building entrance. The parking lot is 5ft-6ft higher than Custer Road with the grade differential accommodated by steep slopes and a concrete retaining wall.

The area east of the existing parking lot has been designated by the installation as a dedicated green field as part of their existing stormwater permit. The "green space" was required as part of another project on the installation as an exchange for impervious surface to promote water quality and stormwater infiltration. This green space has always been an open area on post. It was a historic parade ground during the late 19^{th} century and used for tennis courts throughout much of the 20^{th} century. The tennis courts were recently demolished for stormwater compliance. The permitted green space is delineated on CS100 Site Plan Limits of Work (Appendix B); permitting/pollution reduction requirements are illustrated in site drawings provided in Appendix B.

3.1 EXISTING SITE DESCRIPTION

The existing site consists of the Patton Hall building which is located on the north side of Buffalo Soldier Avenue (Ave) near the intersection with Arlington Road. The existing Patton Hall building is bound by Custer Road (Rd) and parking to the east, residential housing and Buffalo Soldier Ave to the south, the historic Guardhouse and Building 216 to the west, and three existing swimming pools to the north. Custer Road intersects with Arlington Ave at a point slightly east of alignment, resulting in a non-standard, angled junction rather than a traditional four-way intersection. The project site includes the existing Patton Hall building, but also Custer Road, parking lots, and the green space to the east of the Patton Hall building. The site boundary can be seen in Appendix B - CS100 and CS101 – Site Plan (Limits of Work). Native files of the site plan can be provided to the contractor upon request to the Contracting Officer.

The existing parking lot associated with Patton Hall has 107 parking spaces located east of Custer Rd. Access to the parking lot is via two entrances on Custer Rd with one-way traffic flow from south to north with a 23.5-foot-wide drive aisle. It should be noted that the existing parking lot does not meet the current UFC parking lot requirements. There is additional parking space for the facility along Buffalo Soldier Ave via on-street parallel parking and in the parking lot to the southwest of Patton Hall.

3.2 CONSTRUCTION PHASING, STAGING, AND CONTRACTOR ACCESS

There is limited available land for contractor staging and laydown. The contractor staging will be within the LOW as shown on CS100 – Site Plan Limits of Work in Appendix B. Contractor is to submit a construction phasing plan for review and approval by Directorate of Public Works (DPW). Contractors will access Patton Hall through the JBM-HH Visitor Control gate located at 101 Carpenter Rd, Fort Myer, Virginia 22211. After entering the installation, contractors should head northeast, turn left onto McNair Road, then left again on Buffalo Soldier Avenue to reach the project site. Safe pedestrian and vehicular access to the swimming pools is required during the operational months.

3.3 DEMOLITION

Demolition of the existing Patton Hall building with the exception of the Historic Guardhouse is part of this RFP. Along with the demolition of the building, there will be existing utility service demolition. All of the services to the existing building will be removed and cut and capped at the main. Demolition of the exterior electrical equipment (medium voltage pad-mounted switchgear, pad-mounted transformer, generator, and automatic transformer switch) will be removed by Dominion Energy. Demolition of telecommunications duct bank and cabling from existing manhole unknown D to the Patton Hall 5-foot line. New electrical and communications services will need to be established to the guardhouse in order for it to be occupiable. The excavated area from the removal of the Patton Hall building and lower level, should be backfilled with suitable material, compacted, and seeded. The area should be graded with slopes at a maximum of 5:1 with positive drainage towards the existing or proposed storm drainage system. Preserve or improve upon stormwater flow ensuring not to add stormwater flow into the pool area. Depending on the location of the new facility, the existing parking lot may be demolished along with Custer Road and associated concrete retaining wall. Following the demolition of Building 214, it will be necessary to install a new fence to secure the pool area and ensure continuity with the existing fencing. Perform a structural assessment of the existing foundations at the interface between the existing Guardhouse and the Patton Hall building prior to demolition to establish a demolition plan that ensures the Guardhouse foundations will not be undermined at any point during the demolition process. If determined necessary by structural analysis, provide underpinning for the guardhouse foundations. At the completion of demolition, the Guardhouse walls foundations will have the same or higher load bearing capacity as they did prior to demolition. Any foundation or basement wall for which loading is increased after demolition must be analyzed in accordance with International Existing Building Code (IEBC) and strengthened as required to meet code requirements. This includes Guardhouse basement walls that may experience an increase in lateral earth pressure when the adjacent demolished Patton Hall building basement is backfilled with earth fill.

Depending on the location of the new facility, several mature trees are located within the LOW that may need to be removed. Due to the Northern long-eared bat (NLEB) and the Tri-colored bat (TCB) and their status on the Endangered Species List, tree clearing will be limited to the non-active, overwintering season November 15 – April 1.

3.4 NEW CONSTRUCTION SITE LAYOUT GUIDELINES

The new site layout will be required to be within the LOW shown on CS100 Site Plan in Appendix B. According to the Installation Planning Standards and DPW, a minimum separation of twelve feet between buildings is required. The minimum building separation distance will be governed by the DBT. At the conclusion of the project, within the LOW, a minimum of 107 parking spaces must be provided to preserve the existing capability. The proposed parking layout is required to be per UFC 3-201-01 Civil Engineering and Military Surface Deployment and Distribution Command Transportation Engineering Agency (SDDCTEA) Pamphlet 55-17 with 9-foot wide by 18.5-foot-long standard ninety-degree parking spaces. Additionally, parking has to comply with current ABA guidelines. Parking lot pavement markings will be provided in accordance with SDDCTEA Pamphlet 55-17 and the Manual on Uniform Traffic

Control Devices (MUTCD) including the Department of Defense supplement to the MUTCD. Any existing pavement that will be reused to satisfy the parking lot authorization must be milled and overlaid with a minimum 1.5-inch asphalt surfacing.

The parking area may be constructed of asphalt, concrete, or pervious surface. The parking lot will be bound by concrete barrier curb with concrete curb and gutter in areas that provide a stormwater flow path. The pavement thickness has to be in accordance with the project geotechnical report, designed with the Pavement-Transportation Computer Assisted Structural Engineering (PCASE) software, and in accordance with UFC 3-250-01 Pavement Design for Roads and Parking Areas.

A 20-foot-wide drop-off lane is to be provided at the main entrance to the proposed building. The queue length should accommodate a minimum of three vehicles (approximately 60 feet). The drop-off area at the building should have a covered canopy that allows for the drop-off lane and entrance to be undercover. The canopy should be tall enough to accommodate emergency vehicles. The drop-off lane can also serve as emergency vehicle access to the building. The drop-off lane access should be coordinated with the ATO.

Pedestrian circulation must be provided that connects the parking lot to the new building and existing pedestrian facilities. Sidewalks is to be a minimum of 4-inch-thick Portland cement concrete that are a minimum of 4-foot wide in typical pedestrian areas and a minimum of 6-foot wide in main sidewalks with higher pedestrian traffic areas. Sidewalks are to comply with ABA guidelines and provide an accessible path connecting the main entrance, ABA parking spaces, main sidewalk network, and site amenities.

Because the building will include a full kitchen facility, an external receiving area sized to accommodate a large box truck for delivery of goods is required for this facility. The receiving area adjacent drive should accommodate the turning radii of the large box truck. The receiving area will require security with access control for vehicles to be able to drive up to the building.

Betterments - Enlarged drop-off lanes and larger canopy to accommodate more vehicles; longer queue drop-off length, recessed loading dock and dock leveler, landscaping; exterior gathering space with outdoor seating; and additional parking.

3.5 EXISTING FACILITIES

The Guardhouse located on the west side of Patton Hall Building 214 is to remain operational. Demolition of Building 214 will require structural stabilization of the Guardhouse as well as restoration of utility service and building envelope restoration. The Guardhouse must be occupiable at the conclusion of this project. Additionally, the existing swimming pool area is to remain and have access during the operational months.

The green field to the east of the existing parking lot is a Virginia Department of Environmental Quality (VDEQ) permitted green space. If development occurs in this area resulting in reduced permitted green space, the permitted area will require modification or relocation and must be reestablished in an alternative location. This will be in addition to the VDEQ pollutant reduction requirements established for this project. This will need to be coordinated with installation staff and VDEQ. The Pollutant reduction credits obtained in the permitted green space are listed below. The existing Stormwater permitting documents and the Fort Myer Tennis Court Demolition Plan can be seen in Appendix C.

- Total Nitrogen: 5.59 lbs/year
- Total Phosphorus: 1.61 lbs/year
- Total Suspended Solids (TSS): 1,624.40 lbs/year

3.6 GRADING

Grading for the site must direct water towards new or existing drainage features and away from impervious site features. Grading must not result in low spots that hold or direct water towards new or existing site features. Grading for the project has to comply with applicable sections of UFC 3-201-01 Civil Engineering. The Finished First Floor Elevation is required to be 6 inches higher than adjacent ground. The grading should provide positive drainage throughout meeting UFC required minimum and maximum grades as shown in Table 2-2. The grading is to be integral with the storm water management plan.

3.7 UTILITIES

The existing utilities within the LOW consist of water, wastewater, stormwater collection, electric, telecommunications, and natural gas. Existing site utility drawings are provided in Appendix B. Equipment and connection costs for the privatized utilities will be paid directly to the Defense Logistics Agency (DLA) by United States Army Corps of Engineers (USACE); it will not be the responsibility of the Contractor to contract directly with the private utility provider for the utility installations and/or connections in support of this project.

Utilities will need to be restored to the Guardhouse when Building 214 is demolished. Currently there appears to be water and sewer service directly supporting the Guardhouse; all other utilities and building services are provided from Building 214.

Utility	Government or Privatized	POC
Water	Government – DPW	David Mayeda
Wastewater	Government – DPW	David Mayeda
Natural Gas	Government – Washington Gas (WG)	**Contractor to work directly with WG for utility hookup/acceptance – will pay WG directly then transfer to DPW at building acceptance** https://www.washingtongas.com/services/contractors/service-territory Bill Lucas William.f.lucas14.civ@army.mil 703.357.8311
Electric	Privatized – Dominion Energy	James Bakos james.s.bakos@dominionenergy.com 571-271-2180
Telecommunication	Government – Fort Myer NEC Privatized – Verizon	NEC – Jose Molina Delgado Verizon POC TBD
Stormwater	Government – VA Department of Water Quality for Arlington County	

3.8 WATER

The water conveyance system on JBM-HH is owned and operated by the installation. The exiting site is served by a 10-inch water main running parallel along the south side of Patton Hall and the north side of

Buffalo Soldier Ave. The 10-inch main has several service connections to Patton Hall and a 4-inch main that connects the swimming pool area. Existing fire hydrants are positioned along Buffalo Solider Ave to the east and the west of Patton Hall. The fire service and hydrant locations are required to comply with UFC 3-600-01 Fire Protection for Facilities with Change 6.

3.9 WASTEWATER

The gravity sanitary sewer system on JBM-HH is owned and operated by the installation. There is a sanitary sewer main that runs along the south side of Patton Hall then gravity flows to the south along Arlington Ave and ultimately to the treatment plant off the installation. Per discussions with DPW, this main is a 6-inch clay pipe that is aged and not adequate for the new sewer connection. There are several sanitary sewer laterals connecting to the rear of the building (north side). There is an 8-inch PVC sanitary sewer located north of Swimming Pool #3 that can provide a potential gravity sewer connection. There is a manhole located at the curve in Custer Road with an invert of 215.66 and a manhole located north of Swimming Pool #3 near Building 203 with an invert of 210.92. To access the manhole near Building 203, the connection can be directional bored between Swimming Pool #2 and Swimming Pool #3.

3.10 ELECTRIC

The existing exterior electrical equipment consists of a 4-way medium voltage pad-mounted switchgear, pad-mounted transformer, automatic transfer switch, and emergency generator. All of the exterior electrical equipment is owned and operated by the base privatized utility company, Dominion Energy. The sizes of the exterior electrical equipment have been confirmed with Dominion Energy. JBMHH DPW electrical engineer provided an e-letter confirming the medium voltage distribution system is considered reliable in accordance with NFPA 20.

NATURAL GAS

There is a gas main that runs along the north side of Buffalo Soldier Ave. The gas is owned and operated by Washington Gas. Per DWP Utilities contact, the gas main should be adequate to service Patton Hall. Once the gas equipment is specified, a load sheet should be submitted to Washington Gas, and they will determine the meter size. Once the load sheet is submitted, Washington Gas can verify that the main is adequate.

3.11 COMMUNICATIONS

Building 214, Patton Hall is serviced with incoming fiber optic cable from a manhole located North of the building. Two lines are necessary, one Non-classified Internet Protocol Router Network (NIPRNET) and one privatized, coordination with both required.

3.12 EROSION AND SEDIMENT CONTROL, STORMWATER MANAGEMENT, AND CONSTRUCTION GENERAL PERMIT

Erosion and sediment controls will minimize the discharge of pollutants from earth disturbing activities in conformance with VDEQ. Pre-coordination has occurred with VDEQ, Mr. Chantz Ballard during the RFP development.

a. To meet VDEQ water quality requirements, the purchasing of water quality credits may be utilized in accordance with VDEQ standards. The Contractor's DOR must take into account the existing permitted pollution reduction (as noted in Section 3.5) for any development within the Dedicated Green Space in addition to the water quality requirements of the new development. The Contractor and their designer of record are responsible for obtaining Erosion and Sediment Control (E&S) and Stormwater Management Plan (SWM) Plan approval and the Construction General Permit from the Viriginia Department of Environmental Quality (VDEQ). The

Contractor is responsible for all permit and/or review fees. The Contractor and their Designer of Record (DOR) are required to be thoroughly knowledgeable of the latest VDEQ E&S and SWM requirements. Design the site to comply with the most recent version of the *Virginia Stormwater Management Handbook*, Virginia Runoff Reduction Method (VRRM) for water quality, and VA Code Chapter 875 (9VAC25-875-600) for water quantity.

b. During design phase, the Contractor's DOR is to initiate and coordinate with the VDEQ Plan Review Manager to ensure the design will comply with E&S and SWM requirements. USACE and DPW-EMD representatives need to be invited to all meetings with DEQ.

c. Prior to the first VDEQ submission, submit the E&S/SWM Plans and report to the COR for review by USACE and DPW-Environmental Management Division (DPW-EMD). The government will have 14 days to review. Once EMD and USACE comments have been addressed to the satisfaction of USACE and EMD, submit to VDEQ.

d. Submissions to VDEQ Plan Review are electronic. Submission files need to include a transmittal form, submission checklist, cover letter, plans, and report. More information can be found on https://www.deq.virginia.gov/water/stormwater/stormwater-construction/plan-review. Copy USACE and DPW-EMD on all submissions to VDEQ.

e. It is the responsibility of the Contractor to account for all submission review periods within the design schedule. Each submission should be scheduled to allow the reviewing authority time to make comments and request resubmission. It is the Contractor's responsibility to accommodate the review process such that the start of construction is not delayed.

f. If the limit of disturbance is one acre or more, the Contractor has to submit a Construction General Permit (VPDES) Registration Statement and obtain the permit prior to the start of construction. The Construction General Permit web site can be found at: https://www.deq.virginia.gov/water/stormwater/stormwater-construction. Contractor is required to submit all required documentation under "Construction General Permit Fees and Forms".

g. The Construction General Permit requires the Contractor to develop and implement a site-specific Stormwater Pollution Prevention Plan (SWPPP) with the VPDES application. The approved E&S plans are required to be used to the greatest extent possible as part of the SWPPP package.

h. During construction of the permanent SWM facilities, the Contractor is responsible for documenting as-built data to ensure the facilities are constructed per the approved plans and specs. Provide as-built topographic survey and redline drawings on the approved SWM plans. Contractor is required to provide the Engineer's Certification Statement; the Government will not certify the stormwater as-builts. Stormwater as-builts will be submitted to the Government for review prior to submitting to VDEQ with the Notice of Termination process. The permit cannot be closed out until the SWM as-builts are approved by VDEQ.

During construction, Contractor must keep hard copy of approved SWPPP, E&S/SWM plans, report, and permit on site at all times.

3.13 LANDSCAPE DESIGN

3.14 OVERALL LANDSCAPE DESIGN

Preserve and protect the historic landscapes and streetscapes at JBM-HH. Landscape design should be sensitive in accommodating twenty-first-century mission needs while respecting the rich history of the installation.

Betterment – Enhanced landscaping around the Strategic Engagement Platform; additional grading around pool area to create patio seating area.

3.15 TREE PRESERVATION AND REFORESTATION

The designated pattern of tree plantings define outdoor spaces and are important to the character of the installations. Spaces such as the parade fields at both Fort Myer and Fort McNair are listed as contributing features of the National Register historic districts. It is essential to keep the relationship between the buildings and the designed landscape intact to avoid adverse effects. Do not add trees where there were none or remove trees without replacing them with the same species. Tree removal should be avoided if possible. The removal and replacement of trees are to be in accordance with NCPC guidelines which can be found at https://www.ncpc.gov/docs/publications/Tree_Preservation_and_Replacement_Resource_Guide_2020.pdf.

3.16 ENVIRONMENTAL

3.16.1 UNANTICIPATED CULTURAL RESOURCES DISCOVERY

New construction should also take into consideration the potential for archaeological discovery. In the event that buried cultural resources are encountered, the Cultural Resources Manager (CRM) must be consulted. This project does not anticipate any inadvertent discoveries, as defined by the Native American Graves and Protection Act.

3.16.2 ENDAGERED SPECIES

Project construction will need to be sensitive to birds and bats and adhere to the Migratory Birds Treaty Act. Any drilling or blasting that will produce noise or vibrations above existing background levels could negatively impact suitable summer habitat for the Northern long-eared bat and the Tri-colored bat. The NLEB is listed as endangered under the Endangered Species Act (ESA); the TCB has been proposed to be listed as endangered but isn't officially documented on the ESA list at this time. Tree clearing will be limited to the non-active, overwintering season Nov. 15 – April 1. If it is anticipated that these drilling or blasting activities will be loud enough to startle roosting bats in adjacent suitable habitat (~120 dBA), then these activities should be avoided during the pup season from May 15 - July 31. Additionally, construction lighting should be directed into the action area of the construction site, and permanent lighting should be consistent with pre-project lighting in the area.

3.17 PLANT AND TURFGRASS (TEMPORARY AND PERMANENT SEEDING)

3.17.1 SELECTION

All plant and seed selections are required to be updated to reflect new and improved cultivars that are hardier and have greater pest and disease resistance. Maintain and/or establish the maximum amount of vegetation within the LOW including seeding and/or planting of native vegetation for stabilization and landscape purposes after construction is finished.

3.17.2 ESTABLISHMENT AND WARRANTY PERIOD

All new plants and sod must be guaranteed for 12 months. All new plants and seed provided for this project will have an establishment period to commence on the date that inspection shows that the plants installed have been satisfactorily installed.

Maintenance and care are to begin immediately after new plants and seeded areas have been installed. Maintenance of plant material and seeded areas, especially during establishment, should be included as part of 1-year warranty and performed on a regular basis (minimum frequency of monthly). Maintenance is to include but not be limited to: Watering, Weeding/Invasive Species removal, Mowing, Mulching, Debris Removal.

The establishment period is to continue for a minimum period of 12 months for all new plants and seeded areas after date of final acceptance. During the 12-month establishment period, the contractor is to be responsible for regular watering of new plants and seeded areas. Regular watering is required and has to be performed on a consistent basis to ensure successful establishment of plant material and seeded areas. Allowing plant material to die from lack of regular watering because it is cheaper to replace dead plant(s) at end of warranty period is unacceptable. The use of watering aids equal or similar to Tree Gator watering bags is strongly encouraged and should be specified in either the drawings and/or specifications. A minimum 12-month guarantee has to be indicated so that any new plants and seed planting areas which have died (or have grown in a manner uncharacteristic of its type) within the 12-month establishment period will be replaced at the contractor's expense. Seeded areas are judged on the healthiness and fullness of the turf growth. Ninety-five percent of new growth will be present in sodded areas with minimal weed presence. Plants are judged on the vigor, structure, and new growth on the plants. New growth will be present in 85 percent of the plants. In addition to regularly performed maintenance, within 10 days prior to final acceptance, the contractor has to re-mulch all mulched areas; remove invasives/weeds/perform edging and trimming of all planted areas; mow all turfgrass areas; perform tree support adjustments as needed. Guarantee information will be shown in the specifications.

3.18 INVASIVE SPECIES

Invasive species have to be removed before they become incorporated into planting areas. Extensive chemical use is not conducive to water quality improvement and wildlife habitat; mechanical means are preferred with limited chemical application allowed.

3.19 SOIL STABILIZATION

As recommended by the Contractor's Geotechnical Engineer, provide soil stabilization designed to function as required by site conditions in accordance with the State Highway specifications and standards in the state where the project is located. Apply and install geosynthetics in accordance with the manufacturer's written instructions.

3.20 TOPSOIL

The topsoil (existing on-site topsoil and/or imported topsoil) will be tested and modified as necessary to ensure soil and drainage conditions detrimental to plant growth are identified and corrected. Planting soil for turf have to be a minimum 4-inch depth and spread throughout all turf areas. Topsoil is to be fertile, agricultural soil, typical for locality, capable of sustaining vigorous plant growth, and free of subsoil, stones, clay or impurities, plants, weeds and roots. pH value has be between 5.5 and 7.0 and organic content has to be 7 to 10 percent. Topsoil is required to be as specified in specifications and is to be reused from the site as much as possible. If the stored site's topsoil is insufficient or not appropriate for the landscaping needs, additional topsoil will be provided by the Contractor from an off-site source approved by the Contracting Officer Representative (COR).

Topsoil has to be fertile, agricultural soil, typical for locality, capable of sustaining vigorous plant growth, and free of subsoil, stones, clay or impurities, plants, weeds and roots. The soil (existing on-site topsoil and imported topsoil) has to be tested and modified as necessary to ensure soil and drainage conditions detrimental to plant growth are identified and corrected. Topsoil must be free of invasive plants, invasive plant parts, or seeds of invasive plants listed for the state of Virginia. Suitable organic matter of topsoil should range from 7-10 percent and suitable pH should range from 5 to 7.

Subsoil acidity of topsoil should be addressed. Where the subsoil is either acidic or composed of heavy clays, ground limestone are required to be spread at the rate of 4-8 tons/acre (200-400 lbs per 1000 SF) prior to the placement of topsoil. Lime needs to be distributed uniformly over designated areas and worked into the soil. Subsoil needs to be tilled to a minimum depth of 6-inch before placement of topsoil.

Topsoil needs to be uniformly distributed in a 4-6-inch layer and lightly compacted to a minimum thickness of four inches. Topsoil cannot be placed while the topsoil is in a frozen or muddy condition, when the subsoil is excessively wet or in a condition that may otherwise be detrimental to proper grading and seedbed preparation. If the stored site's topsoil is insufficient or not appropriate for the landscaping needs, additional topsoil will be provided by the Contractor from an off-site source approved by the COR.

Prior to delivery, reports have to be provided to COR for any topsoil or backfill delivered to project site and have to include original location of material. Topsoil or backfill material is subject to inspection by the COR and JBM-HH DPW at point of origin and source.

3.21 FERTILIZER

If fertilizer needs to be applied for successful establishment of proposed plants and sod, only use organic fertilizer and do not apply near water, storm drains or drainage ditches. Do not apply if heavy rain is expected. Apply fertilizers only to lawn and planting areas and sweep any product that lands on driveways, sidewalks, or streets, back onto the turfgrass and planting areas.

3.22 SUSTAINABLE MANAGEMENT

Provide landscaping that is low maintenance and weather tolerant.

3.23 SITE FURNISHINGS

The furnishings will be appropriate to their intended use and be part of a coordinated system. They will be of a simple design and provide consistency and continuity in their use. All furnishings will meet ABA requirements. All site furnishings will be economical and low maintenance. If applicable to overall project design, site furnishings may be provided for LEED credit.

3.24 MISCELLANEOUS REQUIREMENTS

Requirements stated in this contract are minimums. Innovative, creative, and life cycle cost effective solutions, which meet or exceed these requirements are encouraged. Any work that is shown elsewhere in this Request for Proposal (drawings, specifications) that is not covered in the written part of the Request for Proposal (RFP) is to be the responsibility of the contractor. In addition, it must be distinctly understood that the failure to mention specifically any work which would normally be required to complete the project will not relieve the contractor of his responsibility to perform such work.

4. APPLICABLE CRITERIA

This building project must conform to the criteria listed herein and in other sections of this RFP. The new structure is required to comply with the latest version of these standards as modified by the current applicable Unified Facility Criteria (UFC). The building structural systems including roof, floors, gravity and lateral load resisting systems must be designed to meet the latest applicable structural codes. All building structural and non-structural components must be stable, securely attached, and able to safely transfer all imposed loads to the foundation. The project is required to meet the service design life requirements for minimum life-safety objective upon completion in accordance with these codes.

4.1 MILITARY CRITERIA

The project must conform to the following criteria. For bidding and design purposes, the Contractor may request exemptions from the Unified Facility Criteria (UFCs) or submit for criteria equivalencies. These requests will not be incorporated into the design until approval received from the Contracting Officer.

Table 1: Military Criteria

DEPARTMENT OF DEFENSE	
DODI5200.48	Controlled Unclassified Information (CUI) *No exemptions or equivalencies will be provided for these criteria
ENGINEERING MANUALS	
EM 385-1-1	Safety and Health Requirements Manual
LAWS, POLICIES, REGULATIONS, AND OTHER CRITERIA	
EISA07	Energy Independence and Security Act of 2007
EO 12770	Metric Usage In Federal Government (a) Metric design and construction is required except when it increases construction cost. Offeror to determine the most cost-efficient system of measurement to be used for the project.
EPACT05	Energy Policy Act of 2005 / Public Law 109-58
TB MED 530	Occupational and Environmental Health Food Sanitation
PROTECTIVE DESIGN CENTER (PDC)	
PDC TR 06-08	Single Degree of Freedom Structural Response Limits

PDC TR 10-02	Blast Resistant Design Methodology for Window Systems Designed Statically and Dynamically
UNIFIED FACILITIES CRITERIA (UFC)	
UFC 1-200-01	DoD Building Code References to applicable International construction codes, such as: International Building Code (IBC), International Mechanical Code (IMC), International Residential Code (IRC), International Plumbing Code (IPC), and International Energy Conservation Code (IECC) are located within this UFC.
UFC 1-200-02	High Performance and Sustainable Building Requirements
UFC 3-101-01	Architecture
UFC 3-110-03	Roofing
UFC 3-120-01	Sign Standards
UFC 3-120-10	Interior Design
UFC 3-201-01	Civil Engineering
UFC 3-210-10	Low Impact Development
UFC 3-220-01	Geotechnical Engineering
UFC 3-230-01	Water Storage, Distribution, and Transmission
UFC 3-240-01	Wastewater Collection and Treatment
UFC 3-250-01	Pavement Design for Roads and Parking Areas
UFC 3-250-03	Standard Practice Manual for Flexible Pavements
UFC 3-250-04	Standard Practice for Concrete Pavement Construction
UFC 3-205-07	Standard Practice for Pavement Recycling
UFC 3-301-01	Structural Engineering

UFC 3-401-01	Mechanical Engineering
UFC 3-410-01	Heating, Ventilating, and Air Conditioning Systems
UFC 3-410-02	Direct Digital Control for HVAC and Other Building Controls Systems
UFC 3-410-06	Cybersecurity
UFC 3-420-01	Plumbing Systems
UFC 3-490-06	Elevators
UFC 3-501-01	Electrical Engineering
UFC 3-520-01	Interior Electrical Systems
UFC 3-530-01	Interior and Exterior Lighting Systems
UFC 3-550-01	Exterior Electrical Power Distribution
UFC 3-570-01	Cathodic Protection
UFC 3-575-01	Lightning and Static Electricity Protection Systems
UFC 3-580-01	Information and Communications Technology Infrastructure Planning and Design
UFC 3-600-01	Fire Protection Engineering for Facilities *No exemptions or equivalencies will be provided for these criteria
UFC 4-010-01	DoD Minimum Antiterrorism Standards for Buildings *No exemptions or equivalencies will be provided for these criteria
UFC 4-010-06	Cybersecurity of Facility-Related Control System (FRCS)
UFC 4-020-01	DoD Security Engineering Facilities Planning Manual

UFC 4-021-01	Design and O&M: Mass Notification Systems
UFC 4-021-02	Electronic Security System
UFC 4-023-03	Design of Buildings to Resist Progressive Collapse *No exemptions or equivalencies will be provided for these criteria
UNIFIED FACILITIES SUPPLEMENTS (UFS)	
UFS 1-300-02	Unified Facilities Guide Specifications (UFGS) Format Standard

5. FUNCTIONAL REQUIREMENTS

5.1 PROJECT & DESIGN REQUIREMENTS

The total area of the new Strategic Engagement Platform facility must not exceed that of the building to be demolished, which is estimated at 70,600 square feet. The maximum height of the new facility must not exceed 65 feet. The program space of the Strategic Engagement Platform includes space for the Pool House. The Pool House program space can be constructed separate from the rest of the Strategic Engagement Platform facility; The Pool House program space is to be located so that it is easily accessible to the pool area.

5.2 PROGRAMMED GROSS AREA OF BUILDINGS

Gross Area of Primary Facilities may not exceed Programmed Gross Areas. Calculate Gross Building Areas per UFC 3-101-01.

5.3 CODE, CRITERIA & LIFE SAFETY REQUIREMENTS

Design and construct project scope and requirements as outlined, or referenced in this RFP, and/or criteria listed in Chapter 4.

5.4 ADJACENCY AND FUNCTIONAL REQUIREMENTS

Building layout is to consolidate and support functional operational performance of spaces/rooms for efficient workflow and functional adjacencies as indicated by Room Data Sheets. Changes to specified functional areas and adjacencies identified should be avoided, however where deviations are unavoidable, Contractor must clearly identify differences within proposals. See space adjacency diagrams provided in Appendix A.

5.5 REQUIRED ASSEMBLIES

5.6 DOORS AND FRAMES

5.6.1 EXTERIOR DOORS AND FRAMES

Primary entrances' finishes are required to be Architectural Class I anodic coating or American Architectural Manufacturers Association (AAMA) 2605 organic coating (color to be determined).

1. Secondary entrances, egress doors and service entrances are required to be hollow metal doors and frames.
2. Provide ABA compatible automatic, overhead power/electric door openers with push button operation only where required and approved by JBM-HH.
3. Glazing is required to have dual pane insulating glass as the standard.
4. Provide tinted glazing to match window glazing as selected by Architectural and Engineering (A/E).
5. Glazing.
    a. Glazing is required to have a transmittance of 45 or less and dual pane insulating glass/glazing as the standard.
    b. Glazing has to be tempered for doors and sidelights at entries.
    c. Limit hollow metal frames to security doors, utility rooms, and other non-public doors.

Hollow metal doors and frames are acceptable for non-entry and service doors. All exterior doors and frames must comply with UFC 4-010-01, Standard 12. Any exterior doors with glazing is required to comply with Standard 10.

5.6.2 INTERIOR DOORS AND FRAMES

1. Solid Wood Core (Factory Stained)
2. Hollow Metal Frames 18 Gauge (Welded). Knock Down Frames are not allowed.
3. Provide Sidelights or vision panels as appropriate.

5.6.3 ARCHITECTURAL BARRIERS ACT (ABA)

Doors and hardware must comply with ABA Standards. Maximum force for pushing or pulling open a door or gate will be no more than 5 pounds. Achieve low force required for opening and closing doors with overhead closers including power assist, and/or blue button handicapped operators with overhead power assist through the entire opening cycle.

5.7 WINDOWS AND FRAMES

1. Provide aluminum windows complying with the American Architectural Manufacturers Association AAMA/NWWDA 101 / I.S. 2. The minimum performance class will be Heavy Commercial (HC).
2. Finish must be Architectural Class I anodic coating or American Architectural Manufacturers Association (AAMA) 2605 organic coating (color to be determined).
3. Provide windows with insulated glass and thermal break necessary to achieve a minimum Condensation Resistance Factor (CRF) of 45.

4. Glass tint, glazing, frames, and connections will be specified by the A/E Service, meeting the minimum requirements in these Standards.

4. Design wind load and resulting design pressure and performance grade will be determined in accordance with the International Building Code (IBC).

5.7.1 GLAZING

1. All exterior glazing must comply with UFC 4-010-01, Standard 10.
2. All exterior glass units are required to be made of a minimum of 1 inch of insulating glass, with 1/4-inch laminated glass on the inside pane and either 1/4-inch annealed or tempered glass on the outside pane. Design Basis Threat as specified by Joint Base Myer-Henderson Hall will impact final glazing design beyond minimum standards.
3. Glazing Design Performance Criteria
    a. U-factors .34 or less
    b. Solar Heat Gain Coefficients (SHGC) .28 or less
    c. Visible Light Transmission (VLT) 30 percent or less
4. Translucent Panels - If used, translucent panels should meet the following design criteria:
    a. U-factors .34 or less
    b. Solar Heat Gain Coefficients (SHGC) .28 or less
    c. Visible Light Transmission (VLT) 30 percent or less
    d. Thermally Broken grid core panels
    e. 2 ¾ inch thick panel

5.8 ASSEMBLY FOR EXTERIOR ENCLOSURES

Assembly will incorporate a continuous air barrier as prescribed in UFC 3-101-01 Architecture and will include but not limited to:

1. The building enclosure functions to control the transfer of the following elements: heat, air, moisture, light/radiation, and noise.
2. Enclosure needs to include a complete and contiguous air/moisture barrier including the roof, exterior walls and foundation systems.

5.8.1 FOUNDATION WALLS

Foundation walls must be designed for the lateral soil loads determined by a geotechnical investigation, in accordance with Section 1803 of UFC 3-301-01. Do not use rubble stone or permanent wood for foundation walls.

5.9 ASSEMBLIES FOR INTERIOR

Provide Interior materials, finishes, and requirements as follows:

5.9.1 INTERIOR FINISHES

For room finishes refer to Room Data Sheets located in Appendix A.

5.9.2 INTERIOR WALLS

The interior walls should be primarily constructed from cold-formed metal studs and 5/8-inch gypboard. Walls with required specific fire and Sound Transmission Class (STC) ratings will vary in assembly construction and meet the appropriate UL standards.

Elevator shafts should be constructed of concrete masonry units.

5.10 PROJECT DESIGN

5.11 ROOM REQUIREMENTS / ROOM DATA SHEETS

Provide room technical and functional requirements as identified by Room Data Sheets, located in Appendix A. These sheets outline the minimum standards for each space, including specifications for finishes, fixtures, fittings, and associated mechanical and electrical systems. The Room Data Sheets serve as the baseline for design and construction, and all proposed solutions must meet or exceed the requirements detailed therein to ensure consistency, performance, and quality across the facility.

5.12 PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

5.13 OCCUPANCY, TYPE OF CONSTRUCTION, AND LIFE SAFETY

5.13.1 ROOM REQUIREMENTS

Refer to room data sheets located in Appendix A.

5.13.2 INTERIOR FLOOR SLAB

Interior floors are required to be concrete construction.

5.14 ROOF ACCESS

All buildings over two stories are required to have a roof hatch with an interior ladder for roof access. Provide ladders up to the roof hatch, ladder up safety posts, and safety rails according to OSHA requirements. Coordinate with UFC 4-010-01, "DoD Minimum Antiterrorism Standards for Buildings."

5.15 SITE COMMUNICATIONS AND SECURITY

5.16 PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

Provide new concrete-encased four-way 4-inch conduits in a ductbank from an existing telecommunications manhole to the new Entrance Facility TR. Provide three empty mesh fabric innerduct in each empty conduit. Provide new 96-strand singlemode fiber Outside Plant (OSP) cable from the new Entrance Facility Telecommunication Room (TR) through the new ductbank and existing telecommunications manhole, continuing through existing ductbank and existing manholes to reach the Fort Myer NEC ITN building to be identified. See Appendix B drawing sheets TS101A and TS101B.

5.17 GUARDHOUSE RESTORATION

Provide new concrete-encased four-way 4-inch conduits in a ductbank from an existing telecommunications manhole to the new Entrance Facility TR. Provide three empty mesh fabric innerduct in each empty conduit. Provide new 24-strand singlemode fiber OSP cable from the new Entrance Facility TR through the new ductbank and existing telecommunications manhole, continuing through existing ductbank and existing manholes to reach the Fort Myer NEC ITN building to be identified. See Appendix B drawing sheets TS101A and TS101B.

A10 FOUNDATIONS

1. SYSTEM DESCRIPTION

Provide building foundation system in accordance with Unified Facilities Criteria (UFC) 3-301-01, Structural Engineering and UFC 3-220-01, Geotechnical Engineering. Design foundations to suit subsurface conditions, capable of transmitting all building loads to the ground. Ground and below construction including foundation systems are required to be composed of durable, corrosion and moisture resistant materials such as reinforced concrete, reinforced solid grouted masonry, weathering or corrosion resistant steel, or other government and industry approved durable and weather resistant construction materials. Reinforced concrete is the preferred construction material. For bidding purposes, it can be assumed that a shallow foundation system will be adequate for design. Refer to Appendix C, Geotechnical Report for RFP, for additional requirements.

See Section B10, Superstructure, for additional loading criteria.

2. GOVERNMENT PROVIDED GEOTECHNICAL INFORMATION

Subsurface soil information is included in other portions of this RFP. Refer to Appendix C, Geotechnical Report for RFP.

The included subsurface information is only for the Contractor's information and is not guaranteed to fully represent all subsurface conditions. The soil borings completed during the 17 June 2025 exploration (included in Appendix C) needs to be used as a basis for bidding; however, the Government will not be responsible for any interpretation or conclusion by the Contractor drawn from the data or information.

The included geotechnical report accompanying the subsurface information is provided only to better convey data (boring logs, testing, and other data) or to document observed site conditions. The assumptions, analysis, and recommendations of any accompanying report were developed for preliminary planning purposes only and may not reflect present project requirements. The Contractor is required to retain a Geotechnical Engineer experienced and licensed in the geographic region of the project to interpret the Government provided information as related to his design concept and develop geotechnical requirements to support design and construction.

Anticipate minor variations in subsurface conditions between borings. The Contractor is responsible for costs associated with the site preparation, ground improvement and foundations except as allowed by Contract Clause Federal Acquisition Regulation (FAR) 52.236-2, "Differing Site Conditions". The Contractor's Geotechnical Engineer must perform additional subsurface investigation/testing as required to adequately determine all applicable geotechnical factors including the type and capacity of the project foundations. The Contractor's Geotechnical Engineer is required to evaluate the information provided and any additional information obtained and prepare a report as described in other portions of this RFP. The minimum requirements for the subsurface investigation and report are as required by EM 1110-1-1804 with associated references.

Contractor’s Geotechnical Engineer should perform the soils investigation at the site for use in the design and construction of the new facility. Perform, at Contractor's expense, subsurface exploration, investigation, testing, and analysis for the design and construction of features such as the building foundations, pavement section(s), stormwater management facility(ies), and utility structure foundations. Prepare a report including laboratory analysis of samples and recommendations for foundation and pavement design by a Professional Engineer as specified and in accordance with UFC 3-220-01, Geotechnical Engineering.

As a minimum, the successful bidder's Geotechnical Engineer must perform additional subsurface exploration and supplementary laboratory testing as necessary to support the design concept.

Provide personnel under the supervision of a registered Professional Engineer to inspect excavations and soil/groundwater conditions throughout construction. The Engineer is required to perform pre-construction and periodic site visits throughout construction to assess site conditions. The Engineer, with the concurrence of the Contractor and the Contracting Officer, is required to update the excavation, sheeting, shoring, and dewatering plans as construction progresses to reflect actual site conditions and is required to submit the updated plan and a written report (with professional stamp) at least monthly informing the Contractor and Contracting Officer of the status of the plan and an accounting of Contractor adherence to the plan; specifically addressing any present or potential problems. The Engineer must be available to meet with the Contracting Officer at any time throughout the Contract duration. Provide the services of the Engineer at no additional cost to the Government. It is important to note that the presence of loose or compressible soils may result in excessive settlement that could impact the performance of surface bearing structures and supporting facilities such as foundations, slabs, pavements, sidewalks, and utilities. The magnitude and duration of consolidation settlement will be dependent on the composition, depth, and thickness of the compressible soils as well as the successful bidder's design concept. The Contractor's Geotechnical Engineer is responsible for evaluating potential global settlement due to designed grade increases and final structural loads. The Contractor's Geotechnical Engineer must develop any settlement mitigation procedures (such as preloading, surcharging, fill monitoring programs, and ground improvement systems) needed to maintain global settlements within tolerable limits. Surcharge material, if required, must remain in place for a minimum of 90 days.

3. SEISMIC DESIGN

Seismic design parameters, including Site Classification and Seismic Design Category, will be determined in accordance with the Geotechnical Engineering report, UFC 3-301-01, ASCE 7, and IBC. Design and construct the building structure to comply with the seismic design parameters and code requirements specific to the construction material selected for this facility.

3.1. EARTHWORK

Prepare the following Unified Facilities Guide Specification (UFGS) as part of the project specification and include the prepared specification section in the design submittal for the project:

UFGS Section 31 00 00 Earthwork

Perform quality assurance for earthwork in accordance with International Building Code (IBC) Chapter 17 and UFGS Section 31 00 00. A competent person, as defined by COE EM 385-1-1, under supervision of a registered Professional Engineer is required to provide inspection of excavations and soil/groundwater conditions throughout construction. The Engineer must perform periodic site visits throughout construction to assess site conditions. The Engineer, with the concurrence of the Contractor and the Contracting Officer, must update the excavation, sheeting, shoring, and dewatering plans as construction progresses to reflect actual site conditions and submit the updated plan and a written report (with professional stamp) at least monthly informing the Contractor and the Contracting Officer of the status of the plan and an accounting of Contractor adherence to the plan; specifically addressing any present or potential problems. The Engineer must be available to meet with the Contracting Officer at any time throughout the contract duration. The Contractor will bear all costs of the Engineer.

3.2. GEOTECHNICAL REPORT

3.2.1. SUBSURFACE SOILS INFORMATION

Any provided subsurface soil information is not guaranteed to fully represent all subsurface conditions. The data included in this RFP is required to be used as basis for bidding. CONTRACTOR-PROVIDED GEOTECHNICAL ENGINEER

Retain a Geotechnical Engineer experienced and licensed in the geographic region of the project to interpret any provided data as related to his design concept and develop requirements for bidding.

Coordinate all work by the Contractor-provided Geotechnical Engineer with the Contracting Officer and ensure that work does not conflict with Base operations. When providing the Foundation Work Design submittal, provide the Contractor's Geotechnical Report (an Adobe Acrobat PDF version on CD and two printed copies) for review and record keeping purposes. The report becomes the property of the Government. Provide the Geotechnical reports generated during construction, such as pile load tests or pile driver analyzer (PDA) results, pile driving results and analysis, to the Contracting Officer (a PDF version and two printed copies) for record keeping purposes.

3.2.2. CONTRACTOR-PROVIDED GEOTECHNICAL REPORT

Submit a written Geotechnical report based upon Government-provided subsurface investigation data and all additional field and laboratory testing accomplished at the discretion of the Contractor's Geotechnical Engineer. The Geotechnical Report must include all requirements listed in UFC 3-220-01, Geotechnical Engineering, paragraph entitled "Section 1803 "Geotechnical Investigations"; in addition, include the following:

The project site description, vicinity map and site map indicating the location of borings and any other sampling locations. Provide 24-hour groundwater observations for at least 20 percent of the soil borings, minimum one boring. Provide notes explaining any abbreviations or symbols used and describing any special site preparation requirements.

Results of all applicable field and laboratory testing, whether Government or Contractor-provided. Address existing subsurface conditions, selection and design of the foundation and floor slab, all underground construction including utility installation and all other site-specific requirements (such as soil stabilization and slope stability).

Engineering analysis, discussion and recommendations addressing:

Settlement analysis. Settlement must be limited as required in EM 1110-1-1904, Settlement Analysis.

Bearing Capacity Analysis.

Foundation selection and construction considerations; dimensions, and installation procedures.

Site preparation (earthwork procedures and equipment), compaction requirements, building slab preparation (as applicable), soil sensitivity to weather and equipment, groundwater influence on construction, mitigation of expansive soils or liquefaction potential, dewatering requirements, slope stability, and other necessary instructions.

Lateral soil pressure coefficients for retaining wall design.

Sheeting and shoring considerations, as applicable.

Pavement design calculations with defined parameters, actual or assumed, and recommended thicknesses and materials, whether for design or for proposed modifications to the RFP provided pavement design.

Haul routes and stockpile locations for earthwork, as applicable.

Calculations to support conclusions and recommendations.

Present recommendations on a structure-by-structure Basis.

Provide the Geotechnical Report signed and sealed by the Contractor-provided Geotechnical Engineer.

Submit report accompanied by a cover letter identifying any report recommendations of the report proposed to be adopted into the design which are interpreted by the Contractor as a change condition to the Geotechnical or Pavement related requirements of the RFP.

3.2.3. GEOTECHNICAL SITE DATA REQUIRED IN DESIGN DRAWINGS

The Contractor's final design drawings must include the Government-provided subsurface data presented in the RFP as noted below, as well as all additional soil borings and laboratory test data results performed by the Contractor. The data provided must include:

Logs of Borings and related summary of laboratory test results and groundwater observations. Provide 24-hour groundwater observations for at least 20 percent of the soil borings, minimum one boring. Provide notes explaining any abbreviations or symbols used and describing any special site preparation requirements.

Indicate locations of all borings on the drawings. Revise applicable design drawings to reference the Contractor's Geotechnical Report as being a basis for design.

3.3. PILE DRIVER ANALYZER (PDA)

If deemed necessary by the Contractor's Geotechnical Engineer, the dynamic wave equation method of analysis, pile driver analyzer, must be used to validate pile and pile hammer compatibility, establish pile driving criteria, establish terminal penetration resistance, or verify as-driven capacity of the pile. Provide PDA or static load test(s) for piles with required allowable design capacity equal to or greater than 40 tons.

4. PERFORMANCE VERIFICATION AND ACCEPTANCE TESTING

Provide verification of satisfactory construction and system performance of the foundations via Performance Verification Testing, and by field inspection, as detailed in this section of the RFP and in Part 2 Section 01 45 00, Quality Control. Provide special tests and special inspections in accordance with Part 2 Section 01 45 00.

5. PILES

If piles are required, perform quality assurance for pile construction in accordance with UFC 1-200-01, DoD Building Code (General Building Requirements). Pile installation procedures and installed piles must be inspected and found to be in compliance with these specifications prior to acceptance of the work.

Provide test piles as directed by the Contractor's Geotechnical Engineer.

Perform pile load tests, if required, in accordance with UFC 1-200-01.

Submit results of the pile test program and final pile installation criteria to the Contracting Officer prior to installation of the production piles.

Submit pile installation survey for integrity and adequacy evaluation prior to superstructure erection.

A1010 STANDARD FOUNDATIONS

As determined by the Designer of Record to be applicable, provide a Standard foundation. "Standard Foundations" are shallow or deep foundations as specifically addressed in International Building Code (IBC) Chapter 18. Treated timber piles may be used if determined acceptable by the Designer of Record.

Column spot footing sizes will be determined by design but must not be less than 24-inches in any direction and a minimum thickness of 12-inches. The spot footings are required to be reinforced according to design requirements with a minimum number and size of 4 bars each way at 12 inches on center bottom reinforcement. Specify footing reinforcement on the drawings by number of bars equally spaced and not by spacing.

Slab-on-ground block-outs for columns which bear below the top of the slab will be detailed on the contract drawings. The top of the footing or pedestal supporting the column is to be at least 8 inches below the top of the slab-on-ground. Unreinforced concrete pedestals are not allowed.

The foundation continuous footings are required to be reinforced concrete with a minimum thickness of 12-inches. The footing width will be determined by design and must be at least 8-inches wider than the foundation wall (minimum 4-inches on each side of the foundation wall). All load-bearing and shear walls, regardless of the wall material, are required to be founded on a continuous footing. Ensure all continuous footings are clearly labeled on the drawings with size and reinforcement.

All foundation system elements including retaining walls must be evaluated for stability using allowable loads with minimum factors of safety of 1.5 against sliding and 2.0 against overturning to withstand combined gravity and lateral loads. Foundation system and retaining wall design must account for surcharge loads from adjacent structures, emergency vehicles (e.g., fire trucks, ladder trucks), and construction vehicles (e.g., rollers, cranes) where applicable. Surcharge load must account for the actual load or AASHTO HL-93 truck load and minimum 5 feet of fill with maximum unit weight based on Geotechnical report. Foundation settlements must be within acceptable limits specified in the Geotechnical Engineer recommendations. Shallow foundations must be located on undisturbed/natural soils and compaction requirements with backfill materials must be in accordance with Geotechnical Engineer and ASTM D1557. Exterior wall and column footings must be placed at or below a frost depth below the final exterior grade determined by the Contractor's Geotechnical Engineer. See Section B20, Exterior Enclosure, for additional requirements for exterior walls used as non-load-bearing walls.

A101001 WALL FOUNDATIONS

Provide foundation walls as required in accordance with the requirements of this section and other portions of this RFP.

A101002 COLUMN FOUNDATIONS AND PILE CAPS

Provide column foundations or pile caps and grade beams as required in accordance with the requirements of this section and other portions of this RFP.

A1020 SPECIAL FOUNDATIONS

As determined by the Designer of Record to be applicable, provide a Special foundation. "Special Foundations" are any foundations that are not specifically "Standard Foundations", or a combination of Standard Foundations and a site improvement/ground modification system. Examples of site improvement/ground modification systems include surcharging, stone columns, rammed aggregate piers, impact densification, compaction grouting, vibroflotation, and other similar systems. As "Special Foundation" techniques or systems typically require the use of specialty contractors, a Professional Engineer must establish installation and acceptance criteria and supervise the installation. The Designer of Record (DOR) must submit justification for use, including acceptable evidence of previous successful installation in similar conditions, methods and equipment used in their installation, proposed testing and inspection to be used, supporting test data, calculations and any other information related to the structural properties and load capacity of such system. The allowable stresses for piles/piers must not exceed those limitations specified in UFC 1-200-01.

A102001 PILE FOUNDATIONS

Where piles are required, design, install, and test piles (including sheet piles, as applicable) in accordance with UFC 1-200-01, except as noted otherwise. Provide piles in accordance with the requirements of the Contractor's Geotechnical Engineer, and the following paragraphs.

A102003 UNDERPINNING

If required, underpin existing construction as required in accordance with the requirements of this section and other portions of this RFP.

A102004 DEWATERING

Dewater site for foundation construction as required by soil conditions and local subsurface and surface water, including rainfall, and considering any potential adverse impact on adjacent facilities, including settlement. Dewatering requirements and methods must be established by the Contractor's Geotechnical Engineer, based on his subsurface exploration and investigation.

A102006 PRESSURE INJECTED GROUTING

If required, pressure inject grout as required in accordance with the requirements of this section and other portions of this RFP.

A1030 SLAB ON GRADE

As determined by the designer of record to be applicable, provide either a standard or structurally supported concrete ground floor slab. Provide ground floor slab(s) designed in accordance with the American Concrete Institute ACI-360R, Design of Slabs on Grade, for the interior spaces. Provide perimeter and underslab insulation in accordance with UFC 3-101-01, Architecture. When providing a structurally supported slab, provide for support of all utilities that may be adversely affected by soil consolidation or expansive soils.

A103001 STANDARD SLAB ON GROUND

If allowed by site conditions and recommended by the Contractor-provided Geotechnical Engineer, provide standard concrete slab on grade to meet the required loading requirement in accordance with the requirements of this section and other portions of this RFP.

Design and construct floor slab on grade in accordance with Contractor's Geotechnical Engineer's Report guidance, and so that any settlement of the floor slab will not result in harmful distortion of the floor, nor vertical misalignment of the floor with other building components (such as doorways and trenches), building utilities or with pile-supported building elements. If these above conditions cannot be met, provide a pile supported slab.

A103003 TRENCHES

Provide reinforced concrete trenches with waterproof joints and seals to prevent ground water infiltration.

A103004 PITS AND BASES

Provide reinforced concrete pits and bases with waterproof joints and seals to prevent ground water infiltration.

A103005 FOUNDATION DRAINAGE

A perimeter drainage system needs to be provided to remove water away from the foundation of the facility and to be deposited in the storm sewerage system of the site. Provide perforated pipe for the foundation drainage system of the specified type, and of a size sufficient to remove water from the foundation successfully.

A103090 OTHER SLAB ON GROUND

Exterior equipment concrete pad foundation or slab-on-grade with turn-down perimeter edges should extend to bear below the frost depth and should be sized adequately for the weight and size of the supported equipment.

1. BLOCK OR BOARD PERIMETER INSULATION

Provide only thermal insulating materials recommended by manufacturer for perimeter insulation.

A1040 STRUCTURALLY SUPPORTED SLAB

As determined by the Designer of Record to be applicable, provide a structurally supported slab. Provide for support of all utilities that may be adversely affected by soil consolidation or expansive soils. Provide stainless steel supports sized adequately to support the in-service utility. Where the structurally supported slab is below the existing adjacent exterior grade, provide water/damp proofing and a perimeter drainage system to remove ground water from the area immediately adjacent to the buildings. Provide under slab insulation in accordance with ASCE 32 if not providing sufficient foundation frost bearing depth.

END OF SECTION

B10 SUPERSTRUCTURE

1. SYSTEM DESCRIPTION

Provide building framing systems in accordance with UFC 2-200-01, General Building Requirements, UFC 3-301-01, Structural Engineering, UFC 4-010-01, Minimum Antiterrorism Standards for Buildings, and with the standards listed in Chapter 4. Refer to UFC 3-301-01 structural load requirements. This project consists of design and construction of a Strategic Engagement Platform. Structures that are part of this project are listed below.

2. PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

2.1. GENERAL REQUIREMENTS

Unless otherwise noted all structural design is required to be in accordance with the latest version of codes, and standards referenced in this document at the time of RFP issuance. Governing codes for this project must be the Joint Base Myer-Henderson Hall Design Standards, local and Federal codes, and the International Building Code as modified by UFC 3-301-01, Structural Engineering. Where UFCs, Federal standards, State standards, Installation standards and other requirements are in conflict, the most stringent requirement must govern.

The intent is to maximize value in the execution of this project, hence requirements stated herein are minimums. The structural systems must be compatible with the project site geographical characteristics and optimize the interior space as required for the intended functions.

2.2. SERVICEABILITY

The structural system cannot exceed serviceability requirements prescribed in the governing building code or recommended deformation limits of the building material. A three-dimensional structural model of the structure will need to be developed to determine drift and deformation resulting from the applied loads and P-delta effects. Material properties, stiffness and strength of elements, and diaphragm stiffness characteristics are to be incorporated into the model. The building system is required to comply with the maximum allowable drift and deformation considering the designed occupancy and details of construction. Structural separation and expansion between structures need to allow for the maximum calculated displacement sufficient to avoid damaging contact.

2.3. LOADING CRITERIA

In addition, design the structure in accordance with the following loading criteria based on occupancy and usage:

2.3.1. DEAD AND COLLATERAL LOADS

The dead load must consist of the actual weight of all materials of construction for the building system including, but not limited to, metal decking, elastomeric coating, built-up insulation, and collateral loads. Collateral loads include fireproofing, ceilings, conduit, sprinkler, piping, mechanical and electrical equipment, and other roof mounted or suspended equipment.

2.3.2. FLOOR LIVE LOADS

Provide for live loads for occupancies or uses indicated in UFC 3-301-01, Structural Engineering.

2.3.3. ROOF LIVE LOADS

A minimum uniformly distributed roof live load of 20 psf is to be applied to account for occupant and roof maintenance purposes.

2.3.4. OTHER DESIGN LOADS

Design the structures to withstand the effects all applicable loads at the project location, including but not limited to wind, ice, rain, seismic, snow, flood and tornado. Determine the design loads using the ASCE Hazard Tool based on the following input criteria: project location, ASCE 7 version, facility Risk Category, and Site Soil Class. A preliminary ASCE Hazardous Report has been completed and can be referenced in Appendix C.

2.3.4.1. RISK CATEGORY

Determine the Risk Category for the facility in accordance with Table 2-2 of UFC 3-301-01, Structural Engineering. A preliminary assessment (that must be verified) suggests Risk Category III based on the building primarily being a public assembly space with an occupant load greater than 300.

2.3.4.2. SNOW AND ICE LOADS

All structures must be designed for the vertical load induced by the weight of snow and ice. The snow load varies depending on the level and the type of roof system, and the span condition of the roof framing. Several snow load conditions must be specified below based on shapes and types of the roof and framing system. At minimum these load conditions must be verified: minimum low-sloped roof snow load, drift and sliding snow loads on lower roofs, unbalanced snow loads for members spanning from ridge to eave, and partial loading for continuous span members that are parallel to the ridgeline. Atmospheric ice loads due to freezing rain, snow must also be considered in the design. Snow and ice load calculations must be in accordance with ASCE 7 and in compliance with the USACE Cold Regions Research and Equipment Laboratory (CRREL)

Snow load calculations must account for partial snow loading, unbalanced snow load, drifts on lower roofs, sliding snow, roof projections and parapets, rain-on-snow surcharge load, and ponding instability requirements. Separation distance between adjacent structures must also be considered during design.

For adjacent structures separated by less than 20 ft, determine the leeward drift on the lower roof in accordance with ASCE 7-22, Section 7.7.2.

Footing depth due to frost will be provided by the Geotechnical Engineer.

B1010 FLOOR CONSTRUCTION

Ground floor construction may consist of cast-in-place structurally reinforced concrete slabs or concrete slab-on-grade with vapor barrier on dry granular fill according to the UFCs.

Follow ACI specific design requirements for concrete floor construction. Provide crack control measures for shrinkage, contraction, and other potential defects using adequate isolation joint and contraction and construction joint features. These items must be considered in floor construction design, base and subbase materials, moisture protection, crack control, load transfer mechanisms at floor joints, and joint details at dock pit floor.

B101001 STRUCTURAL FRAME

Structural frame elements may include columns, girders, beams, trusses, joists, moment frames, shear walls, and bracing. All structures must be designed to comply with the requirements for the Risk Category of the facility. Materials for structural elements must be composed of either structural steel, reinforced concrete (including Insulated Concrete Forms (ICF) and precast/prestressed), reinforced masonry, or a combination thereof. Wood products and cold-formed, light-gauge metal framing are not permitted for the primary gravity load bearing or for primary lateral force resisting system for the building.

B101002 STRUCTURAL INTERIOR WALLS

Provide structural interior walls as required in accordance with the requirements of this section and other portions of this RFP. Provide full-height interior load-bearing walls of either reinforced masonry or concrete within heated structure spanning from ground floor to roof deck within acceptable deflection limits. See Section C10, Interior Construction, for additional requirements.

B101003 FLOOR DECKS AND SLABS

If required, provide floor decks as required in accordance with the requirements of this section and other portions of this RFP. Elevated floor construction may consist of non-composite concrete slabs on form deck, composite concrete slabs on composite steel deck, or cast-in-place concrete slabs on removable forms. The floor system must be properly connected to support framing and design for combination of gravity and lateral load effects as a continuous diaphragm. Frame openings in floor system to maintain continuous load path, and to safely transfer all loads to primary structural members.

B101007 FLOOR RACEWAY SYSTEMS

See Section D50, Electrical, for floor raceway systems.

B1020 ROOF CONSTRUCTION

The roof construction must consist of steel roof deck on steel joists, steel roof deck on steel beams, steel roof deck on cold-formed metal trusses, non-composite concrete slabs on form deck on steel joists, non-composite concrete slabs on form deck on steel beams, composite concrete slabs on composite steel deck, or cast-in-place concrete slabs on removable forms.

The roof deck must be supported on cast-in place concrete walls, concrete masonry walls, steel columns and steel beams, steel columns and joist girders, or concrete columns and concrete beams.

Maintain the load path and diaphragm, chord, and collector continuity around all floor and roof openings. Stair openings in floor and roof diaphragms should not be detrimental to the structure's vertical and lateral load-resisting capacity. Stair openings are to not create vertical or horizontal structural irregularities. Provide framing, drag struts, and collectors around stair openings that maintain the vertical and horizontal load paths of the structural systems.

B102001 STRUCTURAL FRAME

Structural frame elements may include columns, girders, beams, trusses, joists, moment frames, shear walls, and bracing. See Section B20, Exterior Enclosure, for additional requirements for exterior walls used as load-bearing walls or shear walls.

B102003 ROOF DECKS AND SLABS

Provide roof deck as required in accordance with the requirements of this section and other portions of this RFP. Roof deck system may consist of steel deck, non-composite concrete slabs on form deck, composite concrete slabs on composite steel deck, or cast-in-place concrete slabs on removable forms. The roof system must be properly connected to support framing and design to account for combinations of gravity and lateral load effects. Frame roof openings to maintain continuous load path and adequately transfer all loads to primary structural members.

B102004 CANOPIES

Provide canopies as required in accordance with the requirements of this section and other portions of this RFP. Refer to section B102001 for further requirements.

END OF SECTION

B20 EXTERIOR ENCLOSURE

1. SYSTEM DESCRIPTION

Enclosure system consists of an exterior shell, which includes all vertical and horizontal exterior walls, exterior windows, and exterior doors. Enclosure system includes load bearing exterior walls here, and not in System B10, Superstructure. Enclosure system excludes roofing (See System B30, Roofing). For structural frame elements at exterior such as columns, beams, and spandrels are included in Superstructure, with only the applied exterior finishes (e.g., paint, stucco) being included here. Finishes to the inside face of walls which are not an integral part of the wall construction will be included in System C30, Interior Finishes.

B2010 EXTERIOR WALLS

Provide a single thickness, barrier wall composed of an Exterior Closure assembly indicated below.

Provide exterior wall construction that consists of exterior skin system of non-structural outside face elements with integrated wall systems that include; flashing (embedded, exposed, and thru-wall), a water resistive barrier, moisture barrier/ vapor retarder (if required), air barrier, and insulation systems with interior skin system materials achieving a protective finish on interior face of exterior walls. Provide all components necessary for a shingled water resistive barrier to direct water that would penetrate the wall and direct it outside of the wall to drain away from building. Provide exterior enclosure components and barriers in accordance with UFC 3-101-01, Architecture.

Design all work to comply with UFC 3-101-01, Architecture, and UFC 3-301-01, Structural Engineering, and requirements as follows:

Vapor Pressure and Hygrothermal Analysis - Perform a job specific vapor pressure and hygrothermal analysis in accordance with UFC 3-101-01, Architecture. Conclusion of analysis must indicate if a moisture barrier / vapor retarder is required, with appropriate location of moisture barrier / vapor retarder, and anticipated dew-point locations within exterior enclosure during different critical times of the year.

Wind Loads - Provide wind load calculations for exterior cladding in accordance with UFC 1-200-01 and UFC 3-301-01 with comparative analysis of cladding system to be provided.

Water Penetration - No water penetration must occur at a pressure of 8 psf of fixed area when tested in accordance with ASTM E 331.

Insulating Value – Comply with UFC 3-101-01, Architecture, for the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) standards to determine the minimum insulating value of the complete wall system.

Out-of-plane deflection limits for framing supporting exterior wall finishes must be in compliance with UFC 3-301-01.

Provide watertight expansion and crack control joints with sealant materials in exterior enclosure to prevent moisture infiltration and seepage. Exterior wall envelope including windows, storefront, and glazed doors must also comply with minimum antiterrorism performance level criteria mandated in UFC 4-010-01, Minimum Antiterrorism Standards for Buildings. Exposed sealant used as a watertight joint will not be accepted.

B201001 EXTERIOR CLOSURE

Exterior Building Closure Objectives - The new facility design, and subsequent construction documents, are to utilize durable, low-maintenance materials throughout. Exterior material type and percentage coverage will be consistent with district historical requirements. Provide exterior building closure in accordance with UFC 3-101-01, Architecture, UFC 3-301-01, Structural Engineering, and UFC 4-010-01, Minimum Antiterrorism Standards for Buildings. While materials listed below are listed as acceptable, the final decision on its particular use in the total design will be subject to the previously mentioned Historic Preservation agencies and their approval.

Exterior Walls - Wall composition and design are the sole responsibility of the A/E. Allowable materials for façade finish with approval from governing historic preservation agencies are:

- Masonry (Brick Veneer) can be used as a rain screen to match the vernacular of the historic base.
- Concrete Masonry Units (CMU) can be used for structural components of the wall system or as a rainscreen. When used as a rain screen, ground face CMU is preferred. Split face CMU use must be approved by the governing historic preservation agencies and JBM-HH during design.
- Insulated Metal panels are allowed for accent features.
- Fiber Cement Panels can be used for accent areas. If used, fiber cement panels must be held a minimum of 6 inches from ground level.
- Lightweight cast stone may be used as a rain screen material or as an accent material.
- Exterior Insulation Finish System (EIFS) is allowed, but accent only. Do not use it at grade level where damage is probable.
- Insulated precast concrete wall panels are an acceptable system for exterior walls.
- Stucco – 3 Part is allowed.

B201003 INSULATION & VAPOR RETARDER

Provide continuous insulation, vapor retarder, water-resistive barrier, and air barrier to meet or exceed requirements of project's energy savings requirements as indicated by applicable American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) 90.1 calculations called for in Unified Facilities Criteria (UFC) 1-200-02, High Performance and Sustainable Building Requirements, and meeting minimum building envelope insulation requirements of UFC 3-101-01, Architecture.

Provide a continuous air barrier to control air leakage into, and out of conditioned spaces. Air barrier must encompass all elements of each facility that are exposed to outside environment or outside environmental conditions such as roof, walls, floors, and compartmentalized unconditioned portions of the facility such as garages, and negatively pressurized spaces. Permanently seal penetrations through air barriers, joints in air barriers, adjoining construction, and transitions to different air barrier materials.

Provide a continuous water resistive barrier in accordance with UFC 3-101-01, Architecture. Water resistive barrier must resist liquid (bulk) water from being absorbed into the back-up wall assembly if water leaks, penetrates, or seeps past the exterior enclosure cladding system.

Provide a vapor pressure analysis and hygrothermal analysis in accordance with UFC 3-101-01, Architecture. Determine if a moisture barrier/ vapor retarder is required and where it would be located. Include analysis and conclusion in the design analysis for the project. If required by analysis, provide a moisture barrier/ vapor retarder to restrict the flow of moisture through the exterior enclosure.

Include written and graphic descriptions of exterior enclosure barrier materials and location within the wall as a part of Contractor provided design analysis. Identify in the analysis the continuous boundary limits of the air barrier and of the zone or zones to be field tested for building air tightness.

Provide contract drawings that indicate each exterior enclosure barrier location and materials that make up the barriers. Detail the following barrier conditions:

- Typical conditions at wall sections.
- Barrier treatment at wall openings.
- Intersections with other exterior enclosure assemblies and materials. Include intersections at roof and floors.
- Intersections with counter flashing.
- Inside and outside corners.

- Preservation of air and water tightness at anchors for materials that cover the barrier.
- Treatment to seal barrier penetrations such as conduits, pipes, electric boxes, and fixtures.
- Indicate air barrier perimeter, if facility is segmented into areas that are not within the air barrier envelope.

Provide insulation, air barriers, water resistive barriers, and moisture barrier/ vapor retarders (if required) in the exterior enclosure to control heat loss/gain, air infiltration/diffusion, moisture infiltration/diffusion, and water infiltration.

Provide insulation, air barrier and water resistive barrier on all conditioned facilities and moisture barrier/ vapor retarders when required by exterior enclosure vapor pressure and hygrothermal analysis. These barrier materials may be installed separately or combined if different air barrier, moisture barrier/ vapor retarder, and water resistive barrier functions can be consolidated in one material.

Provide exterior enclosure barriers that are durable and designed to last the life of the facility. Seal continuous air and water resistive barrier in a flexible manner to allow for relative movement of adjacent building enclosure components. Support exterior enclosure barriers to withstand maximum positive and negative air pressure to be placed on the building without displacement or damage and transfer load to the structure. Permanently seal penetrations, joints, holes, and transitions to adjoining construction in air and water resistive barriers as recommended by the material manufacturer. Do not compromise exterior enclosure barrier integrity at electrical boxes, fixture supports, and fasteners with holes through the exterior enclosure barriers that allow air or water leakage. Do not expose exterior enclosure barriers or retarders to environmental conditions longer than is recommended by the manufacturer.

B201005 EXTERIOR LOUVERS & SCREENS

Provide exterior louvers and screens, where required, that match the finish of the existing windows and detailed to integrate with the architecture of the building, as appropriate to the design of the building.

If required, provide louvers, which are not an integral part of the mechanical equipment, exterior closures, grilles and screens, storm shutters, and other materials used for a variety of purposes including screening equipment or as louvers for exterior doors.

Louvers, screens, and grilles must be selected in a color and design that is compatible with the fabric of the exterior architectural character as described below.

B201006 HANDRAILS

Design handrails and anchorage connections to resist loads in accordance with IBC. Provide steel materials in accordance with NAAMM Pipe Railing Systems Manual, with the same size handrail and vertical post. Provide series 300 stainless steel pipe collars.

B201008 WALL FLASHING

Flashing must be aluminum or stainless steel or copper. Incorporate the flashing in the water resistive barrier and seal joints to flashing to form a shingled effect and direct liquid water to the exterior of the exterior enclosure and away from back-up wall assembly.

B201009 EXTERIOR PAINTING AND COATINGS

Provide field applied exterior coatings for all items that are not prefinished, and to prefinished items when required to provide a color other than a standard prefinished color. Provide special coating systems for exposed structural elements associated with any exposed structural elements.

B201009 EXTERIOR PAINTING AND SPECIAL COATINGS

Apply coatings directly to all non-prefinished surfaces of the exterior construction. Comply with Master Painters Institute requirements for surface degradation analysis, surface preparation, paint and coating selection, paint application restrictions for substrate materials, and paint application.

Painting practices must comply with applicable federal, state and local laws enacted to ensure compliance with Federal Clean Air Standards. Apply coating materials in accordance with Society for Protective Coatings (SSPC) PA 1. SSPC PA 1 methods are applicable to all substrates.

B201010 EXTERIOR JOINT SEALANTS

Provide exterior application of joint sealants to seal joints and prepare for finish material installation.

Provide sealant joint design, priming, tooling, masking, cleaning and application in accordance with the general requirements of Sealants: A Professionals' Guide from the Sealant, Waterproofing & Restoration Institute (SWRI).

Joints must include proper backing material for sealant support during application, control of sealant depth, and to act as a bond breaker. Use filler boards, backer rods and bond breaker tapes. Provide priming unless specifically not recommended by the sealant manufacturer. Applied sealant must be tooled. Tooling must not compact sealant too less than the minimum sealant thickness required. Mask adjacent surfaces to control sealant boundaries during sealant application.

B201011 SUN CONTROL DEVICES (EXTERIOR)

Detail sun control devices to integrate with the architectural wall system. Sun control devices must be manufactured devices to provide sun control on exterior windows and storefronts. Provide sun control devices to withstand the wind loads prevailing at the project site.

B201012 SCREEN WALL

Provide screen walls compatible with the exterior architecture of the building. Design rooftop mechanical screens to minimize roofing penetrations.

Screen walls include attached or unattached walls adjacent to the main building. Screen walls must conform to the applicable portions of Section B201001 EXTERIOR CLOSURE.

B2020 EXTERIOR WINDOWS

Provide windows in each area of the building that is regularly occupied, to enhance the working environment, without compromising visual acuity and comfort. Natural daylighting is preferred. Windows must meet Antiterrorism requirements.

Windows must be provided with sills on the exterior and stools on the interior of the opening. Sills must be special shaped or cut unit masonry or precast concrete in masonry exterior construction and extruded aluminum or aluminum-wrapped wood framing or formed metal in other construction. Positively slope sills away from windows. Window stools must be slate or solid polymer for commercial construction.

B202001 WINDOWS

Determine the construction of security windows by evaluating the project program security requirements, using the Military Handbook (MIL-HDBK) 1013/1A, Design Guidance for Physical Security of Facilities, to define window requirements.

Provide glazing in exterior windows in accordance with section B202004 EXTERIOR GLAZING.

B202002 STOREFRONTS

Storefronts must be aluminum. Provide one-story storefront system fabricated from formed and extruded aluminum and glass components for exterior use.

B202004 EXTERIOR GLAZING

Provide glazing with a color matching existing adjacent buildings or as prevalent on the base. Provide glazing as appropriate for its specific use. Provide setting and sealing materials, stops and gaskets as recommended by the glass or acrylic sheet manufacturer.

Provide warranty for insulating glass units for a period of 10 years against development of material obstruction to vision (such as dust or film formation on the inner glass surfaces) caused by failure of the hermetic seal, other than through glass breakage. The Contractor is required to provide a glazing warranty for curtain wall glazing written directly to the Government.

Provide warranty for polycarbonate sheet glazing for a period of 5-years against breakage, coating delamination, and yellowing.

B2030 EXTERIOR DOORS

Provide solid door assemblies other than at the main entrance. Exterior doors and frames must be non-corroding galvanized steel.

Provide heavy duty insulated exterior steel doors and frames for service access. Door frames must be welded. Corner knockdown door frames are not permitted.

Use heavy-duty overhead holder and closer to protect doors from wind damage. Provide kickplates on the inside face of all exterior doors.

Weather-protect all exterior doors and related construction with low infiltration weatherstripping and sealants. Provide threshold with offset to stop water penetration while maintaining accessibility compliance.

Conform to the design criteria of ASCE 7.

See section B203008, EXTERIOR DOOR HARDWARE, for door hardware requirements. For all installations, provide a recessed key box (Knox Box) approximately 7 inches x 7 inches (175 mm x 175 mm) with 4-3/4 inches (120 mm) solid steel door at primary exterior entry for storage of keys and access cards accessible by the fire department.

Selection of hardware for hangars should anticipate that the doors and hardware are going to receive abuse. This is in particular for the doors accessing the hangar bay from the exterior. Hardware should be selected based on this anticipation. All hardware must be grade 1 without exception.

B203001 SOLID DOORS

Provide solid steel door assemblies other than at main storefront entrances including painted, prefinished, zinc coated, galvanized heavy-duty, non-corroding, insulated doors with frames and hardware. B203002 GLAZED DOORS

Glazed Doors - Provide Exterior Glazed Doors and Entrances System. Including factory-finish aluminum framed or stainless-steel framed door assemblies with insulated, tinted glazing, frames, and hardware compatible with other buildings on the base and wall opening elements such as lintels, sills, through-wall flashings, and joint sealers.

B203006 BLAST RESISTANT DOORS

Provide blast resistant doors as required in accordance with UFC 4-010-01, DoD Minimum Antiterrorism Standards for Buildings.

B203008 EXTERIOR DOOR HARDWARE

Provide the services of a certified door hardware consultant to prepare the door hardware schedule.

Provide hardware keying compatible with the existing base-wide keying system. Provide replacement interchangeable cores compatible with the Best Lock system.

Provide the services of an Architectural Hardware Consultant (AHC), Certified Door Consultant (CDC), or an Electrified Hardware Consultant (EHC) to assist the Designer of Record in preparation of the door hardware schedule and product selection. The hardware consultant must sign and seal the door hardware construction submittal. Provide, as far as possible, door hardware of one manufacturer's make. All hardware must be clearly and permanently marked by the manufacturer where it will be visible after installation.

END OF SECTION

B30 ROOFING

1. GENERAL SYSTEM DESCRIPTION

Provide watertight roof systems compatible with facility function, construction, and service conditions. Provide complete roof system design and construction services for each facility roof system, including all ancillary and incidental work necessary for a complete, new, watertight roof system installation.

Submittal Requirements: Components of a minimum roof submittal include roof plans, method of drainage, standard details and details unique to the project, wind load calculations and requirements.

Provide a Pre-Design Roofing Conference and Pre-Roofing Conference to ensure roof design and construction is properly coordinated before construction begins.

Refer to UFC 3-110-03, Roofing, UFC 3-101-01, Architecture, and UFC 1-200-02, High Performance Sustainable Building Requirements for additional roofing requirements.

B3010 ROOF COVERINGS

B301001 STEEP SLOPE ROOFING SYSTEMS

Steep slope roofing systems are preferred over low slope roofing systems, where practical. Steep slope roofing systems that are acceptable include metal.

Provide ice and water shield membrane under finished roofing. Provide ice and water shield membrane as a self-adhering rubberized asphalt adhesive with a high-density cross-laminated polyethylene film, able to seal around fasteners.

Provide manufacturer's no-dollar-limit materials and workmanship warranty. Warranty period must not be less than 20 years from date of Government acceptance of work. Issue warranty directly to the Government. Warranty must provide that within warranty period, if the metal roofing system becomes non-watertight, or, shows evidence of corrosion, perforation, rupture, or excess weathering due to deterioration resulting from defective materials, or, installed workmanship the repair or replacement of defective materials and correction of defective workmanship must be the responsibility of roofing system manufacturer. In addition, provide a 2-year contractor installation warranty.

B301002 LOW SLOPE ROOFING SYSTEMS

1. WIND UPLIFT

Provide a complete roof covering assembly in accordance with Factory Mutual (FM) P7825.

2. FIRE SAFETY

Provide a complete roof covering assembly:

- Rated class A in accordance with FM 4470 or underwriters laboratories (UL) 790; and
- Listed as part of fire-classified roof deck construction in ul roofing materials and systems directory (RMSD) or class I roof deck construction in FM P7825C.

As appropriate, use a ventilating base sheet over any materials which may contain moisture which may need to transpire out of the building.

Low slope roofing systems that are acceptable include three-ply built-up roofing systems with modified bitumen cap sheet surfacing or three-ply modified bitumen roofing. Roofing system will be required to include a top surface finish that meets the criteria for Cool Roof Products.

3. ROOF DRAINS

Provide all roof drains, primary and overflow drains as interior drains. Provide low slope roofs at no less than half an inch for each foot and located drain accordingly. Use of scuppers as primary or overflow drains will not be accepted. Provide interior roof drain waste vent (DWV) throughout as insulated drainpipes.

4. COOL ROOF

Meet the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE) 90.1 Chapter 5 values for cool roofing. If a cool roof is not selected in zones 1-3, meet one of the exception requirements listed in ASHRAE 90.1 Chapter 5 or provide thermal insulation above the deck with an R value of 33 or greater.

B301003 ROOF INSULATION AND FILL

Provide roof insulation values no less than in accordance with UFC 1-200-02, High Performance and Sustainable Building Requirements and UFC 3-110-03, Roofing. Roof assembly requires a minimum thermal value of R30 with Air barrier sealed all around.

For fastening roof insulation on low-slope membrane roofs, place fasteners to withstand and obtain an uplift pressure as appropriate for the geographical region.

B301004 FLASHINGS AND TRIM

Flashing and sheet metal work includes scuppers, splash pans, and sheet metal roofing. Flashings must be copper, sheet and strip - ASTM B 370, cold-rolled-tempered Stainless Steel - ASTM A 167, Type 302 or 304 or Pre-Finished Aluminum. Finish must be baked-on factory applied color coating of polyvinylidene fluoride (PVF2) or other equivalent fluorocarbon coating with a minimum thickness of 0.8 to 1.3 mils.

B301005 GUTTERS AND DOWNSPOUTS

Built-in gutter systems, where collection of drainage is integrated within building roof and wall enclosures or is concealed in the exterior cavity wall is prohibited.

B301090 OTHER ROOFING

Provide lightning protection, without penetrating the roof membrane or flashing components.

END OF SECTION

C10 INTERIOR CONSTRUCTION

1. SYSTEM DESCRIPTION

Interior construction includes interior partitions, interior doors, and fittings. Provide durable construction appropriate for the building function. Acoustic properties of materials, as well as durability, must be considered during material selection.

2. GENERAL SYSTEM REQUIREMENTS

Areas of the Project subject to abuse require that "impact resistant" systems be provided. See “Room Data Sheets located in Appendix A for specific requirements on "Partitions”.

C1010 PARTITIONS

Refer to Section C30, Interior Finishes, for additional information on Gypsum Wallboard and finishes. Provide concrete masonry or gypsum board on metal studs partitions.

Refer to Room Data Sheets located in Appendix A, for partition requirements. Where rooms with different partition requirements adjoin one another, provide a combined wall type that meets Scheduled requirements for STC, Fire Rating, Security, and durability as well as finish requirements of both spaces.

For general use, metal studs with gypsum wallboard (GWB), CMU with prime filler coat, or CMU/cast-in-place concrete with GWB are acceptable, unless stated otherwise in Project Program. Reinforce points where doorknobs can strike a wall and anchorage points for wall mounted equipment.

Provide control joints and installation techniques as recommended by the manufacturer. See Section C30, Interior Finishes, for additional information.

Provide painted GWB with access panels at surfaces furred for HVAC, plumbing and other utility services and controls behind wall surfaces.

Acceptable systems where "IMPACT RESISTANCE" (areas subject to physical abuse or wear) is designated in the project program requirements for impact resistance systems include CMU/cast-in-place concrete with or without plaster or furred impact resistant GWB or surface applied impact resistant textured acrylic architectural coating system.

GWB/metal stud system reinforced for impact resistance with a double layer of gypsum board using at least one layer of impact-resistant gypsum board to resist denting and puncturing on the impact surface. If wall is subjected to impact on both sides, both sides of the stud require a double layer of gypsum board. Structural, mechanical, and acoustic design requirements affect the metal stud/gypsum support configuration.

C101001 FIXED PARTITIONS

Provide fixed interior partitions that extend from finished floor to underside of structure above, except where floor to ceiling demountable or retractable partitions are specifically required by the "Room Requirements."

Provide fixed partitions, except where demountable or retractable partitions are specifically required by the "Room Requirements", to include wood or metal studs, GWB, plaster, masonry and cast-in-place concrete walls. Sound-rated partition assemblies must have a minimum Sound Transmission Coefficient (STC) as required by the project program. Construct sound-rated bulkheads above partition assemblies for continuity to the deck above.

C101002 DEMOUNTABLE PARTITIONS

Demountable partitions include sound rated, full height (floor to ceiling) solid partitions, and wire mesh partitions. Coordinate finishes with Section C30 Interior Finishes.

C101003 RETRACTABLE PARTITIONS

Provide retractable partitions to include operable panel partitions and accordion folding partitions. Coordinate finishes with Section C30 Interior Finishes.

C101004 INTERIOR GUARDRAILS AND SCREENS

Provide balustrades where required by code. Provide screens where required to prohibit view of a particular area.

C101005 INTERIOR WINDOWS

Provide interior windows as a complete factory-assembled unit with glass factory or field installed. Where observation windows are scheduled, provide fixed Steel Framed interior windows, assemblies with clear, low-E, double pane glazing, caulking, and other associated work. Observation windows will require fire-rated and a fusible link counter door.

C101006 GLAZED PARTITIONS & STOREFRONTS

Provide glazed storefront system.

C101007 INTERIOR GLAZING

Provide interior glazing of clear glass, tempered glass. ASTM C 1036, unless specified otherwise. Provide patterned glass where required to obscure view into bathrooms and dressing rooms. Provide setting and sealing materials, stops and gaskets as recommended by the glass manufacturer. Glazing thickness indicated in the following paragraphs is the minimum acceptable thickness. Provide thicker glazing if required by the code or the manufacturer for the given application.

C101008 INTERIOR JOINT SEALANT

Sealant joint design and application must be in accordance with the general requirements of Sealants: A Professionals' Guide from the Sealant, Waterproofing & Restoration Institute. Refer to manufacturers' recommendations for chemical resistance.

C1020 INTERIOR DOORS

C102001 STANDARD INTERIOR DOORS

Refer to Part 3 Section 5.0, Room Requirements for door requirements for individual rooms. Where rooms with different door requirements are connected by a door, provide a door type that meets the security and durability as well as finish requirements of both spaces.

C102002 GLAZED INTERIOR DOORS

Provide vision glazing in doors where it is required by the "Room Requirements" portion of this RFP, or it is deemed advantageous to be able to see through the door, either for safety of pedestrian traffic, or other functional reason.

C102003 FIRE DOORS

Provide interior fire doors. Provide in conformance with National Fire Protection Association (NFPA) 80 an NFPA 105. Fire doors and frames must bear the label of UL, FM or WHI attesting to the rating required. Door and frame assemblies must be tested for conformance with NFPA 252 or UL 10C (for positive pressure). Wood fire doors must also comply with ASTM E 152. Provide stainless steel astragals complying with NFPA 80 for fire-rated assemblies and NFPA 105 for smoke control assemblies.

C102004 SLIDING AND FOLDING DOORS

Not Used.

C102005 INTERIOR OVERHEAD DOORS

Provide interior overhead doors as needed. Refer to RFP, B20 Exterior Enclosure - "Overhead Roll-Up and Overhead Sectional Doors" for interior overhead door requirements. Design for ASCE 7 wind loading is not required for interior overhead doors.

C102006 INTERIOR GATES

Not Used.

C102007 INTERIOR DOOR HARDWARE

Provide special door hardware, such as combination locks, and card key system as determined.

C102090 OTHER INTERIOR SPECIALTY DOORS

Not Used.

C102091 OTHER INTERIOR PERSONNEL DOORS

Not Used.

C1030 SPECIALTIES

C103001 COMPARTMENTS, CUBICLES, & TOILET PARTITIONS

Provide phenolic core toilet partitions in all toilet rooms with more than one water closet or urinal.

C103002 TOILET AND BATH ACCESSORIES

Provide toilet and bath accessories to include toilet tissue dispensers, paper towel dispensers, sanitary napkin disposal units, grab bars, and bathroom mirrors.

C103003 MARKER BOARDS AND TACK BOARDS

Provide marker boards and tack boards as indicated in Chapter 5 "Room Requirements" portion.

C103004 IDENTIFYING DEVICES

Provide interior room identification signs at each entrance to each interior room. Provide signage to identify each space by room number and name. Signage for general office areas must have changeable room name sections to accommodate personnel and functional changes.

Incorporate all necessary interior signage as part of the architectural drawings. Interior signage is not collateral equipment. Interior signage must demonstrate complete coordination with the facility design, Structural Interior Design (SID) and FF&E submittals. Provide interior directional signage as required for facility wayfinding. Provide an identifying device at each interior door. Signs must meet Architectural Barriers Act (ABA) Standards requirements. Refer to Unified Facilities (UFC) 3-120-01, Design: Sign Standards, for more information.

C103005 LOCKERS

Provide high density polyethylene/solid plastic lockers. Lockers are funded as part of the construction contract.

C103006 SHELVING

Provide steel utility and plastic laminate clad shelving. Built-in fixed shelving is funded as part of the construction contract. Assemblies include all types of shelving with brackets and all supporting materials and finishing, if required.

C103007 FIRE EXTINGUISHER CABINETS

Provide fire extinguisher cabinets. Cabinet must be constructed of 16 gauge cold-rolled steel door panel / front, and a 22-gauge cold-rolled steel tub. Cabinet must be fire-rated if located in a fire rated wall assembly, and have a full-length piano hinge, and baked enamel finish. Provide a stainless-steel cabinet door if cabinet is exposed to the environment. Size and locate fire extinguisher cabinets to encase extinguisher as required by NFPA 10 and NFPA 101.

C103008 COUNTERS

Provide acrylic counter tops, back splashes, and side splashes as needed.

C103009 CABINETS

Provide cabinetry and millwork items with associated accessories. Cabinetry must be Architectural Woodwork Institute (AWI) premium grade and have concealed hinges with adjustable standards for shelves. All exposed surfaces must be covered with high pressure plastic laminate clad.

Provide specific cabinetry and storage as noted in Chapter 5 Room Requirements.

C103011 CLOSETS

Provide a complete coat check closet for public use, including service counter, coat storage system, and secure enclosures. Provide heavy-duty coat rods with hangers; numbered tagging system; adjustable shelving for bags and accessories. Lockable doors and secure storage areas. Provide plastic high-pressure laminate (HPL) over medium-density fiberboard (MDF) for shelving and millwork components. Exposed edges to receive high-impact PVC or wood edge banding.

C103012 FIRESTOPPING PENETRATIONS

Provide all penetrations through rated walls and floors with rated material for firestopping penetrations.

C103013 SPRAYED FIRE-RESISTIVE MATERIALS

Provide sprayed fire-resistive materials to the building's structural framing components as required by Building Code to prevent structural failure.

C103014 ENTRANCE FLOOR GRILLES AND MATS

Provide recessed floor mats at main building entrances. Provide recessed floor mats at all other building entrances. Provide entrance mats with carpet surface treads. Provide entrance mats at all entrances to the facility. Comply with Architectural Barriers Act (ABA) Standards for installed entrance mats and frames. Provide recessed entrance mats at building entrances with enclosed vestibule and surface applied entranceway mats or entranceway floor tiles at all other entrances. Entranceway mats and entranceway floor tile require the use of a transition edge where the mat adjoins other floor materials. Mat system must meet ASTM D-2047 coefficient of friction requirements of minimum 0.60 for accessible routes and be structurally capable of withstanding a uniform floor load of 300 lbs/sq. ft. All portions of mat system must comply with ASTM E 648, Class I, Critical Radiant Flux, minimum 0.45 watts/m2 for flammability.

C103015 ORNAMENTAL METAL WORK

Provide ornamental stair handrails. Building components made from ornamental metals. Ornamental stair handrails are included in B1010 EXTERIOR STAIRS and C20, STAIRS.

C103090 OTHER INTERIOR SPECIALTIES

Provide above ceiling mounted motorized projection screen. Provide fixed ceiling mount for computer projector. Coordinate location with Audio/Visual equipment package.

END OF SECTION

C20 STAIRS

Provide stairs, including stair construction and stair finishes as required by the building code to provide egress from the building from above or below grade level floors. Stairs must be in accordance with Unified Facility Criteria (UFC) 1-200-01, DoD Building Code (General Building Requirements).

C2010 STAIR CONSTRUCTION

C201001 INTERIOR AND EXTERIOR STAIRS

Concrete is an acceptable finish for exterior stairs. Provide cast aluminum treads with abrasive surface for all exterior concrete stairs.

Steel stairs must be primed and painted.

C201090 HANDRAILS, GUARDRAILS, AND ACCESSORIES

Provide handrails and guardrails. Handrails and guardrails must present a smooth, unbroken surface throughout the length of the stairs.

Handrails and guardrails must be finished to withstand extreme wear conditions.

Ladders and railings complying with Occupational Safety and Health Administration (OSHA) requirements must be provided for access and protection to any mechanical mezzanines, lofts or other similar spaces.

END OF SECTION

C30 INTERIOR FINISHES

1. SYSTEM DESCRIPTION

Interior finishes include wall finishes, floor finishes, wall base finishes, and ceiling finishes.

Provide aesthetically pleasing, functional, durable finishes appropriate to the building's function. Consider acoustic properties of materials, as well as durability and ease of maintenance during material selection. Maximize the use of sustainable materials.

Color selections require the use of wall and floor finish material accents to enhance the color and appearance of the interior design. Provide a wall and floor color design that includes a minimum of two different accents colors throughout the facility. Submit pattern drawings of the accents design with the interior design submittal.

2. GENERAL SYSTEMS REQUIREMENTS

For finishes, refer to Room Data Sheets, see Appendix A.

Provide paint and coating products certified to meet indoor air quality requirements by UL 2818 (Greenguard) Gold, SCS Global Services Indoor Advantage Gold or provide certification by other third-party programs. Provide current product certification documentation from certification body. All interior paints and coatings are required to comply with LEED v4 Low-Emitting Materials requirements. Products must meet volatile organic compounds (VOC) limits according to the SCAQMD Rule 1113 (or equivalent). Compliance applies to all interior walls, ceilings, trim, and doors and contributes to IEQ Credit: Low-Emitting Materials. In areas with high humidity or frequent water exposure, such as locker rooms and showers, paints are required to also provide mold- and mildew-resistant performance.

The Project requires industrial coatings on GWB walls and ceilings in areas subject to high humidity, frequent water exposure and chemical contact. All Contractors and subcontractors that perform surface preparation or coating application must be certified by the Society for Protective Coatings (formerly Steel Structures Painting Council - SSPC) to the requirements of SSPC QP 1 prior to contract award and must remain certified while accomplishing any surface preparation or coating application.

C3010 WALL FINISHES

All interior wall finish materials must be gypsum board or cement board where porcelain tile is used. Refer to Room Data Sheets, Appendix A, for wall finish locations. Provide moisture and mildew resistant interior wall finishes which are easily maintained, and suitable in accordance with industry standards for the architectural surface being finished. For painted wall finishes, refer to C3040 "INTERIOR PAINTING AND SPECIAL COATINGS".

C301001 CONCRETE WALL FINISHES

Cast-in-place or pre-cast concrete wall finishes include, but are not limited to, abrasive blasted surfaces, colored surfaces, exposed aggregate, grooved surfaces, or tooled surfaces.

C301002 PLASTER WALL FINISHES

Veneer plaster must be gypsum plaster veneer finish on gypsum base finishes, or cement plaster veneer finish on concrete or masonry. Refer to Section C3040 for paint system and gloss level.

C301003 GYPSUM WALLBOARD

Conform to specifications, standards and requirements in accordance with Gypsum Association GA 214, GA 216 and GA 224. Provide asbestos free materials only. Provide all Gypsum Wallboard as 5/8-inch thick, Type X fire rated panels as a minimum. Provide a foil back gypsum board when a vapor retarder is required.

C301004 TILE AND TERRAZZO WALL FINISHES

1. CERAMIC TILE WALL SYSTEM FINISHES

Provide ceramic tile wall systems as defined in the Tile Council of North America (TCNA) handbook for ceramic tile installations suitable for the service requirements listed. Install systems in accordance with Tile Council of North America Handbook and American National Standards Institute (ANSI) A108/A118 series standards. Colored epoxy grout with sealer must be provided. Coordinate with ceramic bath accessories for modularity. Include all trim pieces, caps, stops, and returns to complete installation.

2. TERRAZZO FLOORING WITH INTEGRAL BASE

Provide and install poured-in-place epoxy terrazzo flooring with integral cove base for commercial interior applications requiring durability, aesthetics, and ease of maintenance.

C301005 WALL COVERINGS

Wall coverings must be material designed specifically for the specified use. The wallcovering must contain a non-mercury based anti-microbial. The wallcovering must be the type made without the use of cadmium-based stabilizers. Wallcovering must have a Class A flame spread rating of 0-25 and smoke development rating of 0-50 when tested in accordance with ASTM E84. The wall preparation, trimming, adhesive and application must be according to the manufacturer's printed directions. The manufacturer must approve the installers in writing. The material must be easily cleaned by traditional methods such as washing, wiping, or vacuuming. Primer and adhesive must be of a type recommended by the wallcovering manufacturer and must contain a non-mercury based anti-microbial. Adhesive must be strippable type. Do not apply wall coverings to the interior surface of exterior walls.

C301006 ACOUSTICAL PANELS ADHERED TO WALLS

Acoustical wall treatment must be acoustical panels, sound absorbing wall units, or acoustical wall systems. Acoustical panel system must include manufacturer's standard concealed fasteners, splines, tracks, and other components necessary to complete the installation. Fire rating for the complete composite system must be Class A, 200 or less smoke density and flame spread less than 25, when tested in accordance with ASTM E84.

C301090 OTHER WALL FINISHES

1. SOLID SURFACING WALL FINISHES

Solid surfacing material must consist of 100 percent pure acrylic polymer, mineral fillers, and pigments. The material must be homogenous, not coated or laminated, meeting ANSI Z124.3 and ANSI Z124.6 requirements. Superficial damage to a depth of 0.010 inch (.254 mm) must be repairable by sanding or polishing. Provide manufacturer's full range of colors and patterns. Flammability, ASTM E84: Class I/A, flame spread 25 maximum; smoke developed 30 maximum.

If used in a shower, solid surfacing wall finishes must extend from top of shower pan to a minimum of to the underside of ceiling and must surround the shower enclosure. If used in a kitchen, solid surfacing wall finish must extend from top of kitchen countertop to underside of wall cabinet.

Provide solid surfacing with factory recommended fasteners/adhesives/caulk to complete the installation.

2. PLASTIC LAMINATE WALL FINISHES

Plastic laminate used for wall applications must be commercial grade, high-pressure laminate with a #60 finish, approved for vertical applications. National Electrical Manufacturers Association (NEMA) LD 3.

3. DECORATIVE PANELING SYSTEM

Architectural paneling system applied to interior walls are required to include associated furring, fastening, and trim to complete the installation. Wood paneling system finishes are required to be factory or field applied.

4. WOOD TRIM AND DETAILING FINISHES

Decorative panels, chair rail, standing and running trim, must be of AWI custom grade hardwood with a painted or stained finish. Refer to C3040 "INTERIOR PAINTING AND SPECIAL FINISHES" for finish system. Chair rail must be a minimum of 3-1/2 inches (89 mm) high. The profile of chair rail must be a molded shape. Wood trim must include associated furring, fastening, adhesives and trim to complete the installation.

5. IMPACT RESISTANT PANEL OR WAINSCOT WALL FINISHES

The wall covering panel system, or wainscot, must be an impact-resistant acrylic PVC sheet of a minimum 0.060-inch (1.5 mm) thickness in 4 feet by 8 feet (1219 mm by 2438 mm) sheets. The system must be Class A (ASTM E84), Underwriters Laboratories (UL) listed, and chemical and stain resistant. It must include all accessories, such as top caps, joint covers, and inside and outside corners, necessary for complete installation. A full range of colors and textures must be included. The wall panel system must have coordinating color and pattern options for all components within the system. The wall panel system must offer a 21-ounce (595 g) fabric backed vinyl wallcovering laminated to a 0.020 inch (.51 mm) rigid acrylic/PVC backing capped with 1 mil of protective film.

Impact resistant chair or handrail system must be a formed rigid PVC product. Chair or handrail must be a minimum of 3 inches (76 mm) high and be mounted with concealed hardware. Chair or handrail system must be chemical, stain, and bacteria resistant. Chair rail must be UL-classified, conforming to National Fire Protection Association (NFPA)Class A fire rating and ASTM D256-90b for impact strength of 30.2 ft-lbs/inch thick.

6. CORNER AND WALL GUARDS

Corner and wall guards must be high impact formed polyvinyl chloride a minimum of 0.078 inch (2 mm) with concealed mounting hardware and end closure. If used with an impact resistant panels system, the guards must be from the same manufacturer as the impact resistant wall panel system, chair or handrail system and must include all accessories necessary for a complete installation. A full range of styles, colors and textures must be included.

7. STAINLESS STEEL WALL PANELS

Provide factory-fabricated stainless steel wall panels designed for use in commercial kitchen environments. Panels are required to be corrosion-resistant, smooth, and easily cleanable, with concealed fasteners where possible. Finish must be No. 4 satin/brushed stainless steel for durability and appearance. Panels are required to be installed with tight, aligned joints to ensure sanitary conditions and ease of maintenance.

C3020 FLOOR FINISHES

Provide floor finish materials to meet the following requirements:

A. Provide tufted modular carpet tiles, level loop or multi-level loop.
   - Fiber – Solution-dyed nylon or equivalent commercial grade fiber with stain resistance.
   - Backing – Dimensionally stable, moisture-resistant, minimum 10 percent recycled content
   - Environmental Attributes – Minimum 10 percent pre- and post-consumer recycled content in face fiber or backing; carpet tile is expected to be recyclable through the manufacturer's take-back program, when available.
   - Adhesives – Low VOC, CRI Green Label Plus certified.

B. Carpet tile is required to meet a minimum TARR rating of 3.0 (severe use) suitable for heavy commercial traffic areas.

C302001 TILE FLOOR FINISHES

Provide ceramic tile floor systems as defined in the Tile Council of North America (TCNA) handbook for ceramic tile installation and materials for the service requirements listed. Provide installation and materials in accordance with ANSI A108/A118 series standards, except do not use organic adhesives. Provide manufacturer's full range of colors and styles. Tile must be a minimum of two grades above base grade.

Mortar must be Portland cement, ANSI A108.1A/1B/1C/ A118.1, Latex-Portland cement, ANSI A108.5/A118.4 or Epoxy ANSI A108.6/A118.3.

Grout must be factory sanded Portland cement, ANSI A108.10/A118.6, Latex-Portland cement, ANSI A108.10/A118.7 or Epoxy ANSI A108.6/A118.3. Provide tile joint grout sealer on white, light-colored areas that are routinely exposed to water and liquid cleaning materials, entrance areas, and areas that require a high degree of stain resistance, and as required by the manufacturer. Provide chemical resistant epoxy resin for kitchens and other areas where high resistance to staining and absorption are required, ANSI A118.3.

Slip resistant tile must have a minimum Dynamic Coefficient of Friction (wet and dry) of 0.42, ANSI A137.1-2012. Tile must have smooth, non-slip or textured surface and a glazed or unglazed finish. Non-slip or textured surface required for tile in areas where there is excessive water or grease and oils such as kitchens, dining facilities, shower rooms, toilets, and in industrial and maintenance facilities.

C302004 RESILIENT FLOOR FINISHES

All resilient flooring must meet or exceed applicable Architectural Barriers Act (ABA) Standards horizontal requirements. Install each type of flooring with recommended adhesive in accordance with the manufacturers' written instructions. Installers must be approved by the manufacturer in writing and must have a minimum of three years' experience for each type of flooring to be installed. Provide and store a minimum of two percent total quantity for each type flooring, color and pattern within each building for future replacement and patching. Provide manufacturers full line of color and pattern selections, including multi-color patterns. Use the resilient floor finishes as identified in the Project Program or as directed below.

C302005 CARPETING

Installer(s) must be approved by the manufacturer in writing. Carpet manufacturer must be established and in good standing with the industry. A minimum of 5 percent total quantity for each color and pattern must be provided and stored within the building for future replacement patching.

Reclamation of existing carpet to be determined with potential vendor. When carpet is replaced, submit certification documentation from the reclamation facility to the Contracting Officer.

C302007 WALL BASE FINISHES

Provide a wall base for transition between floor and wall finish. If no other type of base is required, provide rubber or vinyl straight base at carpet installations, rubber or vinyl cove base at exposed concrete or resilient tile floors, and a base to match the floor material at hard surface tile floors, or as required in the project program.

C302010 HARDENERS AND SEALERS

Harden and seal concrete floors in accordance with the finished floor manufacture requirements. Utilize other methods of concrete curing if the floor finish manufacturer does not recommend a chemical hardener or sealer. Concrete floors that can utilize a hardener-sealer and will be exposed to traffic must receive a minimum of two coats of hardener-sealer curing agent for dust protection. These hardener-sealer-cured floors must be finished with a curing agent that must penetrate the concrete to permanently seal the floor against moisture and the penetration of contaminants. The curing agent must be non-toxic, non-flammable, and non-combustible and must be installed in accordance with the manufacturer's printed instructions. The finished floor must be dust-free.

Colored concrete floors must include a colored pigment either applied as a topical dye; or a concrete topping with integral color pigment; or a dry shake pigment application, as required by the project program. Concrete floor must be trowel applied in a pattern or must include grit for slip resistance.

C302011 RAISED ACCESS FLOORING

Design support system to allow for 360-degree clearance in laying out cable and cutouts for service to machines and so that panel and stringer together take up maximum of 2 inches (50 mm).

C3030 CEILING FINISHES

Primary ceiling finish in administrative areas must be 24 inches by 24 inches by 5/8 inches minimum thickness suspended acoustical panel ceiling system, except provide a suspended gypsum board ceiling in entrance lobby, restrooms and showers. Provide acoustical panels with a regular edge.

Paint exposed structural systems in accordance with Section C3040 INTERIOR COATINGS AND SPECIAL FINISHES.

Provide ceiling finishes as indicated in Room Data Sheets portion of this RFP.

Refer to C3040 "INTERIOR PAINTING AND SPECIAL COATINGS" for painted ceiling finishes.

C303001 ACOUSTICAL CEILING TILES AND PANELS

All acoustical ceiling panels must be 24 inches by 24 inches (610 mm by 610 mm), with a minimum light reflectance of .75 (except as noted), Class A, flame spread 25 or less and smoke development of 50 or less, ASTM E84.

For typical open-minded office areas, conference rooms, executive offices, provide non-asbestos mineral composition acoustical ceiling panels of Type III with factory-applied standard washable painted finish or Type IV with factory-applied plastic membrane-faced vinyl, Form: 1, 2, or 3. Provide reveal-edge tiles unless otherwise noted.

For typical humid areas such as toilets, kitchens, fitness and locker rooms, provide non-asbestos mineral or glass composition acoustical ceiling panels bonded with ceramic, moisture resistant thermo-setting resin, or other moisture resistant material with factory-applied standard washable painted finish.

Provide NRC and CAC ratings as follows:

Space Type	Minimum NRC Rating	Minimum CAC Rating
Open Office Areas, Auditoriums	35-39	0.70
Conference Rooms, Classrooms	35-39	0.60
Activity Spaces, Lobbies, Corridors	35-39	0.60
Executive and Private Offices	35-39	0.60
Toilets	35-39	0.50
Kitchens	35-39	0.50
Fitness/Locker Rms	35-39	0.50
All Other Spaces	35-39	0.50

C303002 GYPSUM WALLBOARD CEILING FINISHES

Conform to specifications, standards and requirements in accordance with Gypsum Association GA 214, GA 216 and GA 224. Provide asbestos free materials only. Provide featured edge gypsum board on all gypsum surfaces that flatness of joints will be visible, such as up-lighted ceilings, window lighted ceilings, and as recommended by the manufacturer. Provide Type X gypsum board in fire rated assemblies.

C303005 SUSPENSION SYSTEMS

Provide 12 inch by 12 inch (305 mm by 305 mm) aluminum or steel non-corroding intermediate-duty concealed grid system for lay-in acoustical panels (ASTM C635).

C3040 INTERIOR COATINGS AND SPECIAL FINISHES

Paint all interior exposed surfaces except factory finished items that are not intended for field coating including but not limited to finished metals (copper, stainless steel, aluminum, brass and lead) door hardware, interior grilles, registers, diffusers, access panels, and panel boxes.

Apply coatings directly to all non-prefinished surfaces of the interior construction. Comply with Master Painters Institute requirements for surface degradation analysis, surface preparation, paint and coating selection, paint application restrictions for substrate materials, and paint application.

C304001 GENERAL REQUIREMENTS

All paint must be suitable in accordance with the Master Painter Institute (MPI) standards for the interior architectural surface being finished. The current MPI, "Approved Product List" as of the date of contract award, will be used to determine compliance with the submittal requirements of this specification. The Contractor may choose to use a more current MPI "Approved Product List"; however, only one list may be used for the entire contract. All coats on a particular substrate, or a paint system, must be from a single manufacturer. No variation from the MPI Approved Products List is acceptable.

Select paint systems for the project in accordance with the MPI Architectural Painting Decision Tree available on the Whole Building Design Guide. Use this interactive MPI Decision Tree website to identify applicable paint system(s) for the project. The MPI Decision Tree identifies paint systems for each interior or exterior coated surface in "Normal" or "Aggressive" environmental conditions and generally lists the applicable paint systems in descending order of performance. The paint system at the top of each substrate list generally indicates the highest performing acceptable coating system.

Choose the "Aggressive" environmental conditions in the MPI Decision Tree for exterior systems that are used in moist humid conditions, abrasive conditions, chemical exposure conditions, or within five miles proximity of the ocean or a body of water. Also use "Aggressive " environmental conditions in interior spaces that are exposed to in moist humid conditions, abrasive conditions, chemical exposure conditions, such as bathrooms, shower rooms, kitchens, chemical storage area, swimming pools, laundry, sanitary areas, commercial kitchens, industrial production areas, and hospital operating rooms provide paint systems that comply with the MPI Decision Tree "Aggressive" environmental conditions.

If the MPI Decision Tree does not identify all paint systems applicable to the facility, use the MPI Architectural Painting, Exterior Systems Manual to identify other appropriate paint systems for the project. Utilize the "Premium Grade" systems and comply with all limitations stated in the MPI "Approved Product List" for each paint product. Products having an MPI VOC Range E3 must be given preferential consideration over lower VOC Ranges. Use higher performing paint systems unless the lower performing paint system can be justified based on a lifecycle cost to include surface preparation, application, disposal, environmental impact, and required recoating cycles. Only use paint products that have been tested for MPI'S "DETAILED PERFORMANCE" or "EVALUATED PERFORMANCE ". Do not use products that have only been tested for "INTENDED USE".

If an "Aggressive" environmental condition option is not available in the MPI Decision Tree for a certain substrate, use the "Normal" environmental condition option.

Refer to the Additional Exterior Paint and Coating System Requirements below for further system requirements.

Paints and coatings must comply with Master Painters Institute Green Performance Standard GPS-1-12 which is available at the following website: http://www.specifygreen.com/EvrPerf/EnvironmentalPerformance.html. Provide Interior flat intermediate and topcoats of a maximum of 50 g/L VOC and interior non-flat intermediate and topcoats of a maximum 150 g/L VOC. Choose paints that provide performance and are environmentally friendly by using total VOC budgeting to analyze the total impact of all flat, non-flat and special purpose coatings on the project.

C304007 SPECIAL COATINGS TO WALLS

High Performance Architectural Coatings (HIPAC) must be a durable, organic system applied to a continuous (seamless) high-build film and cure to a hard glaze finish. They must be resistant to continuous heat and humidity, abrasion, staining, chemicals, and biological growth. Coating must be installed as a complete system, and as recommended by the manufacturer and have a flame spread index of not more than 25 and a smoke developed index of not more than 50 when tested in accordance with ASTM E84.

Two-component, epoxy-polyamide must be chemical and corrosion-resistant, adhesive, alkali-resistant, and water-tolerant for metal, wood, concrete, masonry surfaces, and painted surfaces where high gloss or glaze type finish, extreme workability and resistance to abrasion and stains is required. Minimum dry film thickness is 3 mils for each of two coats. Provide Gloss or Semigloss finish. Maximum VOC must be 340 grams/liter.

Single Component, Moisture-Curing Urethane must be a flexible, abrasion- and impact-resistant, use for floors, walls, machinery, equipment and other surfaces where good abrasion resistance, color retention, gloss retention, graffiti resistance and good resistance to acids, alkalis, solvents, strong cleaners and sanitizers, fuel and chemicals are necessary. Can also be used on concrete floors, brick and masonry surfaces (properly conditioned), metals (properly primed), and wood (properly prepared and sealed.) Minimum dry film thickness is 3 mils for each of 3 coats. Use Type I, Aliphatic, for exterior use except for oily or resinous exterior wood surfaces. Use Type II, Aromatic, for interior use.

END OF SECTION

D10 CONVEYING

Conveying System(s) include passenger and freight elevators.

D1010 ELEVATORS AND LIFTS

Design assembly and arrangement of elevator, accessories, and supporting systems in accordance with American Society of Mechanical Engineers/American National Standards Institute (ASME/ANSI) A17.1 and UFC 3-490-06 Elevators. Provide all materials and equipment, including but not limited to elevator cab and hoist equipment, operating and signal fixtures, doors, door and car frames, car enclosure, controllers, motors, guide rails, brackets, and testing. If more than one story is constructed, building is required to include both a passenger and freight elevator.

D101001 GENERAL CONSTRUCTION ITEMS

Provide a machine room for every elevator. Locate the elevator machine and controller in the Elevator Machine Room.

D101002 PASSENGER ELEVATORS

If facility is more than one story, provide a minimum of one elevator sized to comply with the International Building Code (IBC) medical stretcher requirement and also designed to vertically transport the largest movable equipment or furniture used on the project. Provide a second, freight elevator sized to move large equipment, located for direct access between the receiving area and the kitchen.

Derive finishes and fixtures from Manufacturer's selections. Coordinate finishes with the interior architectural design and meet the User's needs and functions. Coordinate the design of the elevator machine room with applicable codes and the elevator manufacturer's requirements.

END OF SECTION

D20 PLUMBING

1. SYSTEM DESCRIPTION

The plumbing system for Patton Hall in JBM-HH consists of all fixtures, potable cold and hot water piping and equipment, piping insulation, water heating equipment, sanitary waste and vent piping systems, and other specialty piping and equipment within 5 feet of the building. Water and sewer utilities are owned and operated by JBM-HH Department of Public Works (DPW) and will be designed and constructed in accordance with JBM-HH design guides and the Department of the U.S. Army Unified Facilities Criteria (UFC) and UFGS requirements. Obtain natural gas pressures from the local gas utility provider, Washington Gas. Provide any applications and permits and provide the complete natural gas system from the load side of the utility meter to the heating equipment. Contract with the local gas utility provider for installation of piping and appurtenances up to the load side of the meter. Tie the gas meter into the Building Automation System (BAS).

2. GENERAL SYSTEM REQUIREMENTS

Provide working space around all equipment. Provide concrete housekeeping pads under all equipment. Provide all required fittings, connections and accessories required for a complete and usable system. Install all equipment in accordance with the criteria of section D20 and the manufacturer's recommendations. Design and install in accordance with the latest edition of the International Plumbing Code (IPC) and UFC 3-420-01, Plumbing Systems. Where the word "should" is used in the manufacturer's recommendations, substitute the word "must".

2.1. PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

This facility will be served by domestic cold water, sanitary sewer, electric, and natural gas utilities. Final equipment will be determined by a Life-Cycle Cost Analysis (LCCA). Domestic cold water is required to be provided to all restrooms, water coolers, janitor's closets, kitchen areas, wall hydrants, domestic hot water heaters, and HVAC make-up water systems. Sanitary sewer is required to be provided from all restrooms, floor drains, and kitchen areas, with a separate sanitary sewer line provided from the kitchen cooking area drains to a grease interceptor located on site.

2.2. GUARDHOUSE

The Guardhouse must be served by existing domestic cold water and sanitary sewer as required for building occupancy.

D2010 PLUMBING FIXTURES

Provide quantity and type of plumbing fixtures required for the occupancy, use, and functions described for this facility. Refer to Room Data Sheets for additional specific requirements for spaces with plumbing fixtures. Provide handicapped fixtures in accordance with the referenced criteria in the Project Program. Provide EPA "WaterSense" labeled fixtures where available.

D201001 WATER CLOSETS

Provide wall mounted flush valve type water closets with automatic control in all public restroom spaces.

D201002 URINALS

Provide flush valve type urinals with automatic control in all public restroom spaces.

D201003 LAVATORIES

Provide countertop lavatories with automatic control in each restroom space.

D201004 SINKS

Provide countertop kitchen sink with three compartments in the kitchen spaces. Provide waste disposer unit. Provide service sink in the kitchen spaces.

D201005 SHOWERS

Provide an acrylic shower stall with acrylic shower floor and shower supply fittings in the locker room spaces.

D201006 DRINKING FOUNTAINS AND COOLERS

Provide electric water coolers with integral bottle filler functions in the corridor spaces.

D2020 DOMESTIC WATER DISTRIBUTION

Perform an updated flow test to determine system requirements.

A flow test was performed with the following results:

Date:	24 July 2025
Time:	1215
Location:	JBM-HH
Static pressure:	72 psig
Residual pressure:	37 psig
Flow:	856 gpm

D202001 PIPES AND FITTINGS

Provide piping and fittings for above ground and buried piping.

D202002 VALVES & HYDRANTS

Provide isolation valves at water supplies to fixtures and to provide ease of maintenance as required in the IPC. Provide hose bibbs in mechanical rooms. Provide wall hydrants along the building exterior such that all points along the perimeter can be reached with a 100 foot (30 meter) long hose. Provide hose bibbs to service rooftop HVAC equipment.

D202003 DOMESTIC WATER EQUIPMENT

Provide backflow preventers of types and at points within domestic water systems as specified by IPC. Locate building backflow preventer inside the mechanical room on service entrance lines where not provided exterior to the building. Provide reduced pressure principal type backflow preventer at all make-up water lines inside the mechanical room and at all make-up water lines to systems containing chemical treatment.

Provide electric, heat pump, or natural gas fired water heater for heating domestic water, as defined by the results of an LCCA. Provide master thermostatic mixing valve. Provide 131°F water to kitchen dishwashers, separate from master mixing valve.

Provide domestic hot water recirculation system with high efficiency recirculation pump and recirculation loop with all associated fixtures, equipment, and appurtenances, Provide in-line circulator for domestic hot water distribution system.

Perform an LCCA to determine the efficacy of using a solar domestic hot water system to offset the building's domestic hot water energy use. Provide a complete solar domestic hot water system including heating panels, roof supports, piping, pumps, hot water storage tanks, heat exchangers and controls. Provide a system designed to furnish a minimum of 30 percent of the annual demand for domestic hot water.

If the solar domestic hot water system is located on the roof, provide a coordinated design of the roof elements in accordance with UFC 3-110-03 Roofing. Organize the roof space necessary to accomplish the functions the roof must provide, minimize roof penetrations, and plan the roof to facilitate future reroofing of the facility. Select the roof type and detail roof mounted equipment to complement the implementation of the functions that must take place on the roof and minimize the need for routine maintenance. Accomplish a Pre-Roof Design Conference prior to the design of the roof.

Design and build each solar domestic hot water heating system meeting the requirements of UFC 3-440-01 Facility-Scale Renewable Energy Systems. Each system must be fully integrated with the building DDC controls system.

Provide a complete solar domestic hot water system designed and built by a single contractor who specializes in solar heated water systems. System must be designed, built, and tested by this contractor who is responsible for the system provided to operate as proposed. This design-build contractor must be endorsed in writing, prior to system design, by the manufacturer of the solar plate collectors provided for this solicitation.

The solar domestic hot water system designer must work with the building designers to optimize the roof area, slope and orientation available for solar domestic hot water. Coordinate with other solar systems utilizing the roof area such as photovoltaic systems. Provide solar liquid flat plate collector systems to comply with the requirements specified in UFGS Section 22 33 30 Solar Water Heating System.

D202004 INSULATION & IDENTIFICATION

Provide insulation with vapor barrier on domestic hot water supply and recirculation piping according to the requirements of UFC 3-420-01. Provide insulation with vapor barrier on domestic cold-water piping to prevent condensation, according to the requirements of UFC 3-420-01.

D202005 SPECIALTIES

Provide washing machine connector box for clothes washers. Provide ice maker connector box for refrigerators.

D202090 OTHER DOMESTIC WATER SUPPLY

Provide piping supports in accordance with the IPC.

D2030 SANITARY WASTE

D203001 WASTE PIPE & FITTINGS

Provide pipe and fittings appropriate for below ground installation. Provide pipe and fittings for above ground installation.

D203002 VENT PIPE & FITTINGS

Above-ground vent pipe needs to conform to one of the standards listed in Table 702.1 of the IPC. Single drainage/vent stack systems (such as Philadelphia system) and mechanical air admittance valves will not be permitted.

D203003 FLOOR DRAINS

Provide floor drains where required and as identified in UFC 3-420-01. Drains are required to include a trap primer connection, trap primer, and connection piping. Primer must meet ASSE 1018. Alternatively, a barrier-type trap seal protection device may be used to protect the floor drain trap seal from evaporation. Barrier-type floor drain trap seal protection devices are required to conform to ASSE 1072. The devices are required to be installed in accordance with the manufacturer’s instructions.

Provide floor sinks in kitchens. Provide floor sinks where required for interior air handling unit condensate drains. Install condensate and drain piping to avoid interference with equipment access and prevent trip hazards. Provide floor sinks to receive condensate from air handling units.

D203004 SANITARY & VENT EQUIPMENT

Provide sump pump in the elevator shafts.

D2040 RAINWATER DRAINAGE

Storm water must not be drained into sewers intended for sewage only.

D204001 PIPE & FITTINGS

Provide pipe and fittings appropriate for below ground installation. Provide pipe and fittings for above ground installation. Size and install piping in accordance with the IPC.

D204002 ROOF DRAINS

Roof drains are required to be compatible with the roofing system and conform to ASME A112.6.4, with dome and integral flange, and need to have a device for making a watertight connection between roofing and flashing.

D204003 RAINWATER DRAINAGE EQUIPMENT

Provide roof drains and expansion joints in accordance with USGS 22 00 00 and the IPC. Where required by building design, provide expansion joint(s) of proper size to receive the conductor pipe. The expansion joint must consist of heavy cast-iron housing, brass or bronze sleeve.

D204004 INSULATION & IDENTIFICATION

Provide the same as domestic water piping.

D204090 OTHER RAINWATER DRAINAGE SYSTEM

Provide a secondary (emergency) roof drainage system that must be completely independent of the primary roof drainage system so that failure in the primary system does not influence the proper functioning of the secondary system. The secondary (emergency) roof drainage system discharge must occur in a location above grade that will be noticed by either the users or service personnel of the building. The discharge point becomes a visual indication that the primary system has failed or become blocked.

D2090 OTHER PLUMBING SYSTEMS

D209001 SPECIAL PIPING SYSTEMS

Not Used.

D209003 INTERCEPTORS

Interceptors and separators are required to be provided to prevent the discharge of oil, grease, sand and other substances harmful or hazardous to the public sewer, the private sewage system or the sewage treatment plant or processes.

END OF SECTION

D30 HVAC

1. SYSTEM DESCRIPTION

Provide a heating, ventilating and air conditioning (HVAC) system for the facilities that attain the following objectives: Occupant comfort, Indoor air quality, Acceptable noise levels, Energy efficiency, Reliable operation, and Ease of maintenance. Design and install in accordance with the International Mechanical Code (IMC), including the IMC supplemental requirements within UFC 3-410-01, and UFC 3-401-01, Mechanical Engineering. According to the requirements of UFC 1-200-01, an LCCA needs to be performed to analyze the most cost-efficient HVAC system over 40 years for each facility. The contractor is required to work with the installation to determine the most preferable systems to simulate in the LCCAs.

According to UFC 3-410-01, when the 1 percent occurrence humidity ratio for the outside air (OA) is greater than the inside cooling load design set-point (78°F, 57.9°F dew-point) humidity ratio, dehumidification of outdoor air is necessary and a Dedicated Outdoor Air System (DOAS) must be one of the solutions included by the designer to address the humidity. The DOAS must also be one of the solutions included when the ventilation air for the building is 1000 CFM or greater. The required DOAS solution must apply to new and renovation projects. The calculations must show the Energy Use Index (EUI) for the DOAS as well as the other solutions.

1.1. PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

For this facility, conditioned air needs to be distributed to the occupied spaces by DOAS units and/or air handling units located in mechanical rooms, as determined by the results of the LCCA. The air handling systems are required to have cooling and heating coils, exhaust heat recovery, and variable frequency drive supply fan(s). Terminal conditioning units, as determined by the LCCA, are required to be located downstream near the zones served. To allow for supply air temperature reset based on OA temperature and ensure that interior zones can provide maximum cooling at all times, terminal units that serve interior zones are required to be sized for 55°F supply air. If reheat coils are required, all reheat coils are sized for 55°F entering air temperature. If determined to be life-cycle efficient, space exhaust air, including restrooms, janitor's closet, and recycling areas, need to be delivered to the air handling unit(s) to pretreat incoming ventilation air prior to exhausting out of the building.

The kitchen areas are required to have a dedicated type I grease exhaust system. The grease exhaust system is required to include commercial exhaust hoods, ductwork, and appropriate exhaust fans. The exhaust system needs to be interlocked to operate when the cooking equipment is energized. Kitchen hoods are required to have integrated standalone fire suppression systems and cleaning systems. The kitchen areas are required to have dedicated type II humid air exhaust systems interlocked with commercial dishwasher operation. The air removed by the exhaust systems are required to be made up by a dedicated make-up air unit (MAU); the MAU needs to be interlocked with the Type I and Type II exhaust systems operation, and the MAU system is to be included in the LCCA. Type I and Type II exhaust systems will not be delivered to any heat recovery devices, but the exhaust fan energy use must be included in the results of the LCCA.

1.2. GUARDHOUSE

The existing ductwork distribution system in the guardhouse will remain; contractor is required to test and confirm these systems are in operation. If found deficient, contractor will repair the ductwork to satisfactory operation. Return air plenums must not be used.

The admin areas will be restored for future occupancy. Conditioned air is to be distributed to the occupied spaces. For this facility, conditioned air is to be distributed to the occupied spaces by DOAS units and/or air handling units located in mechanical rooms, as determined by the results of the LCCA. The air handling systems are required to have cooling and heating coils, exhaust heat recovery, and variable frequency drive supply fan(s). Terminal conditioning units, as determined by the LCCA, are required to be located downstream near the zones served. To allow for supply air temperature reset based on OA temperature and ensure that interior zones can provide maximum cooling at all times, terminal units that serve interior zones are required to be sized for 55°F supply air. If reheat coils are required, all reheat coils are sized for 55F entering air temperature. If determined to be life-cycle efficient, space exhaust air, including restrooms, janitor's closet, and recycling areas, are required to be delivered to the air handling unit(s) to pretreat incoming ventilation air prior to exhausting out of the building.

5. GENERAL SYSTEM REQUIREMENTS

Provide working space around all equipment. Provide all required fittings, connections and accessories required for a complete and usable system. Install all equipment in accordance with the criteria listed in D30 and the manufacturer's recommendations. Where the word "should" is used in manufacturer's instructions, substitute the word "must". Consider each HVAC system component relative to its contribution to the whole building energy performance and energy costs.

Provide air conditioning and heating for spaces as indicated and for the following Design conditions:

2. OUTSIDE CONDITIONS (WMO: 724050 Reagan Airport)

Summer Design Conditions		Winter Design Conditions	
1 percent Cooling DB/ WB	92.0°F DB/ 74.7°F WB	99 percent Heating Dry Bulb	20.7°F DB
1 percent Dehumidification DP/ MCDB	74.5°F DP/ 82.2°F MCDB for Evaporative Equipment	Prevailing Wind	From NW at 5 knots

6. INTERIOR DESIGN CONDITIONS

Area Type	Summer Temperature (°F)	Winter Temperature (°F)*	Summer Max. Humidity (percent RH)	
			Nom.	Tol.
Administrative and Office Spaces	78	68	50 percent	N/A
Communication Rooms	72	72	50 percent	± 5 percent
Ballrooms and Multi-purpose Rooms	78	68	50 percent	N/A
Kitchens	78	68	50 percent	N/A
Mechanical and Electrical Rooms, Guardhouse	+10°F over ambient	40	N/A	N/A
Toilets and Showers	78	68	N/A	N/A
Unoccupied Space (Night set-back)	85	55	N/A	N/A
Entry Vestibules	N/A	50	N/A	N/A

*Provide means of humidification where indoor relative humidity is expected to fall below 30 percent at design conditions.

Provide Ventilation rates and systems in accordance with ASHRAE Standard 62.1, Ventilation for Acceptable Indoor Air Quality. Outside air quantities will be sufficient to meet required ventilation rates and maintain a positive pressure relative to the outdoors except for toilets, lockers, janitor closets, mechanical/electrical rooms, storage rooms, and break and snack rooms which would be under negative pressure relative to adjacent areas. Ventilation air is required to be provided by a separate dedicated air handling unit. Outside air is not to be conditioned through terminal units. Provide each zone with its own limited range of control, as allowed by the control system central workstation.

Submitted calculations are required to include computer generated heating and cooling loads and building air balance to substantiate design guidelines were met and to size the necessary HVAC equipment. A nationally recognized heating and cooling load program, such as BLAST, DOE-2.1E, or other programs that perform 8760 hourly calculations are to be used. Supplemental heat gain from equipment, people, and lights, etc. are not to be used in calculating heating loads and sizing heating equipment. Design is required to include a safety factor of 25 percent for heating loads and 10 percent for cooling loads.

Heat gain information for special equipment needs to be obtained from the manufacturer or, where no information is available from ASHRAE Fundamentals. Equipment may include, but not limited to, vending machines, copiers, printers, computers/monitors, televisions, refrigerators, microwave ovens, and communication equipment. Refer to Appendix A for "Room Data Sheets".

Provide minimum 4-inch-thick concrete housekeeping pads and vibration isolators under all floor-mounted equipment with a minimum of 6-inch clearance.

Provide all mechanical equipment HVAC cooling/heating and condenser coils with a manufacturer approved coating system. The heat transfer rating must be as installed.

Where louvers have been added to cool mechanical/electrical rooms because of internal heating gain from equipment, a preheat coil needs to be provided to temper air being drawn into the space directly from the outside during the heating season.

High efficiency single-phase fractional-horsepower alternating-current motors, corresponding to the applications listed in NEMA MG 11. Select polyphase motors based on high efficiency characteristics relative to the applications listed in NEMA MG 10. Additionally, all polyphase squirrel-cage medium induction motors with continuous ratings must meet or exceed energy efficient ratings in accordance with Table 12-10 of NEMA MG 1. Provide controllers for 3-phase motors rated 1 hp (0.75 kW) and above with phase voltage monitors designed to protect motors from phase loss and over/under-voltage. Provide means to prevent automatic restart by a time adjustable restart relay. For packaged equipment, provide controllers including the required monitors and timed restart. Provide reduced voltage starters for all motors 25 hp and larger.

D3010 ENERGY SUPPLY

D301001 OIL SUPPLY SYSTEM

Not Used.

D301002 GAS SUPPLY SYSTEM

Obtain natural gas pressures from the local natural gas utility provider, Washington Gas. Provide any applications and permits and provide the complete natural gas system from the load side of the utility meter to the heating equipment. Contract with the local natural gas utility provider for installation of piping and appurtenances up to the load side of the meter. Provide gas meter on the building main gas line and tie meter into the Building Automation System (BAS).

D301003 STEAM SUPPLY SYSTEM (FROM CENTRAL PLANT)

Not Used.

D301004 HOT WATER SUPPLY SYSTEM (FROM CENTRAL PLANT)

Not Used.

D301005 SOLAR ENERGY SUPPLY SYSTEMS

Perform an LCCA to determine the efficacy of using a solar hot water heating system to offset the building's heating hot water energy use. Provide a complete solar hot water heating system including heating panels, roof supports, piping, pumps, hot water storage tanks, heat exchangers and controls. Provide a system designed to furnish a minimum of 30 percent of the annual demand for hot water.

If the solar hot water system is located on the roof, provide a coordinated design of the roof elements in accordance with UFC 3-110-03 Roofing. Organize the roof space necessary to accomplish the functions the roof must provide, minimize roof penetrations, and plan the roof to facilitate future reroofing of the facility. Select the roof type and detail roof mounted equipment to complement the implementation of the functions that must take place on the roof and minimize the need for routine maintenance. Accomplish a Pre-Roof Design Conference prior to the design of the roof.

Design and build each solar hot water heating system meeting the requirements of UFC 3-440-01 Facility-Scale Renewable Energy Systems. Each system must be fully integrated with the building DDC controls system.

Provide a complete solar hot water system designed and built by a single contractor who specializes in solar heated water systems. System must be designed, built, and tested by this contractor who is responsible for the system provided to operate as proposed. This design-build contractor must be endorsed in writing, prior to system design, by the manufacturer of the solar plate collectors provided for this solicitation.

The solar domestic hot water system designer must work with the building designers to optimize the roof area, slope and orientation available for solar domestic hot water. Coordinate with other solar systems utilizing the roof area such as photovoltaic systems.

Provide solar liquid flat plate collector systems to comply with the requirements specified in UFGS Section 22 33 30 Solar Water Heating System.

D3020 HEAT GENERATING SYSTEMS

Provide a heating system for this facility consisting of a minimum of 2 heat generation units (determined by results of LCCA), maximum N+1, each providing 60 percent of the load.

Do not account for the use of solar generation systems and heat recovery when determining load for equipment sizing. Water supply temperature is to be reset based on outside air temperature controlled by DDC control system.

D302003 FUEL-FIRED UNIT HEATERS

Not Used.

D302001 BOILERS

According to the results of the HVAC system LCCA, provide either packaged gas fired condensing type hot water boilers or electric type hot water boilers. Pulse type combustion boilers are not acceptable. Provide pre-manufactured, multi-wall boiler stack.

D302002 FURNACES

Not Used.

D302003 FUEL-FIRED UNIT HEATERS

Not Used.

D302004 AUXILIARY EQUIPMENT

Not Used.

D302005 EQUIPMENT THERMAL INSULATION

Provide insulation for hot water pumps and other associated heating equipment.

Insulate hot water pumps and equipment as suitable for the temperature and service in rigid block, semi-rigid board, or flexible unicellular insulation to fit as closely as possible to equipment.

D3030 COOLING GENERATING SYSTEMS

Primary pumps serving the building are required to utilize VFD's to allow for variable pumping. Where possible pumps will utilize ECM motors.

D303001 CHILLED WATER SYSTEMS

If determined by the results of the HVAC LCCA, provide a chilled water system for service to the building HVAC equipment. Chiller(s) must operate in temperatures down to 20°F. If it is determined to be life-cycle cost-efficient, provide heat recovery for domestic hot water. Provide insulation and vapor barrier on all chilled water equipment. Provide chiller controls with BACnet communication protocol. Provide complete start-up and operational testing of chiller equipment.

If required, provide factory assembled galvanized steel with stainless steel basin cooling tower(s) with automatic chemical treatment system(s) to serve the water-cooled chillers. Provide with basin heater(s). The load may be served by a single cooling tower. Provide two variable speed cooling towers, each serving up to 100 percent of the load.

D303002 DIRECT EXPANSION SYSTEMS

Provide a dedicated air-cooled direct expansion (DX) ductless split system cooling only unit for the NMCI/Telecom spaces. If coatings are indicated in Section D30, provide copper tube/copper fin construction or immersion applied, baked phenolic or other approved coating that passes the 3000-hour salt spray resistance test using the ASTM B117 procedure. Field applied coatings are not acceptable.

D3040 DISTRIBUTION SYSTEMS

D304001 AIR DISTRIBUTION, HEATING & COOLING

Provide insulated, galvanized steel ductwork constructed, braced, reinforced, installed, supported, and sealed in accordance with the IMC and Sheet Metal and Air Conditioning Contractors' National Association (SMACNA) standards. Duct liner is not allowed. Consider using double wall ductwork for areas the ductwork will be exposed to the conditioned space. Work must meet ASHRAE 189.1. If determined to be life-cycle cost-efficient, provide a Variable Air Volume (VAV) system using ducted returns and sound attenuators. Locate VAV units above ceilings and allow for maintenance and removal of units through lockable access panels.

D304002 STEAM DISTRIBUTION SYSTEMS

Not Used.

D304003 HOT WATER DISTRIBUTION SYSTEMS

Provide a variable speed pumping system to serve the HVAC hot water equipment throughout the facility. Above grade heating water piping is to be constructed of Type L copper or ASTM33, Schedule 40, black steel with threaded fittings or welded.

D304004 CHANGEOVER DISTRIBUTION SYSTEMS

Not Used.

D304005 GLYCOL DISTRIBUTION SYSTEMS

Depending on the results of the LCCA, confirm with installation DPW if glycol is required in the chilled water system. Glycol can be added to chilled water system as a freeze protection assurance but must be accounted for in chiller operation. Glycol is not added to cooling tower AC systems on the Condenser side. HVAC system(s) are separate from domestic/fire systems and must remain separate with backflow preventors to remove potential contamination of potable water system.

D304006 CHILLED WATER DISTRIBUTION SYSTEMS

For exterior buried chilled water distribution systems, coordinate with Section G30, Site Civil/Mechanical Utilities. Provide a variable speed pumping system to serve the HVAC chilled water equipment throughout the facility.

The chilled water piping system is required to be constructed of Type L copper or ASTM 53, Schedule 40, black steel with screwed, welded, or groove-end fittings. Steel or copper chilled water supply and return piping to serve the HVAC equipment throughout the facility. Insulate piping with cellular glass insulation.

D304007 EXHAUST SYSTEMS

Provide ductwork constructed, braced, reinforced, installed, supported, and sealed in accordance with the IMC and SMACNA standards.

Provide ducted exhaust ventilation systems and exhaust fans to serve all ventilated zones of the facility. Provide in-line rooftop or ceiling centrifugal exhaust fans as needed.

The kitchen areas are required to have a dedicated type I grease exhaust system. The grease exhaust system is required to include commercial exhaust hoods, ductwork, and appropriate exhaust fans. The exhaust system need to be interlocked to operate when the cooking equipment is energized. The hoods are required to have integrated standalone fire suppression systems and cleaning systems.

The kitchen dishwashing area is required to have dedicated type II humid air exhaust systems interlocked with commercial dishwasher operation.

D304008 AIR HANDLING UNITS

Provide central station variable volume air handlers. Provide ultraviolet disinfection system. AMCA 210 certified fans with AMCA seal. Fan bearings must have a minimum average life of 200,000 hours at design operating conditions. Provide bird screens for outdoor inlets and outlets.

D304090 OTHER DISTRIBUTION SYSTEMS

Provide circulating pumps with variable frequency drives.

D3050 TERMINAL & PACKAGE UNITS

D305002 UNIT HEATERS

Provide unit heaters to serve the heating requirements of the mechanical room, electrical room, and vestibule areas of the facility.

D305003 FAN COIL UNITS

If determined to be life-cycle cost efficient, provide 4-pipe fan coil units and controls to serve the heating and cooling requirements of the facility. Provide one fan coil unit for each zone and allow adequate space for maintenance. Provide each fan coil unit with a return filter grille to ease maintenance requirements. Provide horizontal fan coil units in the overhead with a means for removal and maintenance of the units through lockable access panels. Provide auxiliary drain pans below valves and appurtenances to prevent piping leaks and condensate forming on chilled water piping from damaging ceilings.

UL-Listed, factory assembled and tested fan coils, AHRI 440 and AHRI certified.

D305004 FIN TUBE RADIATORS AND CONVECTORS

Not Used.

D305005 ELECTRIC HEATING

Provide factory assembled, UL-1025, unit heaters.

D305006 PACKAGE UNITS

Provide 100 percent Outside Air Makeup Air Conditioning Units to precondition outside air prior to distributing to kitchen areas for hood exhaust makeup.

D3060 CONTROLS AND INSTRUMENTATION

D306001 HVAC CONTROLS

1. DIRECT DIGITAL CONTROLS (DDC)

Provide a complete Direct Digital Control (DDC) system to comply with UFGS 23 09 00 Instrumentation and Control for HVAC, UFGS 23 09 23.02 BACnet Direct Digital Control Systems for HVAC and Other Building Control Systems, UFGS 23 09 13 Instrumentation and Control Devices for HVAC and BACnet communication protocol for the facility.

Provide operator workstation and notebook computers and complete application software with all licenses.

Provide meters, monitored by DDC, on the incoming gas, water, and electric utilities of the building.

For electrical energy monitoring refer to Section D50, Electrical, paragraph entitled, Service Entrance Equipment. For potable water meter refer to Section G30, Site Civil/Mechanical Utilities, paragraph entitled, Potable Water Distribution. Set up trend reports to record data daily and store values in the operator workstation DDC computer. Set up trend reports to record data daily and store values in the ASHRAE Standard 135 building controller for later retrieval by either a notebook computer or an operator workstation DDC computer.

Provide meters, monitored by DDC, on the following subsystems for gas and water: HVAC and processes.

Provide meters, monitored by DDC, on the following subsystems for potable water: cooling tower makeup and blowdown, hot-water boilers.

Provide ASHRAE Standard 135 building controller as the main interface for the building control system.

Provide patch panel in the mechanical equipment room for ease of connection and disconnection of equipment.

Provide equipment. Provide panels with locks and alarms. The alarms must include both a flashing light and an audible alarm inside the mechanical room. The alarms must also be a networked alarm (e.g. switch connected to controller DI) with alarm events recorded remotely for a period not less than one year.

Provide flow rate meters, monitored by the DDC system, for hot water, condenser water, and chilled water flow as necessary.

Provide air handlers and all terminal units, including VAV boxes, with discharge/supply temperature sensors.

Provide central air handler unit outside air CFM air flow monitoring stations.

Provide a DDC option for automatic operation of building circulating pumps whenever outdoor air temperature is below 35°F or when there is a high potential for freeze damage.

Provide control to automatically start back-up pumps (or other HVAC equipment) if the primary device fails. Primary and back-up equipment starter circuits must be wired to prevent both pieces of equipment from operating at the same time. Rotate primary and back-up HVAC equipment monthly (adjustable) with a lead/lag control routine.

The Designer of Record must use UFGS Sections 23 09 00, 23 09 23.02 and 23 09 13 if using BACnet protocol and submit the edited specification sections as a part of the project design submittal.

Design requirements must be in accordance with all specification notes, and the BAS Owner must be identified and designated early in the design documentation.

The system must include standalone digital controllers, a communication network, and a workstation computer with control software. Provide stand-alone control routines that operate without connection to the network during a loss of communication. Provide trending, scheduling and alarm tables (may be included with the sequence of operation). Provide reset routines (based on outdoor air temperature or zone demand) for hot water loop temperature setpoints and supply air static pressure control. Use alarming and trending services during performance testing or commissioning. Alarm every sequence routine when out-of-limits or control/response failure occurs. Display all graphic floor plans, equipment graphics, and sequence of operations graphic pages.

All 120-volt wiring must comply with NFPA 70. All 24-volt wiring must comply with the IMC and terminal device manufacturer's recommendations.

Provide training on the installed system according to the maximum training days in UFGS 23 09 00, UFGS 23 09 23.02 and UFGS 23 09 13. Demonstrate all operator workstation functions requiring BACnet services, i.e., navigating through the graphic displays, trending, alarming and monitoring of the new controls system from the existing operator workstation using only the existing application software and without the need to launch other applications or logon to other vendor applications.

2. ELECTRONIC CONTROLS

Provide electronic controls for the HVAC systems and equipment.

If required, provide programmable thermostats with built in keypads for scheduling day and night temperatures with two setback periods for each day. Provide independent summer and winter programs. Thermostats must have temporary and manual override of schedule and battery backup.

D3070 SYSTEMS TESTING AND BALANCING

Provide complete Testing and Balancing (TAB) of all air and water distribution systems and HVAC equipment in accordance with UFGS 23 05 93, Testing, Adjusting, and Balancing for HVAC.

D307001 WATER SIDE TESTING & BALANCING - HEATING & COOLING

Refer to paragraph D3070.

D307002 AIR SIDE TESTING & BALANCING & HEATING, COOLING & EXHAUST

Refer to paragraph D3070.

D307003 HVAC COMMISSIONING

Refer to Section 01 10 00 Chapter 2 for Building Commissioning requirements. Mechanical systems to be commissioned, if provided, include HVAC systems and controls, refrigeration systems and controls, renewable energy systems, and domestic hot water systems. Conform to the commissioning requirements of UFGS Specification Section 23 08 00 Commissioning of Mechanical and Plumbing Systems.

D3090 OTHER HVAC SYSTEMS AND EQUIPMENT

D309001 GENERAL CONSTRUCTION ITEMS

Comply with the Force Protection Criteria. Provide access to all mechanical equipment rooms through the building's exterior walls. Provide mechanical equipment rooms (other than ground floors) with through the wall access doors on building exterior - crane access - with removable handrails. Provide mechanical equipment rooms in basements with pit access with floor drains and stairs and through the wall access doors on building exterior - crane access - with removable handrails.

D309090 OTHER SPECIAL MECHANICAL SYSTEMS

If determined to be life cycle cost efficient, provide total energy (enthalpy) type energy recovery wheels (heat wheels), plate heat exchangers, or heat pipe energy recovery devices in the air handling system.

END OF SECTION

D40 FIRE PROTECTION

1. SYSTEM DESCRIPTION

1.1. PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

Provide an analog addressable integrated fire alarm/mass notification system throughout the facility. Provide complete sprinkler protection throughout the building. The systems are required to be designed and installed in accordance with the latest versions of UFC 3-600-01, UFC 4-021-0, other referenced documents, and base standards. The systems also meet the current edition of all applicable NFPA codes and standards. The building construction is required to comply with UFC 1-200-01 and the IBC.

1.2. GUARDHOUSE

The existing Guardhouse building will remain; however, the existing fire alarm and sprinkler system are fed from the demolished portion of Building 214. The existing systems are to be abandoned in place as the Guardhouse will not be occupied during or after the construction of this project. The systems will be replaced during future renovations by others to occupy the structure.

2. GENERAL SYSTEM REQUIREMENTS

Provide working space around all equipment. Provide concrete pads under all equipment. Provide all required fittings, connections and accessories required for a complete and usable system. Install all equipment in accordance with the criteria of PTS section D40 and the manufacturer's recommendations. Where the word "should" is used in the manufacturer's recommendations, substitute the word "must". The designs are required to comply with the requirements of the applicable UFGS Specifications for FA/MNS and automatic sprinkler systems.

All Design Documents, (i.e. Building Code/Life Safety Analysis, plans, specifications, and calculations) developed for Section D40 must be prepared by, or under the supervision of the design/build contractor's Qualified Fire Protection Engineer (QFPE) as defined by UFC 3-600-01.

Installation drawings, shop drawings or working plans, calculations, other required pre-construction documentation and as-built drawings must be prepared by, or under the direct supervision of a National Institute for Certification in Engineering Technologies (NICET) engineering technician as specified below. NICET engineering technicians must hold a current certification as an engineering technician in the field of Fire Protection Engineering Technology with level certification in the appropriate subfield as listed below:

Fire Suppression	NICET Level IV
Fire Alarm & Mass Notification System	NICET Level IV

Provide training for the active systems consisting of four two (2) -hour sessions to accommodate all shifts of the base fire department and allow for rescheduling for unforeseen fire department responses.

3. SYSTEM DESIGN

Qualified Fire Protection Engineer (QFPE): The Designer is required to provide the services of a QFPE. The QFPE is to be an integral part of the design team and is required to be involved in all aspects of the design of the fire protection systems as required by UFC-300-01. At a minimum, the QFPE will design the automatic suppression systems and all associated systems (service lateral, storage if applicable, fire pump if applicable); the fire alarm and mass notification system; the fire alarm transmitter/receiver system, and any associated auxiliary equipment (e.g carbon monoxide and gas detectors). The QFPE is required to review all fire protection submittals, stamp all shop drawings, calculations, product data, etc. required for the fire alarm, mass notification systems and fire suppression systems.

3.1. LISTING

All materials, equipment, and appurtenances are required to be listed for use by UL (Underwriters' Laboratories) or need to be FM (Factory Mutual) approved.

3.2. AUTHORITY HAVING JURISDICTION (AHJ) AND DESIGNATED FIRE PROTECTION ENGINEER (DFPE)

Code references to authority having jurisdiction (AHJ) are required to be interpreted to mean the office of responsibility, U.S. Army HQ USACE/CECW-CE.

Code references to the designated fire protection engineer (DFPE) are required to be interpreted to mean the CENAD Fire Protection Regional Center of Expertise (RCX) based out of CENAB-EN-DM.

The contractor should coordinate all communication with the DFPE and/or AHJ through the COR.

4. REFERENCES

The design must comply with the references included in Section 4. If an aspect of the facility is not covered by a listed reference, the QFPE must apply the appropriate NFPA code or standard.

5. FIRE PROTECTION AND LIFE SAFETY ANALYSIS

5.1. BID DOCUMENTS

The QFPE is required to submit a design analysis along with the bid documents. The Design Analysis must address site design, life safety, and fire protection system design considerations.

5.2. DESIGN AND CONSTRUCTION SUBMITTALS

The QFPE is required to submit all documents as required by UFC 3-600-01.

5.3. FIRE PROTECTION DESIGN CONSIDERATIONS

Water flow test data is provided for bidding purposes. The QFPE is required to perform/witness additional flow testing for the system design in accordance with UFC 3-600-01 and NFPA 291.

5.4. LIFE SAFETY DESIGN CONSIDERATIONS

The building must meet life safety requirements of UFC 3-600-01 and NFPA 101. The building must also meet the building construction requirements of the IBC.

Provide Knox boxes at the front and rear entrances of the building.

D4010 FIRE ALARM AND MASS NOTIFICATION SYSTEM

The FA/MNS is required to be a voice evacuation fire alarm system with manual voice paging features to combine both fire alarm and mass notification system features in accordance with UFC 3-600-01, UFC 4-021-01, NFPA 72 and base requirements. The systems are required to be installed in accordance with all applicable codes and standards. It needs to utilize an addressable microprocessor-based type system with manual and automatic alarm initiation. Signal transmissions are required to be a multiplex format and be dedicated to fire alarm service only.

Provide a Monaco radio transmitter tied into the base system for monitoring of the FA/MNS.

The system needs to be provided with speakers and clear strobes and visual alarm indication via textural signs (according to ECB 2018-17).

Spacing and location of speakers, strobes and textural signs are required to be commensurate with the ABA Architectural Barriers Act, NFPA 72, and UFC 3-600-01. All appliances located outside or in areas subject to high humidity must be weather-proof.

Integrate HVAC manual shutdown into fire alarm LOC and shunt the units through the FA/MNS system. Provide a remote annunciator that is locked/password protected.

D4020 FIRE SUPPRESSION WATER SUPPLY AND EQUIPMENT

1. AUTOMATIC SPRINKLER SYSTEM

The building is required to be protected throughout by an approved automatic sprinkler system that complies with NFPA 13 and UFC 3-600-01. The sprinkler systems are required to be hydraulically designed. Design criteria is required to comply with UFC 3-600-01.

2. WATER SUPPLY

2.1. HYDRANT TEST

The QFPE must perform a detailed water analysis of the site. The Designer, in the immediate vicinity of the proposed site plan, is required to conduct multiple fire hydrant flow testing. The flow tests are required to be coordinated with the Joint Base Myer-Henderson Hall Fire Chief and the DPW. The Designer is required to utilize their flow test data and hydraulic analysis as the basis for the water supply design.

For bidding purposes, the Contractor can use the following preliminary data; however, additional testing must be performed or witnessed by the QFPE after award. The test reports are located in Appendix D.

	Test 1	Test 2
Date of Test	11 March 2025	24 July 2025
Location of Test	Buffalo Soldier Ave on hydrants east and west of the existing building	Johnson Ln on the east side of the green-field site.
Static Pressure (PSI)	72	72
Residual Pressure (PSI)	42	37
Measured Flow (GPM)	920	856
Flow (GPM) @ 20 PSI	1,238	1,060

2.2. BACKFLOW PREVENTION

Service laterals serving fire suppression systems must have a double-check style backflow preventer (BFP). The BFP is required to have an Outside Stem and Yoke (OS&Y) valve installed on both sides of each assembly for isolation purposes. Both indicating valves on the check valve needs to be monitored by tamper switches.

2.3. FIRE PUMP

The QFPE is required to determine whether a fire pump is required based on the building configuration and system demand. All aspects of the fire and jockey pump design and construction must comply with NFPA 20 and UFC 3-600-01.

2.4. FIRE EXTINGUISHER CABINETS

Fire extinguishers are required to be provided in all buildings. Fire extinguishers are required to be provided and installed in accordance with applicable codes and standards.

END OF SECTION

D50 ELECTRICAL

1. SYSTEM DESCRIPTION

Demolish the existing electrical system in Patton Hall. The existing electrical system in the Guardhouse is to remain and be reused. The existing feeder circuit to the Guardhouse panelboard from Patton Hall is to be demolished. Coordinate and obtain with Dominion Energy a new electrical service to the existing Guardhouse and Patton Hall – Strategic Engagement Platform.

Provide an interior electrical system consisting of service entrance wiring and equipment, distribution and lighting panelboards and switchboards, dry-type step-down transformers, conduits, feeder and branch circuits, motor control equipment, communications, security systems, alarm systems, emergency lighting and power, grounding, and lightning protection including accessories and devices as necessary and required for a complete and usable system. This section covers installations out to the building 5-foot (1.5 meter) line.

Select electrical characteristics of the power system to provide a safe, efficient and economical distribution of power based upon the size and types of electrical loads to be served. Distribute and utilize voltages of the highest level that is practical for the load to be served.

Provide a minimum of 15 percent empty space and 15 percent spare capacity at all levels of the power distribution system including any emergency power systems.

2. GENERAL SYSTEM REQUIREMENTS

Provide an Electrical System complete in place, tested and approved, as specified throughout this RFP, as needed for a complete, usable and proper installation. Install all equipment according to the criteria of Section D50 and the manufacturer's recommendations. Where the word "should" is used in the manufacturer’s recommendations, substitute the word "must".

This section of the RFP includes all electrical work on or within the building out to the 5-foot line. Electrical site work outside the 5-foot line is covered in section G40.

3. SUSTAINABILITY

Provide electrical systems and components that support project sustainability and energy goals and comply with UFC 1-200-02 High Performance and Sustainable Building Requirements.

4. ANTITERRORISM

Measures for force protection are required to be incorporated into the design of these facilities in accordance with UFC 4-010-02. Transformer locations are required to be selected to be in accordance with UFC 4-010-02 for equipment distances from structures. UFC 4-010-02, B.1.2.1 allows transformers to be placed within 33 feet of unobstructed space as long as the equipment provides no opportunity for concealment of explosive devices.

D5010 ELECTRICAL SERVICE AND DISTRIBUTION

D501001 MAIN TRANSFORMERS

Dominion Energy needs to provide the main pad-mounted transformer(s) including smart metering for the Patton Hall – Strategic Engagement Platform and Guardhouse.

D501002 SERVICE ENTRANCE EQUIPMENT

Provide underground services into the facilities.

Provide service entrance equipment in the Patton Hall – Strategic Engagement Platform. Provide the service entrance equipment with digital metering. Provide service entrance equipment in the existing

Guardhouse. Provide surge protective devices (SPD) at the service entrance equipment. Provide service entrance equipment in accordance with UFC 3-520-01.

Provide energy usage monitoring by using digital metering with current transformers on the incoming service entrance equipment. Monitor the total power usage at the service entrance. Monitored output must report to and be compatible with the Direct Digital Controls (DDC) system.

When a switchboard is required, the Designer of Record must utilize UFGS Section 26 24 13 for the project specification and submit the edited specification section as a part of the design submittal for the project.

When digital metering is required for connection to the Direct Digital Controls (DDC) system, the Designer of Record must utilize UFGS Section 26 24 13 to specify the digital metering requirements and submit the edited specification section as a part of the design submittal for the project.

D501003 INTERIOR DISTRIBUTION TRANSFORMERS

Provide dry-type transformers to step-down voltages for general purpose outlets and other low voltage equipment. Step-down transformers will be installed in each electrical room to provide power to the 208Y/120V systems. Transformers are required to be high efficiency copper wound type (meeting, as a minimum, DOE 2016 efficiency requirements) to minimize transformer load and non-load losses. Provide K-factor rated step-down transformers if standard transformers have to be derated due to nonlinear loads with high harmonic content.

D501004 PANELBOARDS

Provide distribution and branch circuit panelboards to serve loads as required. Panelboards must be of sufficient ampacity to serve the designed loads plus a minimum of 15 percent spare circuits capacity and 15 percent load capacity to provide for future growth. All panelboards will be circuit breaker type with copper bus bar. Provide panelboards that comply with UL 67 and UL 50. UL 869A applies if used as service entrance equipment. Panelboards for non-linear loads must be UL-listed, including heat rise tested, in accordance with UL 67, except with the neutral assembly installed and carrying 200 percent of the phase bus current during testing.

Provide molded case circuit breakers in accordance with UL 489. Ground fault circuit interrupting circuit breakers must comply with UL 943. Arc fault circuit breakers must comply with UL 489 and UL 1699. All circuit breakers are required to be fully rated (series-rated circuit breakers are prohibited). All circuit breakers rated 100 amperes and above are required to be 100 percent rated.

D501005 ENCLOSED CIRCUIT BREAKERS

Provide enclosed circuit breakers as required.

Provide molded case circuit breakers in accordance with UL 489. UL 869A applies if used as service entrance equipment. Provide solid neutral when grounded conductor is present.

D501006 MOTOR CONTROL CENTERS

Provide individual motor starters with disconnect switches, variable speed drives, and manual motor starters for motor controls as required by mechanical equipment. Provide all circuits and connections for motors. The Contractor must determine whether or not to combine required devices in motor control center(s).

D501090 OTHER SERVICE AND DISTRIBUTION

Provide transient voltage surge protective devices (SPD) at the following locations: service entrance equipment, telecommunications room panelboards, and fire alarm panelboards.

Provide switchboards that comply with UL 891. All circuit breakers are required to be fully rated (series-rated circuit breakers are prohibited). All circuit breakers rated 100 amperes and above are required to be 100 percent rated.

Provide automatically controlled receptacles according to the most recent publication of ASHRAE/IESNA 90.1, ASHRAE 189.1, federal energy management requirements, and UFC 3-520-01.

D5020 LIGHTING AND BRANCH WIRING

Provide all required branch circuits consisting of insulated conductors in conduit to supply the required power to all lighting, convenience outlets, equipment connections and or special systems associated with the project. Provide connections for all equipment, plumbing systems, air conditioning and ventilation systems, fire alarm systems, cable TV (CATV) systems, security systems, and all other systems provided in this project.

Provide electrical circuits and connections for all systems requiring electrical service.

General purpose receptacles will be provided in all areas as required according to NFPA 70 and UFC 3-520-01. General purpose receptacles are required to be NEMA 5-20R duplex, grounding type. GFCI protection is specified for all NEMA 5-20 receptacles within 6 ft of a sink, receptacles in mechanical rooms, electrical rooms, and for all receptacles mounted on the exterior of the facility, and as required by NFPA 70.

Provide safety disconnects for all mechanical and plumbing equipment. All disconnects are required to be Heavy Duty Rated.

Special purpose receptacles will all be grounding type rated as required for equipment served.

D502001 BRANCH WIRING

Provide branch wiring consisting of insulated conductors in conduit. Homeruns may have no more than three phase conductors. The sharing of neutral conductors between phases is not allowed on any circuit. All circuits have to be provided with a dedicated neutral conductor. All circuits have to be provided with an insulated equipment grounding conductor.

Raceways for the facilities will be:

- Electrical metallic tubing (EMT) for above ground for interior usage only.
- Rigid polyvinyl chloride (PVC) conduit for underground use.
- Rigid metal conduit (RMC), when transitioning from underground to above ground, and in other areas where conduit is subject to physical damage. Also, where specified as needed in a specific application such as hazardous conditions.
- Flexible metal raceways or metal armored cable systems will be used for final motor connections, light fixture locations, and as specified only. Rigid conduit will be specified otherwise.
- Minimum size conduit is designed to be 3/4-inch for electrical circuits and 1-inch for communications and underground circuits.

Conductors in the facility will be copper. Conductor capacity will be determined at an ambient temperature rating of 30 degrees C. with 60 degrees C. insulation for conductors #1 AWG and smaller and 75 degrees C. for conductors larger than #1 AWG. Minimum size conductors to be used will be #12 AWG unless noted otherwise.

Pre-manufactured armored cable, type AC or MC raceway systems will not be permitted for branch circuit distribution, except for the following conditions: lighting fixture whips not exceeding 6 feet in length; concealed in existing walls that are not disturbed during renovations in the Guardhouse building.

Provide wiring and connections for special outlets where required. All homerun circuits must contain no more than 3 phase conductors. Provide switches that comply with NEMA WD-1 and UL 20.

D502002 LIGHTING EQUIPMENT

1. INTERIOR LIGHTING

Provide a complete lighting system consisting of exit and emergency lighting and area lighting consisting of LED lighting including switches and automatic controls including occupancy sensors, vacancy sensors, daylighting controls, automatic lighting shutoff systems and dimming systems. Life safety illumination must be accomplished with area LED fixtures supplied with internal battery back-up. Lighting systems must comply with the most recent publication of ASHRAE/IESNA 90.1, ASHRAE 189.1, federal energy management requirements, and UFC 3-530-01.

Illumination Level: Goal intensities are required to be based on the IESNA lighting Handbook recommendations for the spaces identified, and UFC 3-530-01, Design: Interior, Exterior Lighting and Controls.

Interior lighting will utilize LED lighting fixtures throughout. Administrative office type areas will utilize a recessed LED troffer with frosted lens (or visual glare control) to reduce glare. Utility type and storage rooms will utilize industrial type LED fixtures. Provide under cabinet lighting at all cabinet locations. Minimum lighting fixture efficacy will be as indicated, but not less than 100 LPW (lumens for each watt).

All rooms will be provided with manual control of all lighting (either for direct or override of auto lighting controls). Wall-mounted switches will be located near the entrance to each area or room. Standard wall switches will be rated 277V, 20A.

Occupancy sensors will be specified to sense occupancy and function to turn OFF lighting upon vacancy after the pre-determined set time. The lighting will be routed through local manual ON/OFF switch(s) in large open areas. In small offices and areas, override will be incorporated into the wall mounted occupancy sensor. Occupancy sensors will be dual technology (ultrasonic infrared) type in large open areas. The occupancy sensors will operate in the Vacancy Mode (manual On, Auto Off) for the administrative type areas.

A network-type lighting control system will be provided for this facility. The large rooms, corridors and other multi-occupant spaces will be tied into the main lighting control system. Small office rooms, and other similar small spaces will be provided with stand-alone controls (i.e., occupancy/vacancy) sensors. The lighting control panel will also be connected to the DDC for scheduling override of the exterior lighting. This will allow remote access and easy coordinated schedule adjustments based on occupancy changes via DDC controls.

Illumination Levels are based on the recommendations of UFC 3-530-01 Design: Interior and Exterior Lighting and Controls, and the IESNA Lighting Handbook, 10th Edition.

AREA ILLUMINATION LEVEL (Average)	
Large Open Offices	30 foot-candles (Ft-C)
Conference Rooms	30 Ft-C
Private Offices	30 Ft-C
Electrical/Mechanical Rooms	30 Ft-C
Corridors	10 Ft-C
Communications Rooms	50 Ft-C
Restrooms	10 Ft-C

AREA ILLUMINATION LEVEL (Average)	
Stairways	10 Ft-C
Break Rooms	30 Ft-C
General Storage Rooms	10 Ft-C

For lighting fixtures installed on suspended ceilings, install in accordance with manufacturer's recommendations and the additional requirements for "Severe Seismic Disturbance" contained in ASTM E 580. Fixture support wires must conform with ASTM A 641/A 641M, galvanized regular coating, soft temper.

2. EXTERIOR LIGHTING

Exterior lighting design must comply with the recommendations of the Illumination Engineering Society of North America (IESNA), NFPA 101, and UFC 3-530-01. Point-to-point lighting calculations must be provided. Calculations will be performed to verify IESNA “Enhanced Security” foot-candles. Calculations for NFPA 101 light level requirements to public way (just outside the egress doors) will also be performed.

Exterior lighting will be LED fixtures. Fixtures are required to be anti-glare cutoff type and will be architecturally coordinated with the building theme. Drivers must have a minimum starting temperature of minus 20 deg C. Igniter case temperature must not exceed 90 deg C. Fixture color temperature will be 4000K maximum. The main entries will be designed for 5 FC average; the secondary entries will be designed for at least 3 FC average to meet UFC 3-530-01 guidance in interpreting IESNA goals.

The exterior building perimeter will be illuminated with LED fixtures (canopy and/or wall mounted type as applicable) to matching appearance, the existing fixtures in the area. These fixtures will meet NFPA 101 emergency lighting levels at the egress doors.

3. LED DRIVERS / POWER SUPPLIES

LED drivers (power supplies) and integral batteries must include a 5-year warranty.

D5030 COMMUNICATIONS AND SECURITY

1. PATTON HALL – STRATEGIC ENGAGEMENT PLATFORM

Provide a complete building entrance facility, telecommunications rooms, backbone distribution system, and horizontal distribution system including, but not necessarily limited to, all wiring, pathway systems, grounding, backboards, connector blocks, protectors for all copper service entrance pairs, patch panels, fiber optic distribution panels, terminators for all fiber optic cables, outlet boxes, telephone jacks, data jacks, cover plates and all testing for the various facilities/building included in the scope of work. All network requirements in the building will be unclassified.

2. GUARDHOUSE

Provide a new telecommunication room in the existing guardhouse. Demolish all telecommunications raceway and horizonal cabling inside the guardhouse. Provide new telecommunications horizonal cabling from existing and new outlets to the telecommunication room. Provide telecommunications raceways throughout the building. All network requirements in the building will be unclassified.

3. PATHWAYS, RACEWAYS, AND CABLES

Horizontal cables from patch panel to jacks will be unshielded Category 6A cables. The cabling, raceway, and outlets will be identical for telephone and data outlets.

4. GROUNDING REQUIREMENTS

Each Telecommunications Room will have a grounding bar for telecommunications equipment bonding.

5. OUTLETS AND TERMINATION

The facility's data and telecommunications cables will terminate on 19-inch floor mounted racks comprising 4-post type equipment cabinets. Telecommunications closets will require a three (3) rack configuration. Three racks are needed to address the additional cable termination requirements for this facility needed for compliance with UFC 3-580-01 requirements. Adequate clearance will be provided around the equipment racks as follows: 3 ft clearance in front and back, with 2 ft clearance at the ends of the rack configuration. The 6-inch side rack mounted wire support system is included in this space and design. The horizontal wire support system is required to be included in the design. The top of the 4-post type equipment racks need to be attached to the ladder cable tray for additional stability. Coordinate with Joint Base Myer-Henderson Hall NEC to determine if the 4-post equipment racks are to be lockable with roof fan units.

The facility's OSP telecommunications cables will terminate at a backboard mounted 110 type block(s), after the protected entrance terminal. The Contractor will provide another 110-type block adjacent to the incoming block(s) for cross connect cabling to be performed by the local Network Enterprises Center. The Contractor will provide CAT 6A cabling from the second 110 block to the Telecommunications rack and terminate on RJ45 patch panels. The internal outlet CAT 6A cabling will terminate at RJ45 patch panels in the equipment racks. Copper and fiber optic patch cables will be provided according to UFC 3-580-01.

The facility's fiber optic cables will terminate, on 'LC' connector patch panels placed at the top of the data rack. The internal outlet CAT 6A cabling will terminate at RJ45 patch panels in the data rack. Patch cords from the incoming patch panel(s) to the LAN switch and from the LAN switch to the facility's outlet patch panels for system activation at each data rack in the facility will be provided under this contract. Fiber optic patch panel(s) will be fully enclosed with a front and back cover, equipment rack mounted and have LC connectors. Ultra-high-density cassettes will be used for all fiber optic cable terminations. The factory polished pigtails used to terminate the fiber optic cable will be a minimum of three (3) meters long. All fiber optic terminations are required to be via fusion splicing in cassettes.

The facility's telecommunication CAT 6A cables will terminate on RJ45 110 type patch panels in the telecommunications rack.

VOIP wall telephone outlets will be single gang with mounting hook. Coordinate with the Joint Base Myer-Henderson NEC for the plate required and the exact dimensions of box that is required for the weatherproof phone outlets. Where a weatherproof enclosure is required, the one (1) inch EMT Conduit will enter the rear of the enclosure, and the contractor will ensure that the opening is watertight. Voice lines/outlets will be provided in all locations where money or alcohol is stored. Voice outlets will additionally be provided where required by UFC 3-580-01.

Duplex CAT 6A telecommunications outlets will be provided in office and administrative areas of the facility according to UFC 3-580-01, and Joint Base Myer-Henderson Hall NEC recommendation. Telecommunications outlets will be double gang faceplate with two blank inserts. Each outlet are required to be provided with dual CAT 6A jacks, unless otherwise indicated.

Provide wireless access point mounting and associated 4 category 6A cables for installation of equipment by the government. All wireless access cables are to be color coded blue, terminated on dedicated rack mounted patch panel(s). Design and installation is required to comply with the requirements of UFC 3-580-01.

Small Offices will be provided with two (2) Telecommunications outlets for each room, as a minimum. VTC outlets will be provided in the Conference Room and Briefing Room. Each VTC outlet will be provided with a 6-strand single mode fiber optic cable and duplex LC connector and one dual CAT 6A outlet. All the fiber optic cables for the VTC outlets are required to be properly terminated in a fiber optic patch panel (with splice tray), in the Telecommunications room equipment rack.

Telecommunications outlets will be double gang faceplate with two category 6A jacks and with two blank inserts, unless otherwise indicated. All jacks will be green. These outlets will be mounted in a 4-11/16-inch square junction box that is 2-1/4 inch deep.

Copiers, faxes, printers, etc. will be provided with data and telephone outlets. Outlet mounting will be coordinated with interiors layout. HVAC DDC system will be provided with a dedicated data outlet. Projectors used in Conference rooms and alike will be plug and play type with junction box outlet in ceiling and on wall for direct connection to a computer.

All labeling will be in accordance with TIA 606-B standards. Outlets and cables will be labeled in accordance with TIA- 606-B. An intelligent labeling scheme will be utilized for labeling of outlets and cables. The labeling scheme will include the telecom room, rack#, patch panel and position on patch panel. The sequence will go from left to right. All outlets will be grouped in the patch panels following a sequential manner in the room, from left to right.

6. TELECOMMUNICATIONS ROOMS/SPACES

This facility will be provided with a main Entrance Facility Room. Rooms will be located to ensure that the total horizontal distribution CAT 6A link does not exceed the 295 feet (90 meter) limitation of TIA 568-C.

Each Telecommunications room is to be sized to allow for three 19-inch floor-mounted racks comprising 4-post type equipment cabinets. Adequate clearance will be provided around the equipment racks as follows: three feet clearance in front and back, with 2 ft clearance at the ends of the rack configuration. The top of the 4-post type equipment racks are required to be attached to the ladder cable tray for additional stability.

Provide two walls of the telecom rooms with full height (8-foot 0-inch) fire retardant plywood backboards painted with fire retardant paint. The plywood will be required to be A-C rated and mounted in accordance with UFC 3-580-01.

Ladder cable trays in the telecom rooms have to be sized to allow for a ten-foot slack of cables to be routed/looped in this room. Cable trays will transition to 4-inch conduit sleeves for penetration through the telecom rooms. This will facilitate fire-rating and sealing of penetrations, and more easily allow for any future cables to be routed in the telecom rooms.

110 type patch panels will be provided for the termination of all the 4-pair cat 6A horizontal distribution cables. Spare capacity will be in accordance with UFC-580-01 requirements.

Fiber optic OSP cables will be terminated into a dedicated fiber optic patch panel, using duplex LC connectors. Fiber optic riser cables will be terminated into a dedicated fiber optic patch panel, utilizing duplex LC connectors. All fiber optic patch panels are to be pre-terminated type provided with 4-meter pigtails for fusion splicing of fiber optic cables.

D503004 TELEVISION SYSTEMS

Provide a complete CATV system including all interior equipment required to provide high quality TV signals to all outlets with a return path for interactive television. System will include, but is not necessarily limited to, amplifiers, splitters, combiners, line taps, cables, outlets, tilt compensators and all other parts, components, and equipment necessary to provide a complete and usable system.

The CATV system will comply with UFC 3-580-01. Individual type F connector outlets will be located as coordinated with the Interiors layout. The outlet cables will be routed back to the CATV enclosure through 1-inch conduit and the cable tray system. The cable will have a 24-inch coiled cable end for final terminations by the CATV vendor. The CATV system will utilize trunk and star topology. Series RG-11 will be utilized for trunk cables and Series RG-6 type cables will be utilized for horizontal coax cables. Coax cables will have solid conductors, foil and braided shields, flame retardant PVC jackets. All CATV outlets will be located adjacent to a power outlet. The contractor is responsible for all interior CATV cabling and devices. The local CATV service provider will provide CATV service to this facility. Exterior CATV service cabling will be routed in NEC controlled manholes and ductbanks. A more detailed plan will be provided to the point of connection designated by Installation NEC. A 4-inch conduit from the manhole will be provided to an enclosure in the telecommunications equipment room. The local CATV company coax cables will have solid conductors, foil and braided shields, flame retardant PVC jackets. All CATV outlets will be located adjacent to a power outlet. The contractor is responsible for all interior CATV cabling and devices. The local CATV service provider will provide CATV service to this facility. Time Warner will provide and install service cabling throughout the project site and terminating in the telecommunications equipment room. The contractor is responsible for all coordination with the CATV provider. Conduct CATV testing for all CATV outlets and cabling. Each CATV horizontal distribution link are to be tested using a TDR.

D503005 ELECTRONIC SECURITY SYSTEMS (ESS)

1. ESS INTRUSION DETECTION SYSTEM

The decision to install an IDS for storing cash and alcohol is based on a risk assessment conducted by the local commander and Physical Security Officer. This assessment is required by **AR 190-13**.

- The risk analysis evaluates the value of the assets, the risk of theft or diversion, and the facility's vulnerability.
- Based on this assessment, the commander determines the required security measures, which could include an IDS, in addition to other physical security controls like safes, locks, and lighting

A design for a complete ESS Intrusion Detection System (IDS) is required. Provide infrastructure and equipment for complete IDS construction, Contractor must be required to provide all required electrical branch circuits for the IDS system. Coordinate the specific requirements with Joint Base Myer-Henderson Hall NEC and Security. It is anticipated the base will require integration of the building IDS with the Integrated Commercial Intrusion Detection System VI (ICIDS-VI) program and equipment.

2. ESS VIDEO SURVEILLANCE SYSTEM

A design for a complete ESS video surveillance system (VSS) is required. Provide all infrastructure and equipment for the VSS. Camera head-end equipment will be located in the telecommunication room. Monitoring and control location will be coordinated with the User. Provide VSS camera coverage at all locations associated with cash (registers and offices) and alcohol (bars and storage). Design monitoring locations in accordance with AR 190-13 and AR 215-1 Morale, Welfare, and Recreation Policies and Programs.

3. ESS ACCESS CONTROL SYSTEM (ACS)

A design for a complete ESS Access Control System is required. Provide infrastructure for a complete ACS construction consisting of empty raceways with pull strings, outlet boxes, cover plates, and associated power outlets to enable system installation by the Contractor’s ESS supplier.

Provide an ACS including equipment and supporting infrastructure complete, tested, and operational. ACS must be compatible with the Installation's central monitoring system and monitored within the protected zone/area and at the Installation central monitoring station.

D5090 OTHER ELECTRICAL SERVICES

D509001 GENERAL CONSTRUCTION ITEMS (ELECTRICAL)

Provide General Construction Items (Electrical) including, but not necessarily limited to, all connections, fittings, boxes and associated equipment needed by this and other sections of this RFP as required for a complete and usable system.

Provide firestopping for conduits, cable trays and busways that penetrate fire-rated walls, fire-rated partitions, or fire-rated floors in accordance with Section C10, Interior Construction.

D509002 EMERGENCY LIGHTING AND POWER

Provide power and wiring for emergency lights and exit lights throughout the facility.

Use battery supported emergency drivers in conjunction with normal lighting fixtures for emergency lighting.

Provide emergency lighting units with self-testing and diagnostic control features.

Dominion Energy is to provide a natural gas emergency generator to power the entire building's electrical loads. The automatic transfer switch (ATS) associated with the generator will be provided by Dominion Energy and installed by the contractor.

D509003 GROUNDING SYSTEMS

Provide a complete grounding system for the facility electrical and telecommunications systems for each of the facilities. Grounding have to comply with the requirements of NFPA 70, NFPA 780, and TIA J-Std 607-B.

A separate green colored grounding conductor sized in accordance with NFPA 70 will be installed in all conduits. Ground rods will be copper-clad steel. Selected steel columns will be grounded using ground rods. An equipment ground will be run in all conduits and with all circuits except the service feeder from the building transformer.

D509004 LIGHTNING PROTECTION

Design and install a complete lightning protection system in accordance with UFC 3-575-01, Lightning and Static Electricity Protection System, with a UL Lightning Protection Inspection Certificate certified to NFPA 780, including, but not necessarily limited to, strike termination devices, conductors, ground terminals, interconnecting conductors, surge protective devices, and other connectors and fittings required for a complete and usable system.

Lightning Protection Systems must not void the roof warranty. When a lightning protection system is required, the designer of record must use UFGS section 26 41 00 and submit the edited specification section as part of the design submittal for the project.

D509005 ELECTRIC HEATING

Provide power wiring and connections as required for all electric heating systems and equipment.

D509006 ENERGY MANAGEMENT CONTROL SYSTEM

Provide power wiring and connections as required for all systems and equipment. Coordinate connection requirements with service entrance energy monitoring equipment.

END OF SECTION

E10 EQUIPMENT

E1010 COMMERCIAL EQUIPMENT

E101005 SECURITY AND VAULT EQUIPMENT

Not Used.

E1020 INSTITUTIONAL EQUIPMENT

E102009 AUDIOVISUAL EQUIPMENT

Coordinate design requirements with the end user/Command Information Technology (IT) personnel and provide Audiovisual (AV) equipment. Provide the services of an audiovisual equipment specialist to design and specify the audiovisual equipment.

AV Equipment including electronics potentially connected to data/IT, must be coordinated with design and construction but planned for and funded by the user or Budget Submitting Office sponsoring the user. AV equipment includes but is not limited to intercom/sound systems, smartboards, flat screens, projectors, video teleconferencing, interactive wall systems and Closed-Circuit-Televisions (CCTVs).

Coordinate design requirements with the end user/Command Information technology (IT) personnel and provide AV equipment. Provide the services of an audiovisual equipment specialist to design and specify the audiovisual equipment.

AV Equipment including electronics potentially connected to data/IT, must be coordinated with design and construction but planned for and funded by the user or Budget Submitting Office sponsoring the user. AV equipment includes but is not limited to intercom/sound systems, smartboards, flat screens, projectors, video teleconferencing, interactive wall systems and CCTVs. The AV equipment will be funded as part of the FF&E Package.

Provide the ceiling mounted hardware for a digital projector, to be coordinated with support blocking (see Section C10, Interior Construction) and electrical and data connections (see Section D50, Electrical).

E1040 GOVERNMENT FURNISHED EQUIPMENT

There is no Government Furnished equipment in this project.

E1090 OTHER EQUIPMENT

E109090 OTHER SPECIALIZED FIXED AND MOVABLE EQUIPMENT

END OF SECTION

E20 FURNISHINGS

1. SYSTEM DESCRIPTION

Fixed furnishings are part of Structural Interior Design (SID). Design and documentation of movable furnishings are part of the FF&E Package.

Design and documentation of both SID & FF&E will be funded as part of the construction contract. Purchase and installation of FF&E Package will be funded separately as part of Collateral Equipment.

Movable FF&E for this facility may include, but are not limited to movable furniture systems, freestanding furniture, area/accent rugs, artwork, appliances, monitors, flat screen TVs (not connected to data), accessories, shop equipment, specialty equipment (specified by the Activity) and other miscellaneous items to support facility functions.

Fixed furnishings (items that are fixed to the structure), such as specialty equipment, drying cages, weapon racks, lockers, motorized projection screens, blinds/shades are part of the construction contract.

The AV Equipment will be funded as part of the FF&E Package. The A/V Equipment Package will be identified as a separate line item and priced separately from the FF&E. Refer to Section E10 of this RFP.

2. GENERAL SYSTEMS REQUIREMENTS

Design and provide fixed and movable furnishings for all areas as developed during Activity programming and as indicated in Room Data Sheets (refer to Appendix A of this RFP). Design a complete FF&E package and prepare supporting plans and procurement data. FF&E items identified in this RFP are to be used as a guideline to assist in establishing the minimum facility requirements and do not relieve the Contractor's Interior Designer from developing a complete design package that incorporates ALL of the Activities FF&E requirements. Design in accordance with specific facility-type Unified Facilities Criteria (UFC) and in conjunction with UFC 03-120-10, Interior Design.

Provide services of an Interior Designer with a minimum of one of the following credentials: National Council for Interior Design Qualification (NCIDQ) certification, or state and/or jurisdiction Interior Design Certification, Registration, or License. This Interior Designer must prepare both the Structural Interior Design (SID) and the FF&E Package and participate in all design charrettes and review meetings to develop the building design and floor plan. When AV, shop or specialty equipment is required in the project, obtain services of equipment specialists to specify the AV, shop, or specialty equipment. Contractor's Interior Designer and any Specialists must not have any affiliation with products specified. Closely coordinate all fixed and movable furnishings selections and fully integrate the FF&E package with building systems and finishes.

Contractor's Interior Designer will be responsible for designing and providing specifications for procurement of all FF&E, to include delivery and installation, for facilities built under this contract. Base the FF&E specifications on General Services Administration (GSA) schedules, and other Federal contracts and complying with priorities found in Federal Acquisitions Regulation (FAR) Part 8.404.

Fully integrate the FF&E package into the design and construction schedule for the building.

Consider furnishings relative Sustainability Reporting for Design Build and UFC 1-200-02, High Performance and Sustainable Building Requirements.

3. INTERIOR DESIGN SUBMITTAL AND MEETING REQUIREMENTS

3.1. STRUCTURAL INTERIOR DESIGN (SID)

The SID submittal process must begin following the RFP award. A SID submittal includes Interior Design programming documents, FF&E Floor Plans, and exterior and interior finish/color and material sample boards.

Provide the following SID meetings and submittals:

- Concept design workshop (CDW) or initial design meeting: contractor's interior designer must meet with the activity to develop the interior design SID programming documents, which will include a preliminary furniture and equipment plan and FF&E summary list. Contractor's interior designer will provide in-depth room by room interviews to confirm activity requirements for the new facility(s). These interviews must occur at the activity's current location, if possible, to include building walk-thru(s). Submit minutes of this meeting to contracting officer within seven days.
- Design development (60 to 65 percent) submittal: contractor's interior designer must provide a conceptual finish schedule, proposing finish materials to be used in all spaces. Furniture and equipment plans, and FF&E summary list should be further developed and included in this submittal.
- SID "over the shoulder" review: prior to prefinal (100 percent) submittal, contractor's interior designer must meet with the installation and user group for an "over-the-shoulder" review meeting to present a minimum of (2) options for building finishes/colors/materials. Finishes must display manufacturer's label/specifications and be presented in a "loose" format for preliminary approval prior to the presentation with the activity.
- Prefinal (100 percent) submittal: contractor's interior designer must present a minimum of two (2) sets of building finishes/color/material options, to the activity for approval. Each of these two finish options must be documented and submitted in a 16-inch by 20-inch presentation board format to contracting officer. In addition, the submittal must include an updated furniture and equipment plan, finish schedule, floor finish plan, and finish legend for review and approval.
- Final submittal: contractor's interior designer must incorporate the final approved furniture and equipment plan, finish schedule, floor finish plan and finish legend into contractor's final drawing set. These drawings and all approved finishes must be submitted in 8-1/2 inch by 11-inch binder format, using heavy duty plastic sheet protectors. Submit three (3) sets of final SID submittal.

3.2. STRUCTURAL INTERIOR DESIGN (SID) CONSTRUCTION SUBMITTALS

Once SID finishes have been approved, substitutions will not permitted. In event that revisions are required due to unforeseen conditions such as discontinued products, such revisions must be submitted for approval.

3.3. FIXTURES, FURNISHINGS AND EQUIPMENT (FF&E)

Development of FF&E process must begin with the 60 percent design submittal. Submittals include fixtures, furnishings, and equipment specifications in accordance with functional activity requirements to produce an optimally functional facility. FF&E are all items that are not fixed to the structure but are fully integrated with the building systems and finishes.

Develop design as described and in accordance with functional activity requirements. Include in design all loose furnishings required to produce an optimum functional facility, consistent with quality commercial design. Project also includes the preparation of specific detailed information for each selected item. Each submittal must demonstrate interaction thoroughly with requirements and complete coordination with the facility design and the SID.

For all projects, including fast track projects, the Contractor is responsible for sufficiently scheduling all SID/FF&E submittals early enough to obtain required Government approvals, and meeting all ordering and installation lead times to complete project by contract completion date.

These are minimum requirements, and Contractor will provide any/all additional meetings and submittals that necessary to support Interior Design effort with FF&E and coordination.

Provide the following FF&E meetings and submittals:

- FF&E requirements meeting: this meeting must occur upon completion of design development (or approximately 60 percent), prior to development of FF&E package. Submit minutes of this meeting within 7 business days.
- FF&E "over the shoulder" review: prior to preliminary FF&E submittal contractor's interior designer must meet with functional activity for an "over-the-shoulder" review to present preliminary FF&E options. These can be presented in a "loose" format for preliminary approval prior to the activity presentation.
- FF&E concept presentation: contractor's interior designer must present approved preliminary FF&E package to the activity for approval. This presentation includes loose format samples and catalog cuts. Sample boards are not required.
- Best value analysis (BVA), "over the shoulder review": prior to issuing the BVA, contractor must meet with functional activity for an "over-the-shoulder" review of the solicitation package. BVA solicitation to include the following:
  - Copy of the BVA cover letter
  - BVA solicitation form & questionnaire
  - Technical specification to establish minimum acceptable FF&E requirements
  - Room and typical
  - Furniture plans with legends (pdf format)
- Preliminary BVA submittal - contractor's interior designer must submit three (3) copies of the preliminary BVA package. Provide this submittal in binder format and include the following:
  - Cover title page (project name, project #, submittal date, submittal title)
  - Table of contents
  - Point of contact list
  - Narrative of interior designer objectives
  - Copy of all information sent to bidders and documentation that all required sources were contacted.
  - BVA spreadsheet
  - Solicitation forms submitted by each bidder (cut sheets/highlighted pricing sheets/technical specifications, pricing, dealer and manufacturer qualification for each product showing that products meets all requirements.)
  - Response from Federal Prison Industries, Inc (UNICOR)
- Preliminary FF&E submittal: preliminary FF&E submittal is due at pre-final (100 percent) and must be presented to the activity and contracting officer. Submit the following in a 3-ring binder (with the exception of the 16x20 color boards) for review and approval:
  - Cover title page (project name, project #, submittal date, submittal title)
  - FF&E list (cost summary)
  - Furniture placement plans coded to the FF&E list and furnishings specifications.
  - Specifications for furniture, furnishings, etc.

    - Catalog cuts and finish samples for all specified items.
    - 16x20 inch color boards of furniture/furnishings and finishes specified for activity presentation to indicate overall design intent.

- Final FF&E submittal: submit final FF&E package within 60 calendar days following receipt of review comments on the preliminary FF&E submittal. Final submittal must incorporate the following:
    - Cover title page (project name, project #, submittal date, submittal title).
    - Table of contents.
    - Manufacturer contact list.
    - Procurement data sheets for each product indicating final finish and fabric selections.
    - Copy of final quote on letterhead from the vendor determined to be the best value.
    - Best value determination guideline sheets (completed and signed by the contactor's interior designer)
    - CD copy of the final FF&E binder.
    - Final finish selections and memo samples for FF&E submitted in 8x10 binder format, using heavy-duty plastic sheet protectors.

3.4. FF&E CONSTRUCTION SUBMITTALS

Submit any revisions or deviations caused by discontinued items to the Contracting Officer for approval by the Interior Designer. List Operation and Maintenance Manuals for seating, systems furniture and keyboard trays.

E2010 FIXED FURNISHINGS (SID)

Fixed furnishings (SID) are funded as part of the construction project and are not funded as part of FF&E. Each submittal must demonstrate complete coordination with the facility design and with the package for movable furnishings.

Develop design as described herein and provide storage shelving, equipment racks, and window treatments.

E201001 FIXED ARTWORK

Not Used.

E201002 WINDOW TREATMENTS

Provide windows and other glazed openings to exterior of the building with manually or electrically operated double-roller sunscreen and room darkening shades, which are considered SID and are funded as part of the construction project. Refer to Room Data Sheets for scheduled locations of window treatment.

Provide interior window coverings, associated hardware, and controls at scheduled window and at any interior view window where privacy may be required. Refer to the Project Program for size, pattern and style of window treatments. At a minimum, functional window coverings such as blinds or solar shades are required on all projects.

E201003 FIXED SEATING AND TABLES

As required, but not limited to, provide fixed locker room benches, fixed tables and chairs, and site furniture. Provide locker room benches.

E201004 INTERIOR LANDSCAPE (FIXED)

Not Used.

E201007 Interior Signage

Provide all necessary interior way finding signage including Room Signs, Safety, and directional signs. Interior signage is not collateral equipment. Interior signage are required to demonstrate complete coordination with the facility design, SID and FF&E submittals.

E2020 MOVABLE FURNISHINGS

Design of FF&E package is funded as part of the construction contract base bid. Procurement, purchase and installation coordination of FF&E is a planned modification to contract and funded separately as part of Collateral Equipment.

Design and provide an FF&E package in accordance with UFC 3-120-10, Interior Design, and other portions of this RFP for all areas as developed during Activity programming to provide a fully usable and complete facility.

FF&E Package must include shipping, freight, handling, and professional installation, project management, handling, and applicable sales tax. Perform a Best Value Determination on a minimum of three manufacturers for orders exceeding a total procurement of $3000 from an individual manufacturer. Provide documentation to the Government with the final FF&E package.

Contractor, as a planned modification, will be authorized by the Government Contracting Officer to procure all furniture/furnishings in the approved final FF&E package using predominately negotiated Federal contracts as directed by Contracting Officer. When the modification for turnkey furniture procurement is exercised, the Contractor's proposed Handling and Administrative Rate (HAR) must not exceed 5 percent of the total cost of the FF&E, shipping, freight, handling, and installation. The HAR includes all of Prime Contractor's effort related to storage, coordination, handling and administration of subcontractors, all other associated costs and profit for procurement of FF&E. No other charges, fees, or markups will be authorized. Establish and submit a fixed percentage figure for administration effort of this modification (HAR), with the initial project proposal as part of the Contractor's Pricing Schedule.

FF&E must include furniture, shop equipment, audiovisual equipment, and specialty equipment. Weapon racks, drying cages, and lockers are not considered FF&E. FF&E must be fully integrated with the building systems and finishes. FF&E may also include specialty items for which the customer activity must be responsible for specifying.

Design and provide as required FF&E for all areas as developed during client programming. Design an FF&E package and prepare supporting plans and procurement data in accordance with the general interior design requirements in UFC 3-120-10.

1. FF&E PACKAGE

FF&E Package: Design and provide a fully usable and complete facility to include an FF&E movable furnishings package from Government supply sources according to Federal Acquisition Regulations. The FF&E will include, but not limited to, systems and modular furniture, training and conference furniture, seating, tables, artwork, decorative window covering, specialty furniture and equipment, dormitory room furnishings, and accessories. The government will provide separate funding for the FF&E package. Construction funds will not be used. The FF&E Package must include shipping, freight, handling, installation and the Handling and Administration Rate (HAR) percentage as applied to the final FF&E total cost.

1.1. AUTHORIZATION

Government will provide separate funding for procurement of FF&E package. Upon receipt of required funding, Contractor must be authorized by Contracting Officer as a planned line-item modification to the contract/task order to procure all FF&E using predominately negotiated Federal contracts. Amount of the modification will be the actual cost of these items from the Federal price schedules, including any freight and installation charges from furniture suppliers as well as Contractor's FF&E Handling and Administration Rate (HAR). HAR includes all prime Contractor's effort related to storage, coordination, handling, administration of subcontractors, and all other associated costs and profit for the procurement of FF&E. Prime Contractor will propose in the contract/task order solicitation the FF&E HAR. Contractor's proposed HAR may not exceed 5 percent of the total FF&E costs, as noted on the bid schedule. No other charges, expenses, fees, or markups will be authorized.

Government will approve the final FF&E submittal. The FF&E package will be presented to Contracting Officer and Contractor will provide the FF&E exactly as specified and approved.

2. PURCHASE AND INSTALLATION

Coordinate building completion date with installation dealer of the FF&E Package. Contractor is responsible for the following: issuing purchase orders, receiving acknowledgements, sending copies of purchase orders to the installation dealer(s) specified in the FF&E package, and providing necessary deposits to furniture manufacturers.

The FF&E installation dealer(s) is responsible for the following: Receiving and installing all FF&E specified in the FF&E package, coordinating delivery and installation with the Contractor, inspecting for damage, providing delivery receipts to the Contractor, filing necessary freight claims, hanging artwork, bulletin boards, etc., removing packaging material, cleaning up the site upon completion, and adhering to Contractor's safety requirements.

2.1. USE OF GSA SCHEDULES AND BLANKET PURCHASE AGREEMENTS (BPAS)

Prime Contractor or FF&E dealer will be authorized to purchase supplies or services from GSA Schedules.

2.2. DEPOSITS

Contractor must anticipate providing a deposit of between 30 percent to 50 percent of furniture costs when placing their order.

Manufacturer price increases must be anticipated. Recommend ordering FF&E product once funds are received to avoid incurring additional costs. Delayed production and delivery dates can be noted at the time of order placement to coincide with building completion dates. Any costs incurred due to manufacturer price increases will be the burden of the Contractor.

2.3. DAVIS BACON WAGES

Davis Bacon wages do not apply to the FF&E installer from the government supply sources. The workforce for the FF&E installation and delivery must be separate and distinct from the labor workforce performing under the construction contract.

2.4. SALES TAX

Exemptions for certain State or Local taxes may be available to the Contractor and/or its subcontractors. The Contractor must take maximum advantage of all exemptions, including obtaining a resale permit, from State and Local taxation authorities whether available to it directly or available to the Contractor based on an exemption afforded the government. The responsibility for paying applicable taxes rests with the contractor. State and local taxes applicable to the FF&E line will be included with the subcontractor's quote, if applicable.

2.5. BONDS

FF&E line item is not considered construction, and the Prime Contractor will not be required to secure any additional bond for the award of the FF&E line item unless otherwise indicated in the RFP. If any additional bond is required for the FF&E line item, it is to be included in the prime Contactor's FF&E HAR.

2.6. UNIQUE ITEM IDENTIFICATION (IUID) AND VALIDATION

Unique item identification and valuation is a system of marking and valuing items delivered to DoD that enhances logistics, contracting, and financial business transactions. The IUID policy is mandatory for all DoD contracts that require the delivery of items. An item is a single article or a single unit formed by a grouping of subassemblies, components or constituent parts. Provide DoD Unique item identification, valuation and delivery of data for all required FF&E items for which the government's unit acquisition cost is $5,000 or more.

2.7. BUY AMERICAN ACT AND TRADE AGREEMENT ACT

All supplies under the FF&E line item are subject to the Buy American Act and Trade Agreement Act (TAA). The GSA contracts and Blanket Purchase Agreements are required to comply with the Buy American Act and TAA.

2.8. SMALL BUSINESS REQUIREMENTS

The FF&E is subject to the Contractor's Small Business Goals; however, the government requires the furniture to be purchased from Blanket Purchase Agreements (BPA). Most manufacturers on the Office Furniture BPA are large businesses and most manufacturers on the Dorm and Quarters BPA are small business. Installation dealers are small business. Under the terms of the BPA, the FF&E must be ordered directly through the GSA manufacturer. Using pass-through companies to achieve Small Business Goals will not provide the Contractor credit unless they manufacturer 20 percent or provide 50 percent of the service purchased. The government will not incur additional costs to use small business.

2.9. INSTALLATION

The FF&E package includes the installation of all furniture and furnishings as specified in the FF&E package. The installation dealer specified in the FF&E package will receive, store, and if required, transport to the project site, off load, inside deliver, unpack, assemble, place/install, clean, if required, and dispose of all the trash for all furniture and furnishings. The Contractor's Interior Designer will be responsible for specifying installation services and warehousing, as required, for all collateral equipment. It is the Contractor's responsibility to coordinate the building completion, occupancy, and furniture installation dates with the installation dealer specified in the FF&E package. Any costs associated with storing or delaying furniture shipments is the responsibility of the Contractor.

2.10. INSTALLATION WARRANTY

Install all movable furnishings in accordance with the manufacturer's instructions and warranty requirements. All movable furnishings must be level and aligned, and all doors, drawers and accessories must be level and aligned to open, close and otherwise operate smoothly and securely. All systems furniture must be installed by the systems furniture manufacturer's dealer of record and not the General Contractor. Repair, to the customer's satisfaction, any/all damage to any facility finish that is a result of the furniture installation and correct all punch list items for the furniture/furnishings.

2.11. ORDERING DOCUMENTATION

Provide two copies of all ordering documentation to the Contracting Officer including Factory Order number (FO) and warranty information.

2.12. POST AWARD CHANGES

After award of the FF&E line-item modification, any request to change the FF&E items must be submitted on the Contracting Officer. The FF&E modification has been accepted, priced, and negotiated based on specific line items as detailed in the final package. Those items have been agreed to considering color, specific type and quality of material, price, sustainability, life cycle, and dealership service. The Government will expect and require the Contractor to provide exactly those items. Should changes become necessary, careful consideration is required to ensure that equivalent quality, price and other aspects of the item are maintained. Otherwise, price adjustments must be negotiated. The Contracting Officer will obtain approval from the Government Interior Designer/Collateral Equipment Manager in consultation with the client for any changes to the FF&E.

Post award FF&E manufacturer's price increases are the responsibility of the Contractor and will not be transferred to the government. Recommend ordering FF&E product once funds are received to avoid incurring additional costs. Delayed production and delivery dates can be noted at the time of order placement to coincide with building completion dates.

3. BEST VALUE DETERMINATION

A best value determination is required by Federal Acquisition Regulation (FAR) 8.404 when placing orders against Federal Supply Schedules for the selection of furniture and furnishings. A Best Value Determination (BVD) must also be provided for FF&E installation services. Best Value is defined in FAR 2.101 as ensuring that the order to be placed under a Federal Supply Schedule results in the lowest overall cost alternative (considering price, special features, administrative costs and client's needs) to meet the government's needs.

The Contractor's Interior Designer is responsible for the following written BVD justifications:

$3,000 or less: For any procurement in the FF&E package with a value of $3,000 or less, the Interior Designer may utilize any BPA holder. If the BPA holders cannot supply the item, then any other manufacturer may be utilized.

Greater than $3,000 and $150,000 or less: for any procurement in the FF&E package with a value greater than $3,000 and $150,000 or less, the Contractor's Interior Designer must always review pricing from at least three manufacturers as well as UNICOR. In addition to the review of published list prices, the Contractor's Interior Designer must confirm the pricing with the vendor. Manufacturer's quotes are NOT required. The BVD form must be completed and submitted for all FF&E procurements greater than $3,000 and $150,000 or less.

Greater than $150,000: The Contractor's Interior Designer must solicit proposals from all BPA holders under the applicable group for FF&E procurements greater than $150,000. UNICOR must always be solicited. The Contractor's Interior Designer must develop performance criteria and project requirements based on a generic design for the BPA holders and UNICOR to develop a price and performance proposal. The BVD form must be completed and submitted for all FF&E procurements greater than $150,000 and manufacturer's quotes and a summary of all proposals must be attached.

UNICOR must be considered as part of all BVDs. This must be done by sending an email with the requirements and evaluation criteria if they are not comparable in one or more areas of price, quality, and time of delivery, the designer can specify product under BPA or GSA schedule.

The best value determination must address issues such as space planning; human factors data related to anthropometrics (reach, clearance, adjustability), space, and acoustics; ergonomics; product quality (including construction and materials); sustainability features, product warranties; history of the product and/or manufacturer; ability to service products through dealers or others within a certain geographical range of the project; price (including freight); aesthetics; appropriateness; and lighting, power and telecommunications systems management and/or coordination as related to the facility (when applicable);

and other project specific factors as identified and/or required. Emphasis must be to create a fully integrated design solution by providing quality products to meet the functional needs of the customer. Customer preferences must be considered. The focus must be on the best overall value. Use the GSA Best Value Determination forms provided in Part 6 of this RFP as guidelines for information to be provided.

E202001 MOVABLE ARTWORK

Provide artwork for wall installation as part of the FF&E Submittal according to the project program. Installation of artwork to be completed by installation dealer specified in the approved FF&E package. Type of artwork to be determined by client requirements and budget as described in the Project Program for the project. Install framed artwork 63 inches (1600 mm) on center above finished floor. Include security mounting hardware as required.

E202002 MODULAR PREFABRICATED FURNITURE

Provide Workstation systems product or modular freestanding workstations as required. Do not use modular or systems furniture in shop spaces. Provide at a minimum, an articulating keyboard tray with left- or right-handed mouse extension for each computer location.

1. FURNITURE SYSTEMS

Provide products that meet the performance specifications for systems furniture. The Government Interior Designer must approve any other systems furniture manufacturer. The typical workstation must maximize each allocated space with worksurfaces and overhead closed storage with a surface to accommodate a Government provided computer. An attached articulated keyboard/mouse tray must be selected or provided. Provide a monitor lift if required by the project program.

Powered raceways that will accommodate data and voice wire management must be completely coordinated with all facility systems. The Contractor's Interior Designer must ensure the coordination of all electrical/data and furniture locations. Use of power poles will not be permitted to power FF&E. Provide and coordinate all telecommunication receptacles and outlet requirements (i.e. RJ 11/45 receptacles and cover plates) with the Contractor's Interior Designer and the systems furniture installer. Hard wire all pre-wired furniture with the building systems and coordinate all Information technology (IT) and telephone connections.

2. MODULAR FREESTANDING FURNITURE AND WORKSTATIONS

Provide products that meet the performance specifications for modular freestanding furniture including wood. Provide modular furniture with electrical/data cable trays and grommet holes for private offices and smaller work areas. An attached articulated keyboard/mouse tray (and monitor lift if required by the project program) must be selected. Provide wood surfaces as appropriate. Include accommodation for a Government provided computer and printer.

E202003 FREESTANDING FURNITURE

Provide ergonomic task seating, lounge, reception and guest seating, storage and filing, tables, as required.

E202005 MULTIPLE SEATING (MOVABLE)

Not Used.

E202006 INTERIOR LANDSCAPING (MOVABLE)

Not Used.

E202090 OTHER MOVABLE FURNISHINGS

Provide waste receptacles, recycling containers, fire extinguishers, clocks, literature racks, microwaves, refrigerators, non-plumbed coffee pots, and other appliances as required.

GC will secure the services of a commercial kitchen designer for food service layout. The following list of equipment, while comprehensive is not all inclusive and is provided only to gage scale of effort.

- (3) double battery fryer
- (2) 10 burner ranges (preferably gas)
- (5) double deck convection ovens
- (2) full size combination ovens
- (3) 6 drawer under counter refrigerated worktables
- (2) tilt skillets (48-inch)
- 60-inch griddle
- (8) double door reach in freezers
- (10) double door reach in refrigerators
- coffee station
- ice maker (500lb capacity)
- (7) warming cabinets
- stainless steel worktables with countertops on casters
- trash receptacles
- dry food storage shelves
- double conveyor dishwasher
- waiter service storage (china, glassware, silverware, linens, chaffing dishes, pots, pans)

END OF SECTION

F10 SPECIAL CONSTRUCTION

Special Construction includes the restoration of the Guardhouse as an occupiable facility.

END OF SECTION

F20 SELECTIVE BUILDING DEMOLITION

1. GENERAL SYSTEMS REQUIREMENTS

Perform all off-site work necessary to meet the requirements of the project, local codes, reference standards, technical specifications and performance criteria.

Identify and obtain permits to comply with federal, state, and local regulatory requirements associated with this work. Complete the Permits Record of Decision (PROD) form with the first design submittal package. Determine correct permit fees and pay said fees. Forward copies of permits, permit applications, and the completed PROD form to the Government's Civil Reviewer. Perform work in accordance with the obtained permits.

Coordinate and obtain approval from the Contracting Officer for proposed haul route(s), work site access point(s), employee parking location(s) and material laydown and storage area(s).

Provide shoring, bracing, reinforcing and stabilization to prevent settlement or other movement of the existing Guardhouse portion of the building weakened by the demolition work, but needing to remain in place and be operational.

Provide weather protection enclosures for portions of the building to remain. Protect the building interior materials and equipment from the weather at all times.

F2010 BUILDING ELEMENTS DEMOLITION

This project includes the demolition of the following: Building 214 and restoration of the Guardhouse.

All demolition materials and appurtenances must be properly disposed of in accordance with all applicable regulations. Maximize the use of deconstruction and recycling services. Before demolition can commence, any hazardous materials must be abated in accordance with the requirements of the parts of the RFP. Provide a Demolition Plan/ Deconstruction Plan that is based on a Registered Engineers Survey in accordance with EM 385-1-1 and has been approved by the DOR. Obtain approval from the Contracting Officer for the proposed demolition plan and work/outage schedule prior to demolition activities.

1. GENERAL DEMOLITION

Remove indicated existing structure to bottom of basement foundations. Contractor’s structural engineer will analyze the east wall of the Guardhouse building prior to demolition to verify an adequate load path after demolition is completed. Provide shoring, lateral bracing, and underpinning, as required, to protect the existing building foundations.

The work includes demolition and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Materials that cannot be removed daily must be stored in areas specified in the approved Demolition Plan as described in Part 2 Section 01 57 19.

2. UTILITIES

All of the services to the existing Patton Hall Building will be removed and cut and capped at the main. Demolition of the exterior electrical equipment (medium voltage pad-mounted switchgear, pad-mounted transformer, generator, and automatic transformer switch) will be removed by Dominion Energy. The existing parking lot has an existing sanitary main, electric, and stormwater piping within the parking lot that may need to be relocated depending on the proposed building location.

Remove existing utilities and terminate in a manner conforming to the nationally recognized code covering the specific utility. Disturbance to utilities cannot cause a failure to utilities to remain operational, unless a planned outage is approved by the Installation and coordinated with on-site personnel.

Coordinate utility demolition, removal, and abandonment with on-site personnel for planned outages.

3. DUST CONTROL

Prevent the spread of dust and debris and avoid the creation of a nuisance or hazard in the surrounding area.

4. TRAFFIC CONTROL

Where pedestrian and vehicle access is endangered, use traffic barricades.

5. WEATHER PROTECTION

For portions of the building to remain, protect building interior, materials, and equipment from weather at all times.

6. BURNING

Burning will not be permitted.

F201001 SUBSTRUCTURE & SUPERSTRUCTURE

Perform substructure or superstructure demolition work in accordance with Section F20.

F201002 EXTERIOR CLOSURE

Perform exterior closure demolition work in accordance with Section F20.

For occupied buildings ensure openings to the exterior are secured by the end of the work shift.

F201003 ROOFING

Perform roofing demolition work in accordance with Section F20.

F201004 INTERIOR CONSTRUCTION & FINISHES

Perform interior construction & finishes demolition in accordance with Section F20.

F201005 CONVEYING SYSTEMS

Perform conveying systems demolition in accordance with Section F20.

F201006 MECHANICAL SYSTEMS

Perform mechanical systems demolition in accordance with Section F20.

F201007 ELECTRICAL SYSTEMS

Perform electrical systems demolition in accordance with Section F20.

F201008 EQUIPMENT & FURNISHINGS

Perform special equipment and furnishing demolition in accordance with Section F20.

F201009 OTHER NON-HAZARDOUS SELECTIVE BUILDING DEMOLITION

Perform non-hazardous selective building demolition in accordance with Section F20.

F2020 HAZARDOUS COMPONENT ABATEMENT

Perform work in accordance with specification and plans provided. See division 01 and 02 specifications provided under separate cover.

END OF SECTION

G10 SITE PREPARATION

1. SYSTEM DESCRIPTION

The site preparation activities consist of site clearing, demolition, relocation, earthwork, and hazardous waste remediation to ready the site for other work associated with the project.

2. GENERAL SYSTEM REQUIREMENTS

Develop the project site and perform off-site work necessary to meet the requirements of the project, antiterrorism criteria, local codes, reference standards, technical specifications and performance criteria.

A topographic survey of the existing site has been performed and is included in Part 6. The topographic survey has been provided to show the location of existing facilities, areas of new work required by this RFP and the character of the sites. Prior to starting work, physically verify the location of all existing utilities and obtain additional survey data required to provide a quality final design. Perform a topographic survey in accordance with USACE EM 1110-1-1005 Control and Topographic Surveying, A/E/C CAD Standards, and/or additional USACE survey criteria included herein. Include the topographic survey in all design submittals. The existence, size, and location of the utilities are not guaranteed by the surveys provided. Verify the location of all utilities prior to construction. Electronic files of the topographic surveys will be provided to the Contractor only after award of the Contract.

Unless otherwise noted, provide new facilities at the locations indicated on the drawings. Minimize the impact of construction activity on operations and neighboring facilities.

Identify and obtain permits to comply with federal, state, and local regulatory requirements associated with the work. Submit a complete Regulatory Compliance Checklist Form 124 with the first design submittal package. Determine correct permit fees and pay said fees. Forward copies of permits, permit applications, and the completed Checklist to the Government's Civil Reviewer and Environmental Reviewer. Perform work in accordance with the obtained permits.

Jurisdictional tidal and non-tidal wetlands have not been identified on the project site.

Coordinate and obtain the Contracting Officer's approval for proposed haul route(s), work site access point(s), employee parking location(s) and material laydown and storage area(s).

Refer to Site Analysis and Building Requirements Sections for additional site preparation functional program information.

3. GOVERNMENT PROVIDED GEOTECHNICAL INFORMATION

Refer to Section A10, Foundations, for additional information on the Government provided geotechnical report, which is included in Appendix C of this RFP.

Observed site conditions which may present a challenge during design/construction include: the soils that may be exposed after completion of stripping and excavation may consist of be fatty clays (CH) which are sensitive to moisture and susceptible to expansion. According to the boring logs, the site has areas with existing fill from 2’-12’. The degree of compaction or inclusion of unsuitable material may be encountered in the existing fill. Anticipate these marginal subgrade support conditions and incorporate measures into the design and construction procedures to obtain required soil support while maintaining progress for completion on schedule. Provide personnel under the supervision of a registered Professional Engineer to inspect excavations and soil/groundwater conditions throughout construction. The Engineer is required to perform pre-construction and periodic site visits throughout construction to assess site conditions. The Engineer, with the concurrence of the Contractor and the Contracting Officer, is required to update the excavation, sheeting, shoring and dewatering plans as construction progresses to reflect actual site conditions and is required to submit the updated plan and a written report (with professional stamp) at least monthly informing the Contractor and Contracting Officer of the status of the plan and an accounting of Contractor adherence to the plan; specifically addressing any present or potential problems. The Engineer must be available to meet with the Contracting Officer at any time throughout the Contract duration. Provide the services of the Engineer at no additional cost to the Government. It is important to note that the presence of undocumented fill material may result in settlement that could impact the performance of surface bearing structures and supporting facilities such as foundations, slabs, pavements, sidewalks, and utilities. The magnitude and duration of consolidation settlement will be dependent on the composition, depth, and thickness of the compressible soils as well as the successful bidder's design concept. The Contractor's Geotechnical Engineer is responsible for evaluating potential global settlement due to designed grade increases and final structural loads. The Contractor's Geotechnical Engineer must develop any settlement mitigation procedures (such as preloading, surcharging, fill monitoring programs, and ground improvement systems) needed to maintain global settlements within tolerable limits. Surcharge material, if required, must remain in place for a minimum of 90 days.

G1010 SITE CLEARING

Install erosion and sediment control devices prior to beginning clearing or grubbing operations. If approved by the Government clearing and grubbing may be allowed to accommodate construction equipment within the designated construction laydown area.

G101001 CLEARING

The project site does not have saleable timber. Burning will not be permitted. Clear trees, shrubs, brush and vegetation for construction of the project. Clearing includes the felling, trimming, and cutting of trees into sections and the satisfactory disposal of the trees and other vegetation designated for removal, including downed timber, snags, brush, and rubbish occurring within the areas to be cleared.

G101002 TREE REMOVAL

Remove trees as required for project construction. Remove and dispose of trees to a depth of at least 18 inches (450 mm) below ground surface. Fill depressions with satisfactory material and compact. Mound fill 2 inches (50 mm) above adjacent surface to allow for settling when not part of a subbase.

G101003 STUMP REMOVAL

Remove all stumps under building foundations, roads, parking areas, hard stands, and storage areas as required for project construction to a depth of 5 feet below the finished grade. Remove stumps to a depth of at least 18 inches (450 mm) below ground surface. Fill depressions with satisfactory material and compact. Mound fill 2 inches (50 mm) above adjacent surface to allow for settling when not part of a subbase.

G101004 GRUBBING

Within the clearing limits, remove and dispose of logs, shrubs, brush, matted roots, roots larger than 2 inches (50 mm) in diameter, and other debris to a depth of at least 18 inches (450 mm) below ground surface. Fill depressions made by grubbing with satisfactory material and compact to make the new surface conform to the adjacent surface of the ground.

G101005 SELECTIVE THINNING

G101006 DEBRIS DISPOSAL

Dispose of clearing and grubbing material in the Installation's sanitary landfill according to its requirements and regulations.

Prevent spillage on pavements, streets, or adjacent areas. Dispose of surplus and unsuitable material off of Government property.

G1020 SITE DEMOLITION & RELOCATIONS

1. GENERAL

Demolition work includes the demolition, removal and legal disposal of existing construction debris to accommodate the new construction. Take precautions to prevent damages to existing utilities, construction and materials not scheduled for demolition, repair or replacement; repair damages to the construction and materials to the satisfaction of the Contracting Officer and at no additional cost to the Government.

2. AUTHORIZATION

Do not begin demolition until the Demolition Plan has been approved by, and authorization is received from, the Contracting Officer.

3. TITLE TO MATERIALS

Whenever possible, salvage or recycle features demolished in lieu of disposing of as waste in a landfill. Existing features to be demolished which are not salvageable or reused become the property of the Contractor and must be removed from the project site. The Government will not be responsible for the condition, loss of, or damage to, such property after contract award. Materials and equipment cannot be viewed by prospective purchasers or sold on the site.

4. REUSE OF MATERIALS AND EQUIPMENT

Remove and store materials and equipment to be reused or relocated to prevent damage and reinstall as the work progresses.

5. SALVAGED MATERIALS AND EQUIPMENT

Not Used.

G102001 BUILDING MASS DEMOLITION

Demolish the existing canopy buildings as shown on the drawings.

Refer to Section F20, Selective Building Demolition, for additional information.

G102002 ABOVEGROUND SITE DEMOLITION

1. DUST AND DEBRIS CONTROL

Prevent the spread of dust and debris to occupied portions of a building or on pavements and avoid the creation of a nuisance or hazard in the surrounding area. Do not use water for dust control if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

2. PROTECTION

2.1. TRAFFIC CONTROL

Where pedestrian and driver safety is endangered in the area of removal work, provide traffic control in accordance with FHWA Manual on Uniform Traffic Control Devices.

2.2. EXISTING WORK

Protect existing work that is to remain in place, be reused, or remain the property of the Government. At no additional expense to the Government, repair items that are damaged during performance of the work to original condition or replace with new. Do not overload pavements to remain.

3. PAVING AND SLABS

Remove concrete and asphaltic concrete paving and slabs as required for construction of project. Remove the existing aggregate base in areas to receive new pavement to the depth of the proposed pavement section below new finish grade. Remove the existing aggregate base in areas not to receive new pavement to a depth of 8 inches (200 mm) below existing adjacent grade and break remaining pavement (if any) to allow drainage. Provide neat sawcuts at limits of pavement removal; protect sawcuts so that new pavement butts against the existing without feathering.

4. ABOVEGROUND STORAGE TANKS

The project site does require aboveground storage tank work. Perform aboveground storage tanks work in accordance with the specifications and plans.

G102003 UNDERGROUND SITE DEMOLITION

Remove or relocate existing utilities within 10 feet (3.0 m) of any new facilities or building additions. Existing utilities include but are not limited to piping, structures and conduits. Remove all appurtenances associated with the utility to be removed so there is no presence of the utility at ground surface.

All conduits to be abandoned must remove the wiring in place.

1. UTILITY TERMINATION

Terminate utilities in accordance with state and local rules and regulations; the nationally recognized code; and the requirements of the utility provider covering the specific utility; UFC 3-201-01, Civil Engineering; and approved by the Contracting Officer.

2. PROTECTION OF EXISTING UTILITIES

Protect existing utilities to remain. Where removal of existing utilities and pavement is required, provide approved barricades, temporary covering of exposed areas, and temporary services or connections. Repair damage to existing utilities to remain at no additional expense to the Government.

3. UNDERGROUND STORAGE TANKS

The project site does not require underground storage tank removal.

G102004 BUILDING RELOCATION

Not Used

G102005 UTILITY RELOCATION

Repair relocated items that are damaged or replace damaged items with new undamaged items at no additional expense to the Government.

G102006 FENCING RELOCATION

Remove and replace post foundations. Repair relocated items that are damaged or replace damaged items with new undamaged items at no additional expense to the Government. Refer to Section G204001 for requirements for new fence systems.

G102007 SITE CLEANUP

Waste materials will become the property of the Contractor; transport, dispose of or recycle waste materials in accordance with Section 02 41 00. Remove rubbish and debris from the installation daily; do not allow accumulation inside or outside the building(s) or on pavements. Store materials that cannot be removed daily in areas specified by the Contracting Officer.

1. SPILLS

In the event of a spill or release of hazardous substances, pollutant, contaminant or oil, notify the Contracting Officer immediately. Take immediate containment actions to minimize the effect of any spill or leak in accordance with the approved spill work plan as described in Section 01 57 19. Perform clean up at no additional expense to the Government.

G1030 SITE EARTHWORK

1. General

This section includes the design and construction requirements for earthwork and grading related to construction of the roadways, parking, paved areas and other related sitework. Refer to Section A10, Foundations, for earthwork related to construction of structures, including building, footings, foundations, retaining walls, slabs, tanks, and utility appurtenances.

The construction site is categorized as Category III. Category III means the site is known to be contaminated or there is a strong suspicion that contamination will be encountered during construction. Contamination is expected to be encountered during construction, and abatement will be included as part of the project. However, based on the soils survey completed, there is no reason to suspect contaminated soils or groundwater in the project area. Reference Site Categorization Memo in Appendix C.

G103001 GRADING

Provide site grading in accordance with the requirements of UFC 3-201-01, Civil Engineering.

1. ELEVATIONS

Establish finish floor elevations in accordance with UFC 1-200-01, DoD Building Code (General Building Requirements), and UFC 3-101-01, Architecture.

2. SITE GRADING

Grade the site such that associated storm water runoff does not adversely affect surrounding sites. Preserve natural topographic features to minimize the impact on the existing drainage patterns at and adjacent to the site.

3. FINISHED SURFACES

Provide finish grading with drainage towards new and existing drainage features and with no resulting low spots that hold water or that direct runoff towards new or existing facilities or site amenities.

4. RODENT AND VEGETATION CONTROL

Prevent and eliminate standing water.

G103002 COMMON EXCAVATION

Preserve natural topographic features to minimize cut and fill requirements. Unsuitable material and surplus excavation become the property of the Contractor and must be disposed of as indicated in the Project Program.

G103003 ROCK EXCAVATION

Hard materials and rock will not be encountered. Blasting will not be permitted. If blasting is allowed, conduct it in accordance with EM 385-1-1 and Federal, state, and local safety regulations. Provide blasting mats and use non-electric blasting caps. Notify the Contracting Officer 24 hours prior to blasting.

Do not make requests for additional compensation for degree of hardness or difficulty encountered in removal of material. Unsuitable material and surplus excavation becomes the property of the Contractor and must be disposed of as indicated in the Project Program.

G103004 FILL & BORROW

Common fill, backfill and fill material, select fill, and topsoil may be available at the project site dependent on the proposed site plan.

Any additional material required will have to be obtained from offsite meeting the requirements of the geotechnical report.

1. REQUIREMENTS FOR OFF SITE SOIL

For each borrow site, provide borrow site testing for hazardous materials characteristics from a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring material onsite until tests results have been received and approved by the Contracting Officer.

1.1. SOURCES

Where sufficient topsoil and satisfactory materials are not available on the project site, provide suitable borrow materials.

1.2. UNSATISFACTORY SOIL MATERIALS

Remove uncontaminated unsatisfactory soil materials from the site. Unsatisfactory materials are materials which do not comply with the requirements for satisfactory materials. Unsatisfactory materials also include man-made fills, trash, refuse, backfills from previous construction or material classified as satisfactory which contains root and other organic matter, frozen material, and stones larger than 3 inches. The Contracting Officer is required to be notified of any contaminated materials.

1.3. TOPSOIL

Refer to Section G2050, "Landscaping". Remove unsatisfactory, existing topsoil from the site in accordance with the Project Program. Provide soil stabilization designed to function as required by site conditions in accordance with the State Highway specifications and standards in the state where the project is located. Apply and install geosynthetics in accordance with the manufacturer's written instructions.

G103005 COMPACTION

Provide compaction in accordance with UFGS Section 31 00 00 Earthwork, according to the recommendations of the Contractor's Geotechnical Engineer.

G103006 SOIL STABILIZATION

As recommended by the Contractor’s Geotechnical Engineer, provide soil stabilization designed to function as required by site conditions in accordance with the State Highway specifications and standards in the state where the project is located. Apply and install geosynthetics in accordance with the manufacturer's written instructions.

G103007 SLOPE STABILIZATION

As recommended by the Contractor’s Geotechnical Engineer, provide slope stabilization through appropriate grading and site design for a minimum factor of safety of 1.5 or slope that does not exceed the maximum slope according to the local code requirements. Design and install manufactured products, gabions, geogrids, rock anchors in accordance with the manufacturer's written instructions.

G103009 SHORING

Provide sheeting, shoring, bracing, cribbing and underpinning in accordance with the U.S. Army Corps of Engineer's Safety and Health Requirements Manual (COE EM 385-1-1), UFC 3-220-01, Geotechnical Engineering, UFC 3-301-01, Structural Engineering, and other Federal, State and local codes and requirements. Provide protection of existing structures.

G103010 TEMPORARY DEWATERING

For bidding purposes, it is to be assumed that significant dewatering efforts will not be required. The design of the temporary dewatering system is required to account for soil conditions, rainfall, fluctuations in the groundwater elevations and the potential settlement impact on adjacent facilities due to dewatering. Provide dewatering in accordance with UFGS Section 31 00 00 Earthwork while the excavation is open, maintain the water level continuously, at least 1.0 foot (0.30 m) below the working level.

French drains, sumps, ditches or trenches are not allowed within 3 feet (0.9 m) of the foundation of any structure without written approval of the Government's Civil/Geotechnical Reviewer.

G103011 Temporary Erosion & Sediment Control

Obtain Erosion and Sediment Control permit required for the proposed work from the VDEQ. Submit permit application to the Contracting Officer for approval prior to submitting to the VDEQ.

G103090 OTHER SITE EARTHWORK

Not Used.

G1040 HAZARDOUS WASTE REMEDIATION

The project site does not require hazardous waste remediation; however, the demolition of Building 214 will require hazardous material abatement. See hazardous material survey located in Appendix F.

1. CONTAMINATED SOIL AND GROUNDWATER

The project site does not require contaminated soil or groundwater work.

2. SPILLS

In the event of a spill or release of hazardous substances, pollutant, contaminant or oil, notify the Contracting Officer immediately. Take immediate containment actions to minimize the effect of any spill or leak. Perform clean up at the Contractor's expense in accordance with the approved spill work plan as described in Section 01 57 19, Temporary Environmental Controls.

3. DISPOSAL

Waste materials will become the property of the Contractor; transport, dispose of or recycle waste materials in accordance with Section 02 41 00 and 01 57 19.

Manage, transport and dispose of waste materials in accordance with specifications and plans.

END OF SECTION

G20 SITE IMPROVEMENTS

1. SYSTEM DESCRIPTION

Provide site improvements that support project sustainability goals of Section 01 33 29.

The site improvements consist of pavements and pavement related features, landscaping and other exterior site development work related to this project. Provide a pavement design by a licensed Professional Engineer familiar with conditions local to the project site. Site design, including but not limited to design of parking and pedestrian circulation, will include coordination with the Civil Engineer and the Landscape Architect.

2. GENERAL SYSTEMS REQUIREMENTS

Provide site improvements as required to make a useable facility that meets functional and operational requirements, incorporates all applicable antiterrorism, force protection and physical security requirements and blends into the existing environment. Provide accessibility in conformance with requirements of UFC 1-200-01, DoD Building Code (General Building Requirements).

Identify and obtain permits to comply with federal, state, and local regulatory requirements associated with this work. Complete the Regulatory Compliance Checklist Form 124 with the first design submittal package. Determine correct permit fees and pay said fees. Forward copies of permits, permit applications, and the completed Form 124 to the Government's Civil Reviewer. Perform work in accordance with the obtained permits.

Minimize the impact of construction activity on operations and neighboring facilities.

Locate new site improvements at locations indicated on the drawings in another part of this RFP. If specific locations are not provided, site the improvements to develop appropriate and positive relationships with other facilities and to conform to existing development patterns.

Refer to Site Analysis and Building Requirements Sections for additional site improvement functional program information.

G2010 ROADWAYS

In addition to requirements stated herein, refer to Appendix C, Geotechnical Report for RFP.

Provide roadways, as required, to allow for safe, convenient and logical circulation, while discouraging through traffic. Design pavements based on the anticipated daily traffic over the life of the project as well as the existing soil conditions at the site.

Provide roadways of bituminous pavement or Portland cement concrete (PCC) pavement.

Provide new roadway and other pavement sections as required by soil conditions and determined by the Designer of Record. Design pavement sections in accordance with UFC 3-201-01 Civil Engineering.

Include adequate space for trees and other landscape material in the design for streets, roads, and parking lots.

1. PAVEMENT DESIGN

Provide geometric and pavement design, including minimum pavement sections, in accordance with UFC 3-201-01, Civil Engineering, and the State Department of Transportation. Utilize PCASE software to justify pavement section designs and provide supporting calculations. Provide any required additional pavement design to provide a complete and useable facility.

2. PAVEMENT AESTHETICS

Provide surfaces consistent in color and finish.

3. LANDSCAPING

Include adequate space for trees and other landscape material in the design for streets, roads, and parking lots.

4. TRAFFIC CONTROL DEVICES

Provide and install new traffic control devices (i.e., signs and markings) in accordance with the United States Department of Transportation Federal Highway Administration's Manual on Uniform Traffic Control Devices and their standard, "Rigid Sign Supports". Also provide new traffic control devices along/in the existing streets adjacent to the project site as necessary to provide complete traffic control to the new facilities.

5. EXISTING UTILITY STRUCTURES

Adjust existing utility structures to meet the new finished pavement grades as required.

G201001 BASES & SUBBASES

In addition to requirements stated herein, refer to Appendix C, Geotechnical Report for RFP.

Prepare subgrade in accordance with Section G10, Site Preparation. Use geotextiles for separation or reinforcement in accordance with manufacturer's instructions. Provide base course under paved areas in accordance with the State Highway specifications (SHS) in the state where the project is located.

Place base course in accordance with the SHS for that particular base course and in layers of equal thickness with no compacted layer more than 6 inches (150 mm) thick. Compact base course at optimum moisture content to 100 percent ASTM D 1557 maximum dry density.

G201002 CURBS & GUTTERS

Provide curb and gutter to tie into adjacent facilities.

Provide concrete curbs and gutters in accordance with the SHS and standards or as specified in UFC 3-201-01, Civil Engineering, whichever is more stringent. Where the SHS do not include concrete materials for curbs and gutters, provide concrete in accordance with the standard mix of the SHS for a minimum compressive strength at 28 days of 3500 psi (25 MPa) concrete.

G201003 PAVED SURFACES

Provide Portland cement concrete with a minimum design flexural strength of 650 psi (4.48 MPa) at 28 days.

Provide concrete curbs and gutters in accordance with the SHS and standards or as specified in UFC 3-201-01, Civil Engineering, whichever is more stringent. Where the SHS do not include concrete materials for curbs and gutters, provide concrete in accordance with the standard mix of the SHS for a minimum compressive strength at 28 days of 3500 psi (25 MPa) concrete.

G201004 MARKING & SIGNAGE

Provide pavement markings in accordance with the MUTCD. Provide signage in accordance with the MUTCD.

Provide temporary pavement markings and signage throughout construction to meet phasing requirements indicated in the project program. Provide temporary signage in accordance with the MUTCD.

G201005 GUARDRAILS & BARRIERS

Provide guardrails and bollards in accordance with UFC 3-201-01, Civil Engineering.

G201006 RESURFACING

Provide resurfacing of existing pavement by bituminous mill and overlay.

Adjust rims of existing utility structures to match proposed grades after resurfacing.

G2020 PARKING LOTS

Provide new parking and other pavement sections as required by soil conditions and determined by Designer of Record. Design pavement sections in accordance with UFC 3-201-01 Civil Engineering.

Provide other parking improvements including parking entrances for two-way traffic, signage, and landscaping.

Provide handicapped parking in accordance with UFC 1-200-01, DoD Building Code (General Building Requirements).

G202004 MARKING & SIGNAGE

Provide permanent and temporary markings (pavement, curb and object), signage (regulatory, warning and guidance) and other traffic control devices as required to facilitate proper utilization of the parking areas.

Provide pavement markings and signage in accordance with the MUTCD. Provide temporary pavement markings and signage to meet phasing requirements and in accordance with the MUTCD.

Mark to denote traffic lanes and parking spaces; mark in accordance with the requirements of UFC 3-201-01, Civil Engineering.

G202005 GUARDRAILS & BARRIERS

Provide guardrails and bollards in accordance with UFC 3-201-01, Civil Engineering.

G2030 PEDESTRIAN PAVING

Provide a network of PCC sidewalks, separated from, but connected to vehicular circulation systems, to allow for pedestrian circulation between various new and existing elements of the project. Interface new pedestrian circulation systems with existing pedestrian circulation systems and include input from the Civil Engineer, Architect, and Landscape Architect.

Locate new sidewalks such that they maintain continuity of pedestrian traffic to and from the existing sidewalks adjacent to the site(s).

G2040 SITE DEVELOPMENT

G204001 FENCING & GATES

Provide fence and gates as needed.

G204003 EXTERIOR FURNISHINGS

All site furnishings are required to conform to the Base Exterior Architectural Plan (BEAP) and the Installation Design Guide (IDG). Provide trash receptacles, benches, and other site amenities as required. At a minimum, provide a trash and ash receptacle at the designated smoking area.

G204004 SECURITY STRUCTURES

Not Used.

G204005 SIGNAGE

Provide signage in accordance with the Activity's BEAP and the IDG. Provide facility signage in accordance with local code, the Installation and Appearance Guide, the BEAP and this RFP.

G204090 OTHER SITE IMPROVEMENTS

Provide dumpster pad and enclosure conforming to the Activity's BEAP and Installation Appearance Plan. Other site improvements will conform to the BEAP or Installation Design Guide (whichever is applicable) and to the requirements of UFC 4-010-01.

G2050 LANDSCAPING

Provide complete landscaping consisting of lawn, groundcover, trees, shrubs, perennials, ornamental grasses, organic mulches, and inorganic mulches to provide a quality, cost-effective, functional and visually appealing landscape program that will enhance the development, while complying with antiterrorism, force protection and physical security requirements. Design the landscape to reinforce the facility entry and complement existing landscapes in the vicinity.

Guarantee landscaping for a period of one year. Provide a one-year Establishment and Maintenance period. Landscaping Guarantee and Establishment and Maintenance periods must commence on the date that the inspection by the Contracting Officer shows that all landscaping under this contract has been satisfactorily installed.

Provide complete landscaping maintenance, including but not limited to, routine lawn mowing, edging, pruning, pest inspection/treatment, re-mulching of mulch products, watering, weeding, fertilizing, and restaking, throughout the guarantee period.

Provide mechanical equipment screening wall on three sides of new equipment.

Provide shrubs or small growing trees for screening of mechanical equipment/walls, dumpster enclosures, and other obstructions that do not present an aesthetic view from the street.

Landscape areas are defined as permeable areas within the project boundaries not covered by buildings, roads, parking lots, sidewalks, and other non-permeable areas. Provide landscape improvements to all site areas disturbed by construction.

G205001 FINE GRADING AND SOIL PREPARATION

Provide 4 inches of topsoil for lawn areas and fine grade.

G205002 EROSION CONTROL MEASURES

Prevent erosion from occurring by providing erosion control measures as required by city, state and federal requirements.

G205003 TOPSOIL AND PLANTING BEDS

Provide a planting soil mixture around root balls of shrubs, trees, groundcovers, perennials, and ornamental grasses.

G205004 SEEDING SPRIGGING AND SODDING

Restore existing turf areas disturbed by Contractor operations that are to remain as turf areas.

G205005 PLANTINGS

Preserve existing trees to the greatest extent possible. Select plant material from Master Plant Lists found within the IDG. Other plants not found on these lists may be used if approved by the reviewing

Government Landscape Architect. Final approval of new plant materials rests with the reviewing Government Landscape Architect.

If used for stormwater management, provide bioretention filters, plants, plant quantities, and soil mix in accordance with the State's Best Management Practices (BMP) Design Manual.

Ensure that the bioretention filters are not compromised by sediment deposits during construction. Install bioretention filters only after upstream areas tributary to the bioretention filter location have been fully stabilized.

Preserve existing trees to the greatest extent possible. Identify preserved trees on the plans with tree species, caliper and dripline. Tag trees to be saved with plastic or vinyl tape tied to the tree caliper.

Do not place trees within 10 feet (3 meter) of above or below-grade utility line or structure. Within roadway sightlines, height of mature shrubs is limited to 3 feet (1 m) and trees must be limbed up a minimum of 6 feet (2 m) so their mature growth does not obstruct views from vehicle intersections or points of vehicle ingress or egress. Coordinate utilities between the Landscape Architect and appropriate disciplines.

END OF SECTION

G30 SITE CIVIL/MECHANICAL UTILITIES

1. SYSTEM DESCRIPTION

The site civil/mechanical utility systems include water supply systems, sanitary sewer systems, storm drainage systems, heating distribution systems, cooling distribution systems, fuel distribution systems and associated appurtenances which are more than 5 feet (1.5 meters) outside the building.

The site mechanical utility system consists of piping and appurtenances for natural gas including accessories and devices as required for a complete and usable system up to 5 feet outside buildings.

2. GENERAL SYSTEM REQUIREMENTS

Develop the site to provide water, fire protection, sanitary sewer, storm drainage, and fuel distribution services that meet the requirements of each regulatory agency that governs, and issues permits for the construction and operation of these systems. Site design is required to comply with UFC 3-210-10, Low Impact Development, as well as state or local stormwater management regulations and project sustainability goals.

Provide each system complete and ready for operation. Physically verify the location of existing above and below ground utilities prior to starting work.

Identify and obtain permits to comply with all federal, state, and local regulatory requirements associated with this work. Complete Regulatory Compliance Checklist Form 124 with the first design submittal package. Determine correct permit fees and pay said fees. Forward copies of permits, permit applications, and the completed form124 to the Government's Civil/Mechanical Reviewer. Perform work in accordance with the obtained permits.

Minimize the impact of construction activity on facility operations and neighboring facilities.

Utility connection points are indicated on the drawings in Appendix C. Obtain final approvals from the Government's Civil/Mechanical Reviewer and the Contracting Officer for utility connection points associated with this work.

Coordinate with the local utility providers and pay fees or charges required to connect to their utility. Disciplines involved in site work design must coordinate utility locations with the Civil Engineer and the Landscape Architect. Refer to Site Analysis and Building Requirements Sections for additional site civil/mechanical utilities information.

Provide fittings, connections and accessories required for a complete and usable system. Install equipment in accordance with Section G30 and the manufacturer's recommendations. Where the word "should" is used in the manufacturer's recommendations, substitute the word "must".

After installation of the equipment and systems, provide individual training courses for two Government personnel for each of the items listed below, covering items contained in the Operations and Maintenance manuals. Provide one copy of the Operations and Maintenance manuals for each two course attendees. Provide one DVD disc of the training courses to be used as refresher courses and to train additional personnel. Conduct training by the same factory trained engineer that supervised the installation of the system. During training include classroom discussion as well as hands on maintenance, replacement of typical components and repair type maintenance training for parts typically replaced or repaired in the field. Submit training plan 30 calendar days prior to training sessions; include scheduling, content, outline, and training material handouts. Provide site civil/mechanical utility systems and components that support project sustainability goals of Section 01 33 29, Sustainability Reporting for Design Build.

Verify the sanitary sewer connection to the sanitary sewer system by visual testing. Leakage test is not required.

G3010 WATER SUPPLY

The new water system is an extension of the existing water system. The existing water system serving the project site is owned by the Federal Government. Provide the new water system and connections to the existing water system in accordance with the utility provider's requirements and UFC 3-230-01 Water Storage, Distribution, and Transmission; whichever is more stringent.

Notify the utility provider of the additional demand generated by the proposed facility. Provide a copy of all correspondence with the utility provider to the Government's Civil/Mechanical Reviewer.

Provide connection to the existing water distribution system at the point indicated on the drawings.

Design the new water system so that water consumption is measured at a minimum, by one meter for each facility. Ensure the meter is easily accessible, but not obvious.

Determine domestic and fire demands for the facility and verify the design of all components of the domestic and fire protection supply systems. Design and construct the water system in accordance with UFC 3-230-01, Water Storage, Distribution, and Transmission, the state waterworks' regulations, and the utility provider's requirements. Design the water supply systems to provide required flows and maintain residual pressures based upon peak demands.

If the new water system is an extension of an existing water system, obtain static pressure, residual pressure and flow characteristics of the existing distribution system by actual field tests. Conduct flow and pressure tests and provide design calculations that show the existing lines are capable of handling the additional flows. Connect the new water system to the nearest existing fitting or water line capable of handling the additional flows.

Design the connections to the water system including the meter assemblies and backflow-preventing devices in accordance with the requirements of the Activity or utility provider and the state waterworks regulations.

Wherever possible, locate valve boxes and other utility access structures out of paved areas.

G301002 POTABLE WATER DISTRIBUTION

A water meter on each proposed service line is required. Provide type of meter and remote reading as required by the utility provider. Water meter must be compatible with and monitored by the Direct Digital Controls (DDC) system. Provide precast concrete vault meter box. Backflow preventers are required on service entrance lines.

1. WATER SYSTEM DESIGN

Provide materials, equipment, labor, testing, and miscellaneous related items for water distribution mains and service lines to the facility and connections to the existing water system in accordance with UFC 3-230-01, Water Storage, Distribution, and Transmission; the utility provider's requirements; and the state waterworks' regulations; whichever is more stringent.

2. WATER DISTRIBUTION MAINS

For underground applications, utilize ductile iron or PVC piping for water mains 12 inches (300 mm) in diameter and less. Utilize ductile iron piping for water mains deeper than 10 feet (3.0 m) or larger than 12 inches (300 mm) in diameter.

For aboveground applications, utilize flanged ductile iron pipe for water mains.

3. CONNECTIONS TO EXISTING WATER LINES

Make connections to existing water lines after approval from the system owner is obtained and with a minimum interruption of service on the existing line. Make connections to existing lines under pressure in accordance with the recommended procedures of the manufacturer of the pipe being tapped.

4. WATER SERVICE LINES

Utilize copper tubing or PVC piping for water service lines less than 4 inches (100 mm) in diameter. Utilize ductile iron pipe or PVC pressure pipe for water service lines 4 inches (100 mm) and 6 inches (150 mm) in diameter; see G301002, "Water Distribution Mains" for CORROSION PROTECTION

5. VALVES

Install valves with the same diameter and have the same joint ends as the mains to which they are installed. Provide each type of valve from one manufacturer.

6. WATER METERS

Provide water meter and remote reading as required by the utility provider and in accordance with American Water Works Association (AWWA) standards.

7. BACKFLOW PREVENTION

Provide backflow prevention and cross connection control in accordance with AWWA M-14 and governing local/state plumbing codes and waterworks' regulations.

8. FIRE HYDRANTS

Provide fire hydrants from one manufacturer and in accordance with UFC 3-600-01, Fire Protection Engineering. Coordinate with the project's fire protection designer of record. Provide protection for fire hydrants located in areas subject to vehicle damage. Provide fire hydrants with National Standard threads on hose and pumper connections. Provide a 6 inch (150 mm) inlet, two 2.5 inch (62 mm) hose connections and one pumper connection sized to accommodate local fire department equipment requirements. Paint hydrants with at least one coat of primer and two coats of enamel paint. Barrel and bonnet colors must be in accordance with UFC 3-600-01. Stencil hydrant number and main size on the hydrant barrel using black stencil paint.

9. THRUST RESTRAINT

Provide thrust restraint for all piping, valves, fittings, and other appurtenances of the water distribution system. Provide thrust restraint using restrained joints in accordance with pipe manufacturer's recommendations, AWWA C600 and if for fire service main, NFPA 24.

10. DISINFECTION

Disinfect new water piping and existing water piping affected by Contractor's operations in accordance with the state waterworks' regulations and AWWA C651

G301003 POTABLE WATER STORAGE

Not Used.

G301004 FIRE PROTECTION WATER DISTRIBUTION

Provide water main piping, service lines, fittings, valves, accessories and other materials in compliance with the AWWA standards for a minimum system working pressure of 200 psi (1380 kPa).

G301005 FIRE PROTECTION WATER STORAGE

Provide a fire protection water storage facility as determined necessary. Obtain approval from the Contracting Officer for shape of tank, the color and pattern of exterior coating. Design and construct Fire Protection Water Storage systems in accordance with UFC 3-600-01 and NFPA 22.

G301006 NON-POTABLE WATER DISTRIBUTION

Refer to G301002; note that system disinfection is not required.

G301007 PUMPING STATIONS

A package booster pump station is to be determined by DOR based on the configuration of the proposed building.

G301008 PACKAGED WATER TREATMENT PLANTS

Not Used.

G3020 SANITARY SEWER

The new sanitary sewer system is an extension of the existing sanitary sewer collection system. The existing sanitary sewer collection system serving the project site is owned by the Federal Government. Provide the new sanitary sewer system and connections to the existing sanitary sewer collection system in accordance with Virgina sewerage regulations and UFC 3-240-01 Wastewater Collection; whichever is more stringent.

Notify the utility provider of the additional wastewater flow generated by the proposed facility. Provide a copy of all correspondence with the utility provider to the Government Civil Reviewer.

Provide connection to the existing sanitary sewer collection system. In identifying a suitable point of connection, evaluate the capacity of the existing collection system, condition of existing pipe, and elevation of existing connection.

G302001 SANITARY SEWER PIPING

Provide exterior corrosion protection on metallic pipelines.

Design and construct the gravity sanitary sewage collection system in accordance with UFC 3-240-01, Wastewater Treatment and Collection, and the state sewer collection and treatment regulations. Connect the new sanitary sewage collection system to the nearest feasible existing sanitary manholes or sanitary line adjacent to the project site. Provide design calculations that show the existing system is capable of handling the additional flows.

In areas where chemicals and other substances may be stored (including mechanical and electrical rooms), eliminate floor drains or make provisions to prevent spills from entering the sanitary sewer system. If there is process flow from equipment, discharge can be hard piped, with air gaps, to the sanitary sewer. Wherever possible, locate manholes and other utility access structures out of paved areas.

G302001 SANITARY SEWER PIPING

Provide materials, equipment, labor, testing, and miscellaneous related items to provide sanitary sewage lines for collection and services from the buildings.

G302002 SANITARY SEWER MANHOLES & CLEANOUTS

Provide precast concrete manholes only. Provide lockable manhole covers. Provide all materials, equipment, labor, testing, and miscellaneous related items for the sanitary manholes in accordance with the following: Set manhole rim elevations flush with finished surface of paved areas or 1 inch (25 mm) above finished grade in unpaved areas. ASTM C 923M (ASTM C 923), resilient connectors for making joints between manhole and pipes entering manhole.

G302003 LIFT STATIONS AND PUMPING STATIONS

A wastewater pump station may be required. Provide a duplex submersible pumpstation in accordance with the utility provider's requirements. Provide pump station wet well of precast concrete construction. Provide adjacent valve vault of precast concrete construction.

Provide automatic control to start and stop the pump system. Provide automatic level control by floatsin accordance with the preferences of the system owner.

Provide an emergency pump connection.

Provide a telemetering system and recording equipment to a location manned 24 hours a day for the transmission and recording of pump operation. Transmit alarms to a location manned 24 hours a day.

Provide electrical connections for a portable emergency generator hook-up sized to start up and maintain the total rated running capacity of the station, including the pumps, controls, lighting, ventilation and other auxiliary equipment.

1. GENERAL REQUIREMENTS

If a pump station is allowed, provide materials, equipment, labor, testing and miscellaneous related items for a packaged lift or pump station system for the facility in compliance with the UFC 3-240-01, Wastewater Collection and Treatment; the state sewerage regulations; and the utility provider's requirements.

G302090 OTHER SANITARY SEWER

Not Used.

G3030 STORM SEWER

The new storm sewer system is an extension of the existing storm sewer system. The existing storm sewer system serving the project site is owned by the Federal Government. Provide the new storm sewer system and connections to the existing storm sewer system in accordance with the utility provider's requirements, UFC 3-201-01 Civil Engineering; UFC 3-210-10 Low Impact Development, VDEQ, local stormwater management laws and regulations and project sustainability goals; whichever is more stringent.

Provide connection to the existing storm sewer collection system d. Confirm that the existing outfall has adequate capacity to receive the additional stormwater flow generated by the project.

Provide materials, equipment, labor, testing, and miscellaneous related items to provide storm drainage collection system to drain the site. Design and construct the storm sewer collection system in accordance with UFC 3-201-01, Civil Engineering; the utility provider's requirements; and the state stormwater management laws and regulations. Design project site to prevent stormwater runoff in excess of the capacity of the existing utility system.

G303001 STORM SEWER PIPING

The following materials for storm sewer piping are not allowed: corrugated steel corrugated aluminum.

G303002 STORM SEWER STRUCTURES

Provide precast or cast-in-place concrete drainage structures Provide materials, equipment, labor, testing, and miscellaneous related items for the drainage structures. Set structure rim elevations flush with finished surface of paved areas or 1 inch (25 mm) above finished grade in unpaved areas. Provide resilient connectors for making joints between manhole and pipes entering manhole in conformance with ASTM C 923M (ASTM C 923).

G303003 LIFT STATIONS

Stormwater pump stations are not allowed.

G303004 CULVERTS

Provide reinforced concrete for culverts 12 inches (300 mm) and larger in diameter; PVC, polyethylene and polypropylene pipe may only be used when written approval is received by the Government's Civil Reviewer or indicated in another part of the RFP. See G303001 for material and installation requirements.

Provide flared end sections of the same material as the pipe material. Provide erosion control in accordance with the State Erosion and Sedimentation Control Standards or EPA guidance where State Standards are unavailable.

G303005 HEADWALLS

Provide cast-in-place concrete headwalls in accordance with the SHS and standards where the project is located.

G303006 EROSION & SEDIMENT CONTROL MEASURES

Refer to Section G103011.

G303007 STORM WATER MANAGEMENT

Provide stormwater management that complies with state stormwater regulations, UFC 3-201-01 Civil Engineering, UFC 3-210-10 Low Impact Development, and Energy Security and Independence Act (EISA) Section 438; whichever is more stringent. Provide Low Impact Development (LID) features in accordance with UFC 3-210-10 Low Impact Development.

The Contractor must obtain regulatory permits required for the proposed work from the VDEQ. Coordinate reports, submittals, and permit applications through the Contracting Officer.

1. STORMWATER COLLECTION AND STORAGE

Provide permanent stormwater management (i.e., detention and retention ponds, LID and other drainage features) to control stormwater runoff in accordance with UFC 3-201-01, Civil Engineering, UFC 3-210-10, Low Impact Development, EISA 438, State and local stormwater management Laws and Regulations and project sustainability goals. Integrate permanent stormwater management features into the site design in accordance with UFC 3-201-01, Civil Engineering.

Parking areas, roads, walks, courtyards, training areas and similar site features may not be used to detain or retain stormwater. Manage stormwater within detention or retention ponds and the LID features indicated in Part 3 of this RFP. Prevent upstream and downstream property damage.

G303090 OTHER STORM SEWER

An oil/water interceptor is not required.

G3040 HEATING DISTRIBUTION

G304002 OVERHEAD STEAM SYSTEMS

Not Used.

G304003 UNDERGROUND HOT WATER SYSTEMS

Not Used.

G304004 UNDERGROUND STEAM SYSTEMS

Not Used.

G304005 REINFORCED CONCRETE MANHOLES & VALVE BOXES

Provide concrete manholes for valves and building supply taps. Provide solid plate cover with access and ventilation openings. Provide electric operated sump pump in each manhole.

G304090 OTHER HEATING DISTRIBUTION

Provide a sacrificial anode cathodic protection system for the underground metallic piping systems.

G3050 COOLING DISTRIBUTION

G305001 Overhead Cooling Systems

Not Used.

G305002 UNDERGROUND COOLING SYSTEMS

Not Used.

G3060 FUEL DISTRIBUTION

G306001 LIQUID FUEL DISTRIBUTION PIPING

Not Used.

G306004 LIQUID FUEL DISPENSING EQUIPMENT

Not Used.

G306006 GAS DISTRIBUTION PIPING NATURAL GAS)

Provide natural gas piping system. See ESR D30 for gas meter requirements associated with direct digital control (DDC) system.

G306007 GAS STORAGE TANKS

Not Used.

G306009 OTHER GAS DISTRIBUTION

Not Used.

G306090 OTHER FUEL DISTRIBUTION

Not Used.

G3090 OTHER SITE MECHANICAL UTILITIES

Not Used.

END OF SECTION

G40 SITE ELECTRICAL UTILITIES

1. SYSTEM DESCRIPTION

Demolish the existing electrical system and provide a new electrical system. The existing exterior electrical equipment (medium voltage pad-mounted switchgear, pad-mounted transformer, generator, automatic transfer switch) is owned and operated by Dominion Energy. Dominion Energy will remove this equipment. The new exterior electrical equipment (medium voltage pad-mounted switchgear, pad-mounted transformers, natural gas emergency generator, automatic transfer switch) is to be provided by Dominion Energy. The exterior equipment must be located in an area not visible from the front of the facilities. Dominion Energy ROM is provided in Appendix D.

The site electrical utility system consists of all power and telecommunications (fiber optic and copper) from the existing distribution system point of connection including all connections, accessories and devices as necessary and required for a complete and usable system. This section covers installations up to within 5 feet (1.5 meters) of new and existing building locations.

2. GENERAL SYSTEM REQUIREMENTS

Provide an Electrical System complete in place, tested and approved, as specified throughout this RFP, as needed for a complete, usable and proper installation. Install all equipment in accordance with Section G40 and the manufacturer's recommendations. Where the word "should" is used in the manufacturer's recommendations, substitute the word "must". The site power distribution system must be in accordance with UFC 3-550-01 Exterior Electrical Power Distribution.

Provide site electrical utilities systems and components that support project sustainability goals of Section 01 33 29.

G4010 ELECTRICAL DISTRIBUTION

Connect to the new 480Y/277 volt, three phase, four wire, 60 hertz electrical power system. The connection point must be underground in concrete encased ductbank from the service entrance equipment to the Dominion Energy pad-mounted transformer.

Dominion Energy calculated the preliminary available fault current at the primary side of the pad-mounted transformer to be 7,225 symmetrical amperes. Dominion Energy will determine the finalized available fault current based on the Dominion Energy Non-Residential Load Letter submitted by the contractor.

G401001 SUBSTATIONS

Not Used

G401002 TRANSFORMERS

Dominion Energy will provide three phase pad-mounted transformers including smart metering to feed each facility, including the Guardhouse and Patton Hall – Strategic Engagement Platform. Dominion Energy ROM is provided in Appendix D.

G401003 SWITCHES, CONTROLS AND DEVICES

Dominion Energy will provide the medium voltage pad-mounted switchgear configured with the existing underground distribution system. Dominion Energy ROM is provided in Appendix D.

G401004 OVERHEAD ELECTRIC CONDUCTORS

Not Used

G401005 TOWERS, POLES, CROSSARMS AND INSULATORS

Not Used.

G401006 UNDERGROUND ELECTRIC CONDUCTORS

Provide a 600-volt secondary underground electrical distribution system to meet the connection requirements. Dominion Energy is to provide the medium voltage underground electrical distribution system. Dominion Energy ROM is provided in Appendix D.

Route underground cables to minimize splices. Cable pulling tensions must not exceed the maximum pulling tension recommended by the cable manufacturer.

G401007 DUCTBANKS, MANHOLES, HANDHOLES AND RACEWAYS

Provide a system of concrete encased ductbanks, handholes and manholes for all underground power wiring Concrete manholes and handholes to be standard type precast concrete. Composite/Fiberglass handholes must be polymer concrete reinforced with heavy weave fiberglass reinforcing. Provide manholes and handholes with load ratings suitable for the location installed. Dominion Energy is to provide the manholes and concrete encased ductbanks for the medium voltage electrical distribution system. Dominion Energy ROM is provided in Appendix D.

G401008 GROUNDING SYSTEMS

Provide a complete grounding system for the electrical power distribution system.

G401009 METERING

Not Used.

G401010 CATHODIC PROTECTION SYSTEMS

Provide cathodic protection systems in accordance with UFC 3-570-01 and UFC 3-570-06, and based on performance UFGS specification, 26 42 14.00 10, following National Association of Colleges and Employers (NACE) guidelines.

G4020 SITE LIGHTING

Dominion Energy is to provide site lighting for exterior parking areas, including underground distribution, handholes, grounding, poles, fixtures and controls as required for a complete and usable system.   Dominion Energy ROM is provided in Appendix D.

G402001 EXTERIOR LIGHTING FIXTURES AND CONTROLS

Not Used.

G402003 OTHER AREA LIGHTING

Not Used.

G402004 LIGHTING POLES

Not Used.

G402005 UNDERGROUND ELECTRIC CONDUCTORS

Not Used.

G402006 DUCTBANKS, MANHOLES AND HANDHOLES

Not Used.

G402007 GROUNDING SYSTEMS

Not Used.

G4090 SITE TELECOMMUNICATIONS UTILITIES

Provide a new telecommunications ductbank from the designated existing telecommunications manhole to the Patton Hall Strategic Engagement Platform, terminating in the Entrance Facility Telecommunications Room.

Provide a new telecommunications ductbank from the designated existing telecommunications manhole to the existing/remaining Guardhouse, terminating in the new Entrance Facility Telecommunications Room.

G409007 PHOTOVOLTAIC ENERGY SYSTEM

Not Used.

END OF SECTION